

THE LIMITS OF VERIFIABILITY: CREDIBILITY AND FLEXIBILITY IN COMMUNICATION

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Abstract

We compare verifiable and unverifiable communication in a sender–receiver setting with partially aligned preferences, where the sender knows more than what her evidence can prove. We identify a credibility–flexibility tradeoff: tying the sender’s claims to her evidence enhances her credibility, but limits how flexibly she can communicate when her evidence does not accurately reflect her private information. This tradeoff changes the economics of verifiable disclosure. When preferences are sufficiently aligned, full evidence disclosure is neither sustainable in equilibrium nor efficient. We show that verifiability facilitates communication when preferences are sufficiently misaligned, but hinders it when they are sufficiently aligned, suggesting that institutions that impose verification need not always improve information transmission. Finally, we study how verifiable and unverifiable communication can complement one another when both are available: unverifiable communication can contextualize verifiable evidence when the latter is misleading.

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1 Introduction

The efficiency of decision making in economic, political, and legal institutions often depends on the accuracy of information transmission from experts to decision makers. Organizations rely on information from their local managers, regulators on information from agencies, courts on information from witnesses. These institutions have evolved a wide range of rules to govern communication. For instance, some organizations require managers to report through standardized performance metrics, while others rely more heavily on informal qualitative assessments; some federal agencies must support proposed regulations with audited cost-benefit analyses, whereas others advise policymakers through more informal briefings; adversarial legal systems impose strict evidentiary rules, whereas inquisitorial systems give the court broader latitude in considering testimony and opinions. This variation is partly a matter of institutional design. An institution can establish procedures for collecting and verifying evidence and restrict admissible communication to claims supported by that evidence, thereby creating a regime of *verifiable* communication. When no such procedures are in place, communication is *unverifiable*: claims are not tied to evidence, and their content is disciplined only by equilibrium incentives.

Although these two communication regimes have largely been studied separately in economics, verifiable communication has commonly been viewed as superior, in the sense that it facilitates information transmission and thereby alleviates information asymmetries. Indeed, in many settings, verifiable communication provides the sender with powerful incentives to voluntarily reveal her private information, even when that information is unfavorable to her (see, e.g., [Milgrom, 2008](#)). These incentives can persist even when the sender's and receiver's preferences are partially aligned ([Seidmann and Winter, 1997](#)). More generally, like commitment, verifiability is often viewed as a device through which the informed party can overcome the credibility limitations of cheap talk ([Sobel, 2013](#)).¹

This paper qualifies this perspective. We introduce a model of verifiable disclosure in which the sender knows more than her evidence can prove. This framework identifies a credibility–flexibility tradeoff that governs the comparison between verifiable and unverifiable commu-

¹This perspective is perhaps even more widespread in accounting, where verifiability has long been regarded as a central criterion of measurement ([Ijiri, 1975](#)) and remains prominent in modern accounting guidance. The [International Accounting Standards Board \(2018\)](#)'s conceptual framework, for instance, calls for maximizing verifiability to the extent possible.

nication. Verifiability facilitates information transmission when sender and receiver are sufficiently misaligned, but hinders it when their preferences are sufficiently aligned. The idea rests on a simple observation. Verifiability shapes communication in two opposing ways. On the one hand, it strengthens the sender’s *credibility* by giving her claims a meaning that is, at least partly, determined by exogenous evidence. On the other hand, it limits the sender’s *flexibility* by preventing her from freely tailoring her message to what she privately knows. Whether verifiability ultimately improves communication depends on the balance between these two forces. The same forces also creates the scope for studying how verifiable and unverifiable communication interact, and potentially complement one another, when both are available.

To formalize these ideas, we study a model in which a sender communicates with a receiver in order to influence her behavior. The sender’s and receiver’s payoffs—which depend on an unknown payoff-relevant state and on the receiver’s action—are partially aligned, as in [Crawford and Sobel \(1982\)](#): a bias parameter governs the degree of misalignment. The sender privately observes both the state, which is unverifiable, and the realization of an exogenous signal that is partially informative about the state. She communicates with the receiver either by announcing which signal she has received or by remaining silent.

We first compare two communication regimes, captured by two alternative versions of this model. In one, signal disclosure is *unverifiable*, meaning that the sender can announce any signal, regardless of whether it was in fact realized. This version of the model corresponds to a standard cheap-talk environment with a restricted message space. In the other, signal disclosure is *verifiable*, meaning that the sender can announce that she received a given signal only if that signal was in fact realized. To make the comparison as transparent as possible, we abstract from additional frictions—including costly disclosure, endogenous evidence acquisition, and moral hazard—that could independently affect the role of verifiability (see, e.g. [Milgrom, 2008](#)).

Our model of verifiable disclosure differs from typical disclosure models in two important respects, which drive many of our results. First, the sender knows more than her evidence can prove: she observes the state, but can verifiably disclose only the realization of a signal about it. Second, the sender’s preferences are partially aligned with those of the receiver. These two ingredients change the economics of verifiable disclosure. Standard disclosure models typically assume that the sender can verifiably disclose her private information, or that her bias is sufficiently strong that she always prefers a higher receiver action regardless of the state,

or both. In such settings, verifiability is valuable because the state-dependent restrictions it imposes on the sender’s strategy discipline her claims and thereby enhance her credibility. The same force is present in our setting, but it is now opposed by a second one. Because evidence is noisy, its realization need not accurately represent what the sender privately knows about the state. When preferences are partially aligned, the sender would benefit from using language flexibly to convey information that the evidence does not fully capture. Verifiability limits that flexibility and can thus hinder rather than facilitate communication. This creates a tradeoff absent in the traditional literature: a tension between credibility and flexibility.

To understand how this tradeoff reshapes verifiable disclosure and to lay the groundwork for our later comparison with unverifiable communication, we begin by characterizing the equilibrium set under verifiability, which is richer than in standard disclosure models. A natural point of departure is to ask whether an analogue of the unraveling argument holds in our setting: can the outcome in which the receiver learns the entire verifiable content of the sender’s private information—here, the realized signal—be supported in equilibrium? We call this the *full-evidence-disclosure* outcome. Proposition 1 shows that it can be sustained in equilibrium if and only if the sender’s bias is sufficiently large. Thus, when the sender and receiver are sufficiently aligned, there is no equilibrium in which the receiver learns exactly the realization of the sender’s signal. The obstacle to full disclosure is finding an interpretation of silence that makes every sender type prefer disclosing her evidence. When the bias is large, a sufficiently pessimistic interpretation accomplishes this. When preferences are sufficiently aligned, however, whatever action silence induces is close to the ideal action of some sender types: these types prefer to withhold their noisy evidence because it is particularly misleading.

This intuition suggests that silence may be used to reveal more than the evidence itself. Proposition 2 formalizes this idea. When preferences are sufficiently aligned, every receiver-optimal equilibrium under verifiable communication is strictly more informative than the full-evidence-disclosure outcome: the receiver learns strictly more than the content of the sender’s evidence. Therefore, even if full evidence disclosure could be externally enforced, it would be inefficient. In contrast to the standard literature, silence in these equilibria is not used to conceal unfavorable evidence, but to withhold evidence that would be misleading in light of the sender’s private information. A direct policy implication is that disclosure mandates can backfire: by eliminating the informational role of silence, they reduce information transmission.

Next, we compare receiver-optimal equilibria under the two alternative communication regimes. Proposition 3 formalizes the tradeoff between credibility and flexibility introduced above. When preferences are sufficiently misaligned, verifiable communication is more informative than unverifiable communication: by restricting the sender’s strategy, verifiability curbs imitation and enhances her credibility, facilitating information transmission. Conversely, when preferences are sufficiently aligned, the ranking is reversed: the restrictions imposed by verifiability become an obstacle to information transmission, preventing the sender from using language flexibly to communicate her private information.

To understand the tradeoff in Proposition 3, we ask whether the value of verifiable communication comes primarily from the fact that the evidence is intrinsically informative about the state or from the fact that verifiability restricts the sender’s strategy. To disentangle these two forces, we first study our disclosure model when evidence is statistically uninformative, thereby shutting down the first channel. We find that verifiable communication can still be informative. Perhaps more surprisingly, it can outperform unverifiable communication for an intermediate range of biases. The reason is that, although the realized signal reveals nothing about the state, its stochastic availability restricts which messages each sender type can send, thereby limiting imitation and enhancing credibility. Second, we show that making evidence more informative in the Blackwell sense need not improve communication: for intermediate biases, it can instead reduce equilibrium informativeness. Together, these results show that what fundamentally drives the value of verifiability is not only the information contained in the evidence itself, but also the constraints that it imposes on the sender’s strategy.

Finally, we study how verifiable and unverifiable communication interact when both are available. This communication mode arises naturally in many applications, yet has received very little attention in the literature. One reason for this gap is that, in classic disclosure models, verifiable disclosure leads to full disclosure of the sender’s private information, leaving no scope for meaningful interaction with cheap talk. Our model suggests otherwise. The same credibility–flexibility tradeoff discussed above creates scope for nontrivial interactions: because the sender knows more than her evidence can prove, unverifiable communication can *complement* verifiable disclosure by conveying what the evidence cannot certify.

We therefore extend our baseline model to a hybrid game in which the sender sends both a verifiable and an unverifiable message. Two lessons emerge. First, the two channels can be

complements. For every bias level, there exists a class of equilibria in which the sender always discloses her evidence and then uses unverifiable communication to qualify its interpretation (Proposition 4). The ability to contextualize evidence that would otherwise be misleading is precisely what makes its disclosure sustainable, in contrast with Proposition 1. Second, despite this complementarity, disclosure mandates need not be innocuous. Requiring disclosure can alter the beliefs under which unverifiable communication takes place and thereby undermine informative communication that would otherwise be sustainable (Remark 2). Absent a disclosure mandate, however, the availability of both channels benefits the receiver, since the hybrid game can reproduce the equilibrium outcomes attainable under either mode alone (Remark 3).

Empirical Applications. One application of our framework concerns the design of internal reporting protocols within organizations. Lower-level managers often possess information about employee performance or operational problems that formal metrics cannot fully capture. Some organizations choose to transmit this information through relatively flexible and informal protocols—such as qualitative updates and subjective assessments; others rely more heavily on standardized and formal protocols, such as key performance indicators (KPIs) and other evidence-based metrics. Our model predicts that flexible protocols should be more prevalent in smaller organizations, where incentives tend to be more closely aligned, whereas standardized protocols should be more prevalent in large firms and bureaucracies, where incentive conflicts tend to be greater. A partial illustration of this pattern is documented by [Liberti and Mian \(2009\)](#). They find that greater hierarchical distance between loan officers and the supervisors who approve loans reduces reliance on soft information and increases reliance on hard information. Similarly, [Qian et al. \(2015\)](#) show that, following a reform that decentralized decision-making in Chinese banking, approval decisions became more informative when branch president–loan officer pairs had worked together for longer periods. A related literature studies how geographic distance shifts decision-making from soft to hard information: [Petersen and Rajan \(2002\)](#) show that small firms increasingly borrow from more distant lenders and interact with them in more impersonal ways, while [Agarwal and Hauswald \(2010\)](#) show that proximity facilitates the collection of soft information.

A second application concerns financial reporting by private firms. In the United States, these firms are not legally required to produce audited financial statements, and indeed a large share do not. In this setting, [Lisowsky and Minnis \(2020\)](#) show that, controlling for other firm

characteristics, the prevalence of audits rises with ownership dispersion, a feature plausibly associated with greater incentive misalignment. This pattern is broadly consistent with the predictions of our model.

Related Literature. Our paper contributes primarily to the literature on verifiable disclosure (see Ben-Porath et al., 2026a; Dranove and Jin, 2010; Beyer et al., 2010; Milgrom, 2008, for reviews). Standard disclosure models feature a sender with monotonic preferences in the receiver’s action and evidence that can fully certify the sender’s type (see, e.g., Grossman, 1981; Milgrom, 1981). Under these conditions, unraveling is the unique equilibrium outcome. Subsequent work asks how far unraveling extends beyond this benchmark. One strand relaxes monotonicity by allowing for more general, state-dependent preferences (e.g., Seidmann and Winter 1997), while another considers settings in which the sender’s type is only partially provable (e.g., Mathis 2008). More generally, Okuno-Fujiwara et al. (1990) and Hagenbach et al. (2014) characterize conditions for unraveling in Bayesian games with certifiable information. Unraveling can fail when the sender’s bias changes sign across types (see also Giovannoni and Seidmann, 2007), or when evidence is probabilistically unavailable (e.g., Dye, 1985). The baseline disclosure model has also been enriched along several dimensions by introducing additional frictions and design margins. The literature studies settings in which disclosure is costly (e.g., Jovanovic, 1982; Verrecchia, 1983), evidence is acquired endogenously (e.g., Matthews and Postlewaite, 1985; Shavell, 1994; Lizzeri, 1999), the evidence structure is itself a design variable (e.g., DeMarzo et al., 2019; Ali et al., 2022; Shishkin, 2026), or disclosure affects endogenous productive decisions (e.g., Ben-Porath et al., 2018).

By contrast, we study a deliberately parsimonious disclosure model that abstracts from these additional frictions, allowing us to isolate how two basic ingredients of the problem—noisy evidence and partially aligned preferences—shape the value of verifiability. Several papers study disclosure when evidence is noisy (e.g., Fishman and Hagerty, 1990; Di Tillio et al., 2021; Bull and Watson, 2019; Ali et al., 2022; Farina et al., 2025).² These papers typically assume that the sender always prefers a higher receiver action. As a result, whether the sender also observes the underlying state is inessential for the receiver-optimal Perfect Bayesian equilibrium, whereas it plays a central role in our analysis.

One exception that combines partially aligned preferences with noisy evidence is Haghtalab

²Ball and Kattwinkel (2025) and Ben-Porath et al. (2026b) study noisy evidence in mechanism design.

et al. (2024). In their model, the sender observes n noisy signals and must disclose exactly one of them. When $n = 1$ —the case closest to our model—disclosure of the realized signal is effectively mandated. Thus, the credibility–flexibility tradeoff we study is absent. Less closely related in terms of the model, but perhaps closer in spirit, is Bertomeu and Cheynel (2026), who study communication with partially aligned preferences, noisy private information, and an imperfect language. In their model, as in ours, what can be communicated is an imperfect representation of the sender’s private information. As a result, literal truthfulness need not coincide with accurate communication, and requiring closer adherence to the facts can reduce welfare.

Finally, our paper contributes to a limited literature on the interactions between verifiable and unverifiable communication. Eső and Galambos (2013) study a sender with Crawford–Sobel preferences who chooses either to communicate through cheap talk or to provide costly, fully revealing hard evidence. Dasgupta (2023) studies a similar either-or structure, but with costless disclosure and state-independent sender preferences, asking whether the availability of cheap talk disrupts the classic unraveling result. Closer to our setting, Bertomeu and Marinovic (2016) allow the sender to provide both hard and soft information and show, among other results, that mandatory disclosure of hard information can reduce the transmission of soft information; their mechanism operates through costly misreporting and uncertainty about whether the sender’s report is verified, under state-independent preferences and therefore differs substantially from ours.

2 Model

We study a model in which a sender privately observes a state and sends a message to the receiver, who then chooses an action that affects both players’ payoffs. The sender is partially biased, in that she seeks to induce an action that is higher than what the receiver would like. The sender communicates by deciding whether or not to disclose a noisy signal whose realization is imperfectly informative about the state. We will compare two communication regimes: in the first, signal disclosure is verifiable; in the second, it is not.

More formally, let $\theta \in \Theta \triangleq [0, 1]$ be an unverifiable state drawn from an atomless prior distribution F with full support. Let (S, π) be an *evidence structure*, where $S = \{s_1, \dots, s_N\}$ is

a finite and ordered set of possible signal realizations, and $\pi : \Theta \rightarrow \Delta(S)$ is Borel measurable and specifies the conditional distribution of signals given the state. We assume $|S| = N \geq 2$. For every θ , the conditional signal distribution has full support, that is $\pi(s \mid \theta) > 0$ for all s . In other words, every signal can arise from every state. For expositional convenience, we also assume that (S, π) satisfies the monotone likelihood ratio property (MLRP). Let F_s denote the posterior distribution of θ conditional on signal s . We say that (S, π) is *informative* if there exist $s, s' \in S$ such that $F_s \neq F_{s'}$, and *uninformative* otherwise.

The sender privately observes (θ, s) , her *type*. We will often refer to the realized signal s as the sender's *evidence*. Conditional on her type, the sender sends a message $m \in M$ to the receiver. The message space is $M \triangleq S \cup \{\circ\}$, where $\circ$ is a shorthand symbol we interpret as "silence." After observing m , the receiver chooses an action $a \in A \triangleq \mathbb{R}$.

We study our model under two alternative communication technologies. Evidence is *verifiable* if a type- (θ, s) sender can either disclose s or remain silent; that is, $m \in M(\theta, s) \triangleq \{s, \circ\}$. Evidence is *unverifiable* if a type- (θ, s) sender can send any message in M ; that is, $m \in M(\theta, s) \triangleq M$. We refer to the resulting games as the game with verifiable evidence and the game with unverifiable evidence, respectively.

We now describe the players' payoffs, which follow the standard specification of [Crawford and Sobel \(1982\)](#). Let $u : A \times \Theta \times \mathbb{R}_+ \rightarrow \mathbb{R}$ be twice continuously differentiable, with subscripts denoting partial derivatives. The receiver's payoff is $u^{\mathcal{R}}(a, \theta) \triangleq u(a, \theta, 0)$ and the sender's payoff is $u^{\mathcal{S}}(a, \theta, b) \triangleq u(a, \theta, b)$. Thus, when $b = 0$, preferences coincide. We make the following assumptions on u . First, assume $u_{11} < 0 < u_{12}$, so that both players' payoffs are strictly concave in the receiver's action and supermodular in (a, θ) . Let $a^{\mathcal{R}}(\theta)$ and $a^{\mathcal{S}}(\theta, b)$ denote the players' unique preferred actions. Without loss of generality, we normalize u so that $a^{\mathcal{R}}(0) = 0$ and $a^{\mathcal{R}}(1) = 1$. This reduces the effective action space to $[0, 1]$. Second, assume $u_{13} > 0$. Then $a^{\mathcal{S}}(\theta, b)$ is strictly increasing in b , and hence $a^{\mathcal{S}}(\theta, b) > a^{\mathcal{R}}(\theta)$ whenever $b > 0$. That is, b captures the sender's *upward bias*, with larger values corresponding to greater preference misalignment. Finally, let $\lim_{b \rightarrow \infty} u_1(a, \theta, b) > 0$ for all (a, θ) , so that for sufficiently large bias the sender prefers higher actions regardless of the state.

A sender's strategy is a map $\sigma : \Theta \times S \rightarrow \Delta(M)$. When evidence is verifiable, the sender's strategy is restricted to satisfy the verifiability constraint introduced above: for all (θ, s) , $\sigma(m \mid \theta, s) = 0$ whenever $m \notin \{s, \circ\}$. When evidence is unverifiable, the sender's strategy is unre-

stricted. A receiver's strategy is a map $\alpha : M \rightarrow \Delta(A)$. Without loss of generality, we focus on pure strategies.³ A belief system is a map $\mu : M \rightarrow \Delta(\Theta \times S)$. The triple (σ, α, μ) is an assessment. The solution concept is Perfect Bayesian Equilibrium (PBE), which we often simply call an equilibrium. The definition of a PBE in our setting is standard and is therefore relegated to Appendix B.2.

To illustrate the model, we now discuss a simple example of our setting that we will use throughout the paper.

Example 1. The prior distribution F is uniform, and payoffs are quadratic with an additive sender bias: $u^R(a, \theta) = -(a - \theta)^2$ and $u^S(a, \theta, b) = -(a - \theta - b)^2$. Thus, the receiver's preferred action is $a^R(\theta) = \theta$, whereas the sender's is $a^S(\theta, b) = \theta + b$. Evidence is binary, $S = \{s_L, s_H\}$, with $\pi(s_H | \theta) = k\theta + \frac{1-k}{2}$ for $k \in [0, 1]$, and $\pi(s_L | \theta) = 1 - \pi(s_H | \theta)$. For every k , the two signals occur with equal ex ante probability. A higher k corresponds to more informative evidence in the Blackwell sense. $\triangle$

Discussion. We briefly pause to highlight some aspects of our model.

Our game with unverifiable evidence is a variation of a standard cheap-talk game with an exogenously restricted alphabet of $N + 1$ messages.⁴ This restriction is expositional: it makes the comparison with the game with verifiable evidence particularly transparent, as the two games differ only in the sender's flexibility in choosing which message to send, not in the richness of the available language. Under unverifiable evidence, moreover, the evidence structure (S, π) , despite possibly being informative about the state, does not shape the equilibrium set beyond pinning down the size of the message alphabet (see Appendix B.2).

Our main innovation in the game with verifiable evidence is to combine two ingredients that are critical for our results. First, the sender's and receiver's preferences are partially aligned, as captured by the bias parameter $b \geq 0$. Second, the sender's verifiable evidence is a noisy signal of her private type. As a result, the sender cannot fully certify her type, but only the realization

³For the receiver, this follows from the strict concavity in a of her payoff. For the sender, we show that, under both communication technologies, every equilibrium outcome can be induced by an equilibrium in which her strategy is pure. See Lemmas A.1 and A.6 in the Appendix.

⁴Jäger et al. (2011), Blume and Board (2013), and Sobel (2016), among others, study cheap-talk communication with an exogenously restricted language.

of the signal s .⁵ For simplicity, we restricted verifiable disclosure to be all-or-nothing, i.e., $M(\theta, s) = \{s, \circ\}$, an assumption that isolates the credibility–flexibility tradeoff in its simplest and starkest form; we discuss richer verifiable languages in Section 6.

The two games can be interpreted as polar communication protocols. The game with unverifiable evidence captures the familiar case in which certification is unavailable or prohibitively costly. Our game with verifiable evidence requires a more careful interpretation. In a standard disclosure model—where the sender’s bias is large—a supplementary cheap-talk channel would be uninformative, so omitting it is without loss. In our setting, by contrast, cheap talk could be informative. Its absence must therefore be understood as an institutional restriction. This restriction need not mean that the sender is literally unable to make unverifiable claims. Rather, it can reflect an institutional commitment, which is especially relevant in large organization such as bureaucracies, to base decisions only on verifiable information. [Milgrom and Roberts \(1988\)](#) provide a rationale for this type of restriction: organizations may deliberately limit the information to which decision makers can respond, even at the cost of foregoing valuable information, because greater responsiveness would encourage influence activities that divert time and effort from more productive tasks.

Our main interest is in the *comparison* between these two modes of communication, which allows us to address our central question: does making information verifiable necessarily facilitate information transmission? Sections 3 and 4 develop this comparison. Our framework also allows us to study how the two modes *interact* when they are available together, which we turn to in Section 5.

2.1 Outcomes and Informativeness

Before beginning the analysis, we establish some general terminology that will be used throughout the paper.

An *outcome* of the game, $\lambda \in \Delta(\Theta \times A)$, is a joint distribution over the state and the action, and describes the result of the interaction between the sender and the receiver. Each strategy

⁵Many real-world disclosure settings share these features. A manager may know more than standardized metrics capture; an agency may know more than a cost-benefit analysis can quantify; and an expert witness may know more than admissible evidence conveys. In each case, the verifiable evidence certifies a coarse summary of what the sender knows, and the degree of preference alignment can vary.

profile induces an outcome. An outcome is *uninformative* if the receiver's action is independent of the state. It is said to be *informative* otherwise.

Given an outcome λ , let $U(\lambda) \triangleq \mathbb{E}_\lambda[u^\mathcal{R}(a, \theta)]$ denote the receiver's expected payoff. We use the receiver's expected payoff as our measure of equilibrium informativeness.

Definition 1. Let λ and λ' be two outcomes. We say that λ is **more informative** than λ' if $U(\lambda) \geq U(\lambda')$.

In both versions of the game—with verifiable or unverifiable evidence—there are typically multiple equilibria.⁶ An equilibrium that is of special interest to us is the one that is most informative from the receiver's perspective. We call such an equilibrium **receiver-optimal**.⁷ Receiver-optimal equilibria exist under both communication technologies (see Appendix B.2).

Finally, we define another important outcome that we will use extensively. It is the analogue, in our setting with noisy evidence, of the unraveling outcome in the standard disclosure model. We call it *full evidence disclosure*: the sender always discloses her realized signal, and the receiver best responds to it. Before stating the formal definition, recall that $a^\mathcal{R}(\theta)$ denotes the receiver's unique preferred action in state θ , and that F_s denotes the posterior distribution of θ induced by signal s . Let us extend the notation and write $a^\mathcal{R}(F_s)$ for the receiver's optimal action upon observing s .

Definition 2. Suppose the sender always discloses her realized signal and the receiver best responds to it. The outcome induced by this strategy profile is the **full-evidence-disclosure (FED) outcome**. An equilibrium is a **FED equilibrium** if it induces this outcome.⁸

⁶With unverifiable evidence, equilibrium multiplicity is standard. With verifiable evidence, less so. It arises here because the sender knows more than what her evidence can prove; in particular, an uninformative equilibrium always exists. Multiplicity can also arise without this informational gap under other preference conditions (see Ali et al. (2026)).

⁷In settings with quadratic preferences, such as in Example 1, the sender's and receiver's ex ante payoffs differ only by a constant. Hence, a receiver-optimal equilibrium is Pareto optimal.

⁸Formally, let the sender's strategy be $\sigma(m = s \mid \theta, s) = 1$ for all (θ, s) , and let the receiver's strategy satisfy $\alpha(s) = a^\mathcal{R}(F_s)$ for every s . The outcome induced by this strategy profile is the full-evidence-disclosure outcome. Note that multiple strategy profiles may induce this outcome.

3 Communicating with Verifiable Evidence

This section analyzes the game with verifiable evidence. Unlike in standard disclosure models, its equilibrium set is quite rich. Characterizing this set reveals the forces that shape information transmission under verifiability and lays the groundwork for our later comparison between verifiable and unverifiable evidence.

We begin by characterizing under what conditions a full-evidence-disclosure equilibrium exists. That is, we ask whether an analogue of the unraveling result also holds in our setting with noisy evidence. Our first result establishes that the existence of a full-evidence-disclosure equilibrium critically depends on the sender's bias. To state this result, let $\bar{b}$ denote the unique sender's bias that satisfies $u^S(0, 0, \bar{b}) = u^S(a^R(F_{s_N}), 0, \bar{b})$. In other words, for any $b \geq \bar{b}$, even the lowest-state sender weakly prefers the highest action induced under full evidence disclosure, $a^R(F_{s_N})$, to the lowest feasible action, 0.

Proposition 1. *Consider the game with verifiable evidence. Suppose that evidence is informative. A full-evidence-disclosure (FED) equilibrium exists if and only if $b \geq \bar{b}$.*

The positive part of Proposition 1, namely the existence of a FED equilibrium for large b , is in line with the disclosure literature, which has largely focused on settings in which the sender's bias is sufficiently large that her payoff is increasing in the receiver's action at every state. Our existence result follows a similar logic. We show that when $b \geq \bar{b}$, a full-evidence-disclosure equilibrium can be supported by “punishing” deviations with off-path beliefs that are sufficiently pessimistic, for instance by specifying that the receiver's belief after silence induces action 0, the lowest receiver-optimal action.

The more novel insight of Proposition 1 is its negative clause: when the sender's and receiver's preferences are sufficiently aligned, there is no equilibrium that induces the full-evidence disclosure outcome. To build intuition, consider the extreme case of perfectly aligned preferences, $b = 0$, and suppose by contradiction that a FED equilibrium exists. Let $a_o \in [0, 1]$ denote the receiver's action following silence. Since $a^R(\theta)$ is continuous and ranges from 0 to 1, there exists a state θ_o such that $a^R(\theta_o) = a^S(\theta_o, 0) = a_o$. Since evidence is informative, there exists a signal s such that $a^R(F_s) \neq a_o$. Moreover, every signal can arise from every state. Under the FED outcome, a type- (θ_o, s) sender must induce action $a^R(F_s)$. But her pre-

ferred action is exactly $a_\circ$, so she strictly prefers deviating to silence, a contradiction.⁹

This argument reflects a more general “separating” force: when $b < \bar{b}$, there is no way to assign a single action in response to silence that is uniformly unattractive to all sender types and thus deters all deviations from the FED outcome. The reason is not only that the sender knows more than what evidence can prove, but also that, conditional on any signal, different types still prefer different actions. Whatever action $a_\circ$ silence induces, some types find it attractive relative to the action they would induce under FED. This separating force prevents the FED outcome from being sustained, but also creates scope for equilibrium communication to distinguish types that share the same evidence and thereby transmit information beyond what is encoded in the evidence itself. In other words, when preferences are sufficiently aligned, the receiver-optimal equilibrium outcome may be strictly more informative than the full-evidence-disclosure outcome.

The following result formalizes this intuition. To state it, we introduce a new threshold $\bar{B}$, which is strictly higher than $\bar{b}$, and is the unique bias level at which $a^S(0, \bar{B}) = 1$. In other words, for any $b \geq \bar{B}$ and every state θ , the sender’s payoff is strictly increasing on $[0, 1]$, the relevant range of receiver actions. That is, the sender always prefers a higher receiver action, regardless of the state, as in the standard disclosure setting.

Proposition 2. *Consider the game with verifiable evidence.*

1. *If $b \geq \bar{B}$, every receiver-optimal equilibrium is as informative as the full-evidence-disclosure outcome.*
2. *If b is sufficiently small, every receiver-optimal equilibrium is strictly more informative than the full-evidence-disclosure outcome.*
3. *For every $b < \bar{B}$, there is a prior F and an evidence structure (S, π) satisfying the assumptions of Section 2 such that every receiver-optimal equilibrium is strictly more informative than the full-evidence-disclosure outcome.*

The first clause in this proposition is the analogue of the classic unraveling results in **Milgrom (1981)** and **Grossman (1981)**. When $b \geq \bar{B}$, every sender type prefers higher receiver actions

⁹By continuity, there is in fact a positive-measure set of types (θ, s) with θ sufficiently close to $\theta_\circ$ who strictly prefer deviating to silence, which induces $a_\circ$, rather than inducing the FED action $a^R(F_s)$.

over the relevant action range. In this case, not only does a full-evidence-disclosure equilibrium exist, as already established by Proposition 1, but it is also receiver-optimal. This leads to the familiar conclusion that policies mandating disclosure are, in principle, ineffective when the sender and the receiver are strongly misaligned: a receiver-optimal equilibrium already achieves the same outcome implemented by such policies.

The rest of Proposition 2 establishes a converse to this result. The second clause shows that when the sender's and receiver's preferences are sufficiently aligned, there exist equilibria that are more informative than the full-evidence-disclosure outcome. To prove this result, we first analyze the $b = 0$ case, in which the sender and the receiver have common interests. We study an auxiliary problem in which the sender can commit to a verifiable communication rule to maximize the receiver's expected payoff, thus abstracting from interim incentive constraints. We show that the optimal commitment solution induces an outcome that is more informative than the full-evidence-disclosure outcome. The proof argument highlights that, when $b = 0$, the sender exploits the available language more flexibly than under FED. Finally, we show that this commitment outcome can be supported in equilibrium. A continuity argument then shows that the receiver-optimal equilibrium payoff is right-continuous at $b = 0$, extending the conclusion to positive but small b .

Finally, the third clause shows that for every $b < \bar{B}$, there exist a prior F and an evidence structure (S, π) such that every receiver-optimal equilibrium is strictly more informative than the FED outcome. In the proof, we use an evidence structure that is statistically uninformative, so the FED outcome itself is uninformative. Even then, verifiability can sustain an informative equilibrium because the realized signal constrains which types can send which messages.

A policy implication of Proposition 2 is that, in settings in which the sender's bias is not extreme, mandated disclosure can be detrimental to information transmission. The sender can exploit her knowledge of the state, together with the flexibility afforded by the silence message, to communicate more than what is encoded in her evidence. This is often, though not always, accomplished by putting all messages on the equilibrium path. In particular, silence can be used productively by the sender, which explains why allowing the sender discretion over whether to remain silent can improve communication.

We conclude this discussion by emphasizing that the results in this section depend on the assumption that the sender knows more than what her evidence can prove. To see this, consider

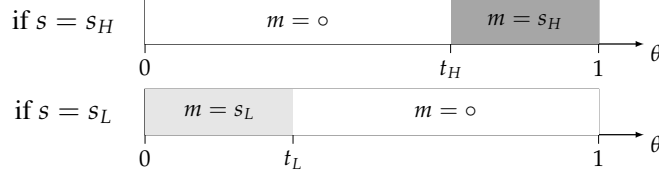


Figure 1: A depiction of a sender's evidence-concordant strategy.

a variant of our model in which the sender only observes s and not θ , so that evidence disclosure exhausts the sender's private information. In this case, for all $b \geq 0$, a full-evidence-disclosure equilibrium exists and is receiver-optimal, in stark contrast with Propositions 1 and 2 (see Proposition B.3 in the Online Appendix).

Example 2. To illustrate the results so far, we return to the setting of Example 1 and let $k = 1$. That is, the prior is uniform, the payoffs are quadratic, the bias is additive, the signals are binary, and their distribution is linear, i.e., $\pi(s_H|\theta) = \theta$. In the FED outcome, the receiver's actions are $a_{s_L} \triangleq a^{\mathcal{R}}(F_{s_L}) = \frac{1}{3}$ and $a_{s_H} \triangleq a^{\mathcal{R}}(F_{s_H}) = \frac{2}{3}$, with equal probability. The receiver's expected payoff under this outcome is $-\frac{1}{18}$. By Proposition 1, the FED outcome can be supported in equilibrium if and only if $b \geq \bar{b} = \frac{a_{s_H}}{2} = \frac{1}{3}$. Moreover, by Proposition 2, the FED outcome is receiver-optimal for $b \geq \bar{B} = 1$, whereas for sufficiently small b it is neither sustainable in equilibrium nor receiver-optimal.

We next introduce a class of equilibria that are more informative than the FED outcome for low b and converge to a FED equilibrium as b increases. Fix cutoffs $t_L, t_H \in [0, 1]$. Consider a sender who discloses s_H if $\theta \geq t_H$ and discloses s_L if $\theta \leq t_L$; otherwise, she remains silent. Thus, evidence is disclosed only when it is “concordant” with the state, while silence pools the discordant realizations: good evidence when the state is low and bad evidence when the state is high. We call such a strategy *evidence concordant* (see Figure 1).

To illustrate, suppose $b = 0$. Let $t_H \cong \frac{2}{3}$ be the unique feasible root of $8t^2 + 5t = 7$ and let $t_L = 1 - t_H$. The evidence-concordant strategy then induces three on-path actions: $a_H = 2t_H - \frac{1}{2} \cong \frac{5}{6}$, $a_L = 2t_L - \frac{1}{2} \cong \frac{1}{6}$, and $a_o = \frac{1}{2}$. This strategy constitutes an equilibrium and gives the receiver an expected payoff of approximately -0.0174 , substantially higher than the FED payoff $-\frac{1}{18}$.

More generally, we can show that an evidence-concordant equilibrium exists and is unique for every $b \geq 0$. As b increases, both cutoffs (t_L, t_H) decrease, so the sender becomes less likely to disclose s_L and more likely to disclose s_H . At $b = \frac{1}{2}$, both cutoffs reach zero: s_H

is disclosed, while silence perfectly reveals s_L . Hence, the evidence-concordant equilibrium becomes a FED equilibrium. It remains a FED equilibrium for all $b \geq \frac{1}{2}$. These comparative statics also imply that the receiver's payoff, and hence equilibrium informativeness, decreases with b , and strictly so whenever $b < \frac{1}{2}$ (see Figure 2).

Even in this simple environment, there can be eight classes of informative equilibria, each of which can be characterized analytically.¹⁰ Thus, it is feasible to compare them numerically to show that the evidence-concordant equilibrium is receiver-optimal. Beyond this example, however, receiver-optimal equilibria need not be evidence concordant, uniqueness within an equilibrium class may fail, equilibrium cutoffs need not vary monotonically with b , and the structure of receiver-optimal equilibria may itself change with b . These observations illustrate how complex the game with verifiable evidence can be in general. $\triangle$

4 Comparing Verifiable and Unverifiable Communication

The previous section shows how silence can improve information transmission. Because silence is available to every sender type, it provides the sender with some flexibility even within the game with verifiable evidence. This observation raises a natural question: what happens when the sender has unrestricted flexibility over the message she sends, as in the game with unverifiable evidence? We answer this question in Section 4.1. Then, in Sections 4.2 and 4.3, we study how the answer depends on the informativeness of the verifiable evidence.

4.1 Flexibility vs Credibility

We begin by comparing receiver-optimal equilibrium outcomes in the games with verifiable and unverifiable evidence. Recall that, under verifiable evidence, a type- (θ, s) sender must choose a message in $M(\theta, s) = \{\circ, s\}$, whereas under unverifiable evidence she can freely choose any message in $M = \{\circ\} \cup S$. Thus, unverifiable evidence gives the sender greater flexibility in how she communicates.

¹⁰Lemma A.1 in the Appendix shows that, in the general model, every equilibrium under verifiable evidence is outcome-equivalent to one in which the sender follows a cutoff rule: for each signal, she either discloses it below some cutoff, discloses it above some cutoff, or never discloses it. With binary evidence, this leaves at most eight classes of informative equilibria.

Our next result highlights the tradeoff created by this additional flexibility in the sender's communications.

Proposition 3. *There exist thresholds $0 < \beta_0 < \beta_1 < \beta_2 \leq \infty$ such that*

1. *For all $b \leq \beta_0$, receiver-optimal equilibria of the game with verifiable evidence are strictly **less informative** than those of the game with unverifiable evidence.*
2. *For all $b \in (\beta_1, \beta_2)$, receiver-optimal equilibria of the game with verifiable evidence are strictly **more informative** than those of the game with unverifiable evidence. Moreover, if the evidence structure is informative, then $\beta_2 = \infty$, so the comparison holds for all $b > \beta_1$.*

We first focus on the simpler case in which the evidence is informative, deferring the uninformative case to Section 4.2. Proposition 3 shows that neither communication regime is uniformly more informative across bias levels: verifiability facilitates communication when the bias is large, but hinders it when the bias is small. This result qualifies the common view that verifiability necessarily improves information transmission. Importantly, by holding the message alphabet fixed and varying only whether the sender's access to each message is constrained by her realized evidence, our comparison isolates the strategic effect of verifiability itself, abstracting from additional frictions—such as costly disclosure, endogenous evidence acquisition, and moral hazard—that could independently shape its role in communication. The result also gives the choice of communication protocol an institutional-design dimension: even when certification is feasible, whether a verifiable protocol is desirable depends on the degree of preference alignment. This provides a rationale for the observed variation in reporting protocols discussed in the introduction.

We briefly discuss the proof intuition. When the sender's and receiver's preferences are sufficiently misaligned, that is, when $b > \beta_1$, flexibility in the sender's language is a liability. The proof shows that, for every $b > \beta_1$, all equilibria in the game with unverifiable evidence are uninformative. By contrast, the game with verifiable evidence can still sustain information transmission. Indeed, when evidence is informative, we show that a strictly informative equilibrium exists *for every* bias level b .¹¹

¹¹To construct such an equilibrium, let senders disclose their evidence only if they observe s_N —the most

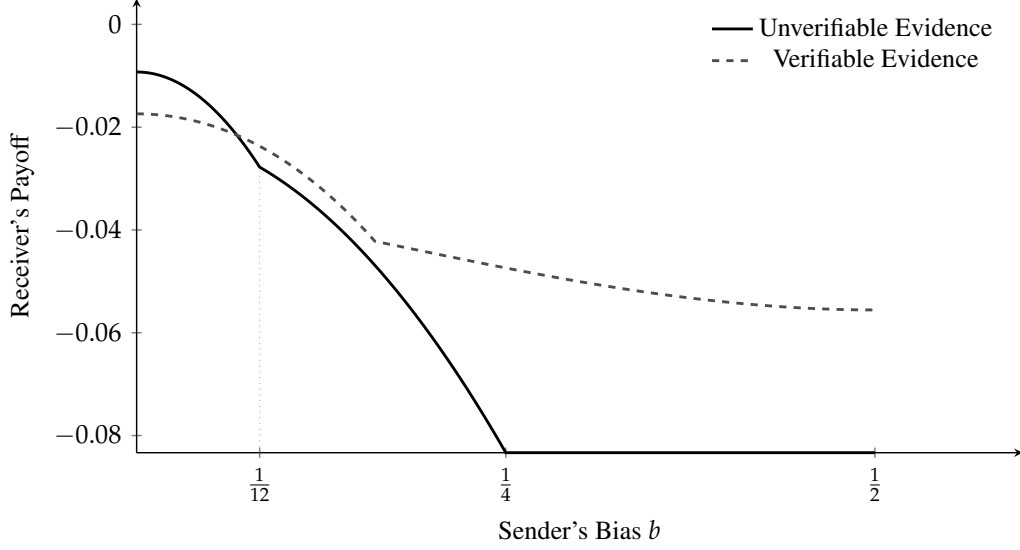


Figure 2: Receiver-optimal equilibrium payoffs of the game with verifiable (dashed) and unverifiable (solid) evidence in the setting of Example 1 with $k = 1$.

Conversely, when the sender's and receiver's preferences are sufficiently aligned, that is, when $b < \beta_0$, flexibility in the sender's language becomes a boon for communication. This is particularly salient at $b = 0$, when preferences coincide. With unverifiable communication, the sender can flexibly use the available $N + 1$ messages and convey as much information as this restricted message alphabet permits. With verifiable evidence, by contrast, the sender's strategy is constrained by the evidence she happens to observe. The key step in the proof is to show that this restriction binds at the optimum for all games: Lemma A.9 proves that, at $b = 0$, every verifiable equilibrium yields *strictly* less information than the optimal $N + 1$ -message cheap-talk equilibrium. A continuity argument then shows that the receiver-optimal equilibrium payoffs under both communication technologies are right-continuous at $b = 0$, so this strict ranking persists for all sufficiently small positive b .

Example 2 (continued). To illustrate, we return to the setting of Example 2. Recall that the prior is uniform, payoffs are quadratic with an additive bias, signals are binary, and their distribution is linear, with $\pi(s_H | \theta) = \theta$. The game with unverifiable evidence is a textbook, uniform-quadratic, cheap-talk example. Its equilibria are characterized by finite monotone partitions, with more cells corresponding to greater informativeness. Here, $\beta_1 = \frac{1}{4}$ (see Equations

favorable signal—and the state is sufficiently high. Let all other sender types remain silent. It can be shown that such a strategy can be supported in equilibrium and is informative.

tion (3) in Appendix A.3). An equilibrium with at least two cells—that is, an informative equilibrium—exists if and only if $b < \frac{1}{4}$. An equilibrium with three cells—the maximal number permitted by the language—exists if and only if $b < \frac{1}{12}$.

Figure 2 plots the informativeness of receiver-optimal equilibria under unverifiable communication together with that under verifiable communication from Example 2. It illustrates Proposition 3: verifiability leads to more informative communication when the bias is large, whereas unverifiability does so when the bias is small. $\triangle$

4.2 Is Uninformative Evidence Useful?

Our discussion so far has shown that verifiability can improve communication when the sender’s bias is sufficiently large. We have, however, focused on the case in which the evidence itself is informative. Thus, the advantage of verifiability may reflect not only the credibility created by restricting which messages each sender type can use, but also the statistical information contained in the evidence itself. Can verifiability improve communication even when the evidence is statistically uninformative, purely by restricting which messages each sender type can send?¹² Proposition 3 shows that this is possible for intermediate levels of bias, $b \in (\beta_1, \beta_2)$.

A typical informative equilibrium in this region has the following structure: senders who observe some signal $\bar{s}$ disclose it only when the state is sufficiently high, while all other senders remain silent. In this sense, uninformative evidence acts as an exogenous source of randomization. Although the realization of $\bar{s}$ carries no information about the state, only senders who happen to observe $\bar{s}$ can use that message. This restriction limits imitation across types and thereby creates the credibility needed to sustain informative communication.¹³

This result therefore points to a connection between verifiable communication and the familiar idea that exogenous noise in cheap-talk can improve information transmission by limiting imitation (see, e.g., Myerson (1991), Krishna and Morgan (2004), Blume et al. (2007),

¹²Recall that the evidence structure (S, π) is uninformative if there exists $(p_s)_{s \in S}$ with $p_s > 0$ and $\sum_{s \in S} p_s = 1$ such that $\pi(s | \theta) = p_s$ for all (θ, s) . That is, the signal realization is independent of the state.

¹³Such an equilibrium remains informative only up to the upper bias threshold β_2 . For larger b ’s, the receiver optimal equilibrium is full evidence disclosure (FED). But since the evidence itself is uninformative, the FED outcome is also uninformative. Thus, for large biases, the games with verifiable and unverifiable evidence therefore yield equally informative receiver-optimal equilibrium outcomes.

and Goltsman et al. (2009)). In these papers, noise is typically embedded in the communication channel itself and operates after the sender has chosen an otherwise unverifiable message. In our setting, by contrast, noise arises from the stochastic availability of evidence and determines which messages are available to the sender.

The following uniform-quadratic example provides a fully tractable illustration of how uninformative verifiable evidence can facilitate communication.

Example 3. As in the previous examples, consider a setting with uniform prior, quadratic payoffs with an additive sender's bias. Unlike in the previous examples, let (S, π) be an uninformative—but otherwise arbitrary—evidence structure. We can characterize receiver-optimal equilibria for every bias level. In particular, Remark B.1 in the Online Appendix shows that there exists $\beta^\circ \in (\frac{1}{8}, \frac{1}{4})$ such that:

1. If $b < \frac{1}{12}$, receiver-optimal equilibria of the game with verifiable evidence are strictly less informative than those of the game with unverifiable evidence.
2. If $b \in [\frac{1}{12}, \beta^\circ]$, receiver-optimal equilibria are equally informative regardless of whether evidence is verifiable.
3. If $b \in (\beta^\circ, \frac{1}{2})$, receiver-optimal equilibria of the game with verifiable evidence are strictly more informative than those of the game with unverifiable evidence.
4. If $b \geq \frac{1}{2}$, receiver-optimal equilibria are uninformative regardless of whether evidence is verifiable.

Therefore, even when evidence is uninformative, verifiable communication can be strictly more informative than unverifiable communication.

△

4.3 Does More Informative Evidence Help Communication?

The previous subsection shows that informative evidence is not necessary for verifiability to facilitate communication. This naturally raises the question of whether making the evidence *more* informative—in the Blackwell sense—nonetheless makes verifiable communication more effective.

The next result shows that more-informative evidence facilitates communication when preferences are either perfectly aligned or sufficiently misaligned, but may hinder communication for intermediate values of the bias.

Remark 1. *Consider the game with verifiable evidence.*

1. *Suppose (S, π) is Blackwell more informative than (S', π') . If $b = 0$ or $b \geq \bar{B}$, then receiver-optimal equilibria under (S, π) are more informative than those under (S', π') .*
2. *For intermediate values of b , there exist two evidence structures (S, π) and (S', π') , with (S, π) Blackwell more informative than (S', π') , such that receiver-optimal equilibria under (S, π) are strictly less informative than those under (S', π') .*

The first clause is immediate when $b \geq \bar{B}$. In that case, the receiver-optimal equilibrium is FED (Proposition 2). Because the receiver's value from observing the realized signal is monotone in the Blackwell order, the conclusion follows.¹⁴ When $b = 0$, sender and receiver have common interests and a receiver-optimal equilibrium under verifiable evidence can be characterized through an auxiliary optimization problem in which the sender commits ex ante to a verifiable disclosure policy, as in our discussion of Proposition 2. The proof shows that the value of this optimization problem, and thus the expected payoff of the receiver, is increasing in the Blackwell order.¹⁵

The second clause shows that, for intermediate b 's, the receiver may be worse off in settings where the evidence is more informative. The intuition is that a Blackwell-more-informative evidence structure can make the availability of evidence more predictable conditional on the state. This can weaken the strategic discipline created by the random availability of verifiable messages, as discussed in the previous subsection. At intermediate biases, this discipline can be an important source of informative communication. We present an example in Appendix A.4.¹⁶

¹⁴Bertomeu et al. (2021) and Lichtig and Weksler (2023) contain results in a similar vein.

¹⁵This is not immediate. With verifiable communication, a sender who observes a more-informative signal cannot in general reproduce, message by message, the outcome generated by a less-informative signal, because the feasible disclosures depend on the realized evidence. The proof shows how to overcome this difficulty by first handling deterministic garblings and then extending the argument to arbitrary garblings.

¹⁶In a different setting with no strategic communication, Kuvalekar et al. (2023) show that more precise verifiable information can reduce welfare in decision making under the shadow of ex post liability.

5 Interacting Verifiable and Unverifiable Communication

So far, the paper has compared verifiable and unverifiable communication, considering the two regimes in isolation. In this section, we bring the two together and study how they interact.

The interaction between verifiable and unverifiable communication has received little attention in the literature. One possible reason is that, in classic disclosure models, the sender can verifiably reveal all of her private information, leaving no meaningful role for additional cheap talk. Our disclosure framework instead creates scope for nontrivial interactions: because the sender knows more than her evidence can establish, unverifiable communication may convey information that verifiable disclosure alone cannot—and may therefore complement it. For example, a sender may disclose verifiable evidence that would be misleading in isolation—as when, in our model, a high- θ sender observes a very low signal s —knowing that she can later qualify and contextualize it through unverifiable statements.

More formally, this section considers a variant of our model, which we refer to as the *hybrid game*, that retains all the primitives of Section 2 and modifies only the communication technology. In the hybrid game, a sender of type (θ, s) sends *two* messages: a verifiable one, $m^V \in M^V = S \cup \{\circ\}$, subject to the same verifiability constraint as before, and an unverifiable one, $m^U \in M^U$. The receiver observes both messages before choosing her action. Strategies, beliefs, and equilibrium are otherwise defined as in Section 2.

To illustrate how verifiable and unverifiable communication may interact, we introduce a class of equilibria, which we call **disclose-and-refine**, in which the interaction between the two modes is particularly simple. In a disclose-and-refine equilibrium, every type- (θ, s) sender discloses her realized signal, so that $m^V = s$, and uses the unverifiable message to further refine the receiver’s posterior conditional on s . On-path communication can then be decomposed into two steps: disclose s and then—conditional on the induced distribution F_s —play a standard cheap-talk equilibrium.

In addition to providing a convenient way to illustrate how the two modes of communication may interact, disclose-and-refine equilibria are also salient for two other reasons. First, they are the natural analogue in the hybrid game of full-evidence-disclosure equilibria introduced in Section 2.1, since the receiver learns the realized signal. Second, they capture what would happen if disclosure of verifiable evidence were mandated by law while the sender remained free to additionally communicate through cheap talk. This gives this class of equilibria a natural

policy interpretation.

The next example illustrates a two-way interaction between the two communication channels in a disclose-and-refine equilibrium. Disclosing the evidence reshapes the receiver's interim posterior and, with it, the scope for cheap talk: depending on the realized signal, it can make informative cheap talk easier or harder to sustain. Conversely, cheap talk allows the sender to qualify and contextualize the disclosed evidence, which can in turn make disclosure itself sustainable.

Example 4. We return to the setting of Example 1 and let $k = \frac{3}{5}$ and $b = \frac{1}{5}$. That is, payoffs are quadratic, the prior F is uniform, the evidence structure is binary, $S = \{s_L, s_H\}$, and $\pi(s_H | \theta) = \frac{1}{5}(1 + 3\theta)$. The interim posterior density induced by signals s_i is $f(\theta | s_i) = 2\pi(s_i | \theta)$, for each $i \in \{L, H\}$. Thus, upon observing the signal, the receiver would take actions $a^{s_L} \triangleq \mathbb{E}(\theta | s_L) = \frac{2}{5}$ and $a^{s_H} \triangleq \mathbb{E}(\theta | s_H) = \frac{3}{5}$. Finally, suppose the unverifiable message space M^U has at least two elements.

We begin with two preliminary observations. First, in the game with only verifiable communication, the FED outcome cannot be sustained in equilibrium because $b < \bar{b} = \frac{1}{2}a^{s_H}$. Second, in the game with only unverifiable communication, the receiver-optimal equilibrium is a two-cell partition with cutoff $t = \frac{1}{10}$, and induced actions $a_1 = \frac{1}{20}$ and $a_2 = \frac{11}{20}$.

We now construct a disclose-and-refine equilibrium of the hybrid game. Suppose first that s_H is disclosed. The most-informative cheap-talk continuation has two actions separated by a cutoff t_{s_H} .¹⁷ The induced actions are $a_1^{s_H} = \mathbb{E}(\theta | \theta \leq t_{s_H}, s_H) = \frac{t_{s_H}(1+2t_{s_H})}{2+3t_{s_H}}$ and $a_2^{s_H} = \mathbb{E}(\theta | \theta \geq t_{s_H}, s_H) = \frac{2t_{s_H}^2+3t_{s_H}+3}{3t_{s_H}+5}$. Indifference of the threshold type t_{s_H} requires $\frac{a_1^{s_H}+a_2^{s_H}}{2} = t_{s_H} + b$, which reduces to a cubic equation with a unique feasible root $t_{s_H} \cong 0.17$. Hence, $a_1^{s_H} \cong 0.09$ and $a_2^{s_H} \cong 0.65$.

Now suppose that s_L is disclosed. In this case, the interim posterior is sufficiently pessimistic that every cheap-talk equilibrium is babbling. Hence, the receiver learns nothing beyond the evidence itself and chooses $a^{s_L} = \frac{2}{5}$.

It remains to verify that full disclosure of the verifiable evidence is incentive compatible. Assign the off-path action $a_o = 0$ following any deviation to $m^V = o$. A type θ prefers an equilibrium action $a > 0$ to a_o if and only if $a \leq 2(\theta + b)$. Since the right-hand side is

¹⁷A three-action partition can be shown to be infeasible; we omit the argument.

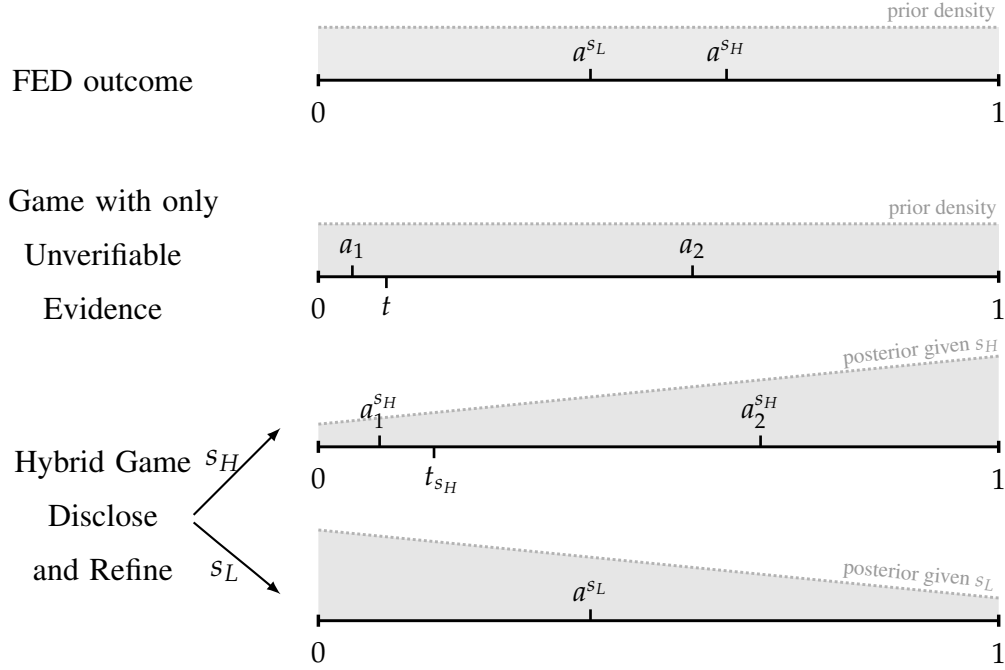


Figure 3: The communication outcomes from Example 4. The shaded areas represent density functions.

increasing in θ , it suffices to check the lowest type assigned to each equilibrium action. For a^{s_L} this type is $\theta = 0$, and $a^{s_L} = 2b$. For $a_1^{s_H}$ it is again $\theta = 0$, and $a_1^{s_H} < 2b$. Finally, for $a_2^{s_H}$ it is $\theta = t_{s_H}$, and $a_2^{s_H} < 2(t_{s_H} + b)$. Thus no sender benefits from withholding her verifiable evidence, and the strategy profile constitutes a disclose-and-refine equilibrium. $\triangle$

In the disclose-and-refine equilibrium of Example 4, the sender discloses her evidence and then uses cheap talk to qualify its interpretation. For example, when the state is sufficiently low, signal s_H is particularly misleading. Nonetheless, the sender discloses it, but also qualifies it downward with a cheap-talk message, which decreases the receiver's action from $a^{s_H} = 0.60$ (the action that would have been taken following disclosure in the verifiable only game) to $a_1^{s_H} = 0.09$.

The example above illustrates that the interaction between verifiable and unverifiable communication can materially change the way in which the sender communicates. Without the ability to use unverifiable communication, the sender would not be able to implement full evidence disclosure in equilibrium. Yet, with unverifiable communication, this becomes possible. This is not a special feature of Example 4, but a more general property of our setting.

Proposition 4. *Suppose $M^U = [0, 1]$. Disclose-and-refine equilibria exist for all $b \geq 0$.*

This result stands in contrast with Proposition 1, which shows that when only verifiable communication is available, no full-evidence-disclosure equilibrium exists when b is small. Instead, once the sender can supplement her verifiable evidence with unverifiable statements, full evidence disclosure can always be sustained.

The key to Proposition 4 is that the unverifiable channel allows sender types with the same signal to induce different actions while still disclosing their evidence. To see why this matters, suppose $b > 0$. Conditional on each signal s , consider an interval-partition equilibrium of the cheap-talk game with prior F_s that has the maximal number of steps, and denote its lowest induced action by a_1^s . Such an equilibrium satisfies the “no-incentive-to-separate” (NITS) criterion of Chen et al. (2008): the lowest state $\theta = 0$ weakly prefers a_1^s to $a^{\mathcal{R}}(0) = 0$, that is, $u^S(a_1^s, 0, b) \geq u^S(0, 0, b)$. We can therefore support disclosure of s by letting silence induce action 0. Applying this construction after every signal sustains full evidence disclosure.¹⁸

The above discussion raises the question of whether the interaction between verifiable and unverifiable communication always benefits the receiver, relative to settings in which only one mode of communication is available. We provide a qualified answer. Within the class of disclose-and-refine equilibria, the answer is negative: the interaction can sometimes be detrimental.

Remark 2. *The receiver-optimal equilibrium within the class of disclose-and-refine equilibria can be strictly less informative than the most informative equilibrium under either unverifiable or verifiable communication alone.*

The reason is that disclosing the verifiable signal may tie the sender’s hands when using the unverifiable message. When the verifiable signals are themselves insufficiently informative, the sender may be better off avoiding verifiable communication altogether. This observation connects to the paper’s broader theme: verifiability is not necessarily a positive force in communication. A concrete policy implication is that mandating disclosure can reduce informativeness even when the sender retains access to an unrestricted unverifiable channel.¹⁹ We illustrate the

¹⁸At $b = 0$, however, the lowest state strictly prefers action 0 to the action induced by any nondegenerate lowest pool. That is where we use the assumption that M^U has the cardinality of the continuum (normalized to $M^U = [0, 1]$): unverifiable messages can then fully reveal θ .

¹⁹This observation is related to Bertomeu and Marinovic (2016), who, in a different setting, show how mandating the disclosure of hard information can crowd out the transmission of soft information.

mechanism underlying Remark 2 by continuing the analysis of Example 4.

Example 4 (continued). Consider again the equilibria derived in Example 4. We now compare their informativeness. Under quadratic loss, the receiver's expected payoff is $U = -\frac{1}{12} + \text{Var}(a)$, so we use $I \triangleq \text{Var}(a)$ as an equivalent measure of informativeness in this example. The most informative equilibrium of the game with unverifiable communication induces actions $a_1 = \frac{1}{20}$ and $a_2 = \frac{11}{20}$ with probabilities $\frac{1}{10}$ and $\frac{9}{10}$, respectively. Hence $I^{\text{CT}} = \text{Var}(a) \cong 0.023$. Next, consider instead the most informative disclose-and-refine equilibrium. By the law of total variance, we can decompose its informativeness into the informativeness of the FED outcome and the additional informativeness generated by the cheap talk components: $I^{\text{DR}} = I^{\text{FED}} + I_{s_L}^{\text{CT}} + I_{s_H}^{\text{CT}}$. The first term is $I^{\text{FED}} = \frac{1}{2}(a^{s_L} - \frac{1}{2})^2 + \frac{1}{2}(a^{s_H} - \frac{1}{2})^2 = \frac{1}{100}$. The additional contribution following s_L is zero, while the contribution following s_H is $I_{s_H}^{\text{CT}} = \frac{1}{2} \text{Var}(a \mid s_H) \cong 0.012$. Therefore, $I^{\text{DR}} \cong 0.022$. Hence, the most informative disclose-and-refine equilibrium is less informative than the most informative equilibrium under unverifiable communication alone.

This example also shows that disclose-and-refine can be dominated by verifiable communication alone. Consider the equilibrium under verifiable communication in which s_L is never disclosed, while s_H is disclosed if and only if $\theta \geq t$. It is easy to check that the indifference condition yields $t \cong \frac{1}{3}$, and the induced actions give $I^V \cong 0.028 > I^{\text{DR}}$.

Hence, in this example, the most informative disclose-and-refine equilibrium is less informative than the receiver-optimal equilibria under either communication technology alone. $\triangle$

The negative result of Remark 2, however, hinges on restricting attention to disclose-and-refine equilibria. Once we consider the most informative equilibrium in the *entire* equilibrium set, the conclusion is reversed: combining verifiable and unverifiable communication cannot reduce the receiver's informativeness.

Remark 3. *Receiver-optimal equilibria in the hybrid game are weakly more informative than receiver-optimal equilibria under either verifiable or unverifiable communication alone.*²⁰ *Moreover, if $M^U = [0, 1]$, the hybrid game can be strictly more informative than both communication modes in isolation only if $b \in (0, \bar{B})$.*

²⁰Existence of receiver-optimal equilibria in the hybrid game is established in the Online Appendix B.3.

The first part of the remark follows from the observation that the set of equilibrium outcomes of the hybrid game contains the union of the equilibrium outcome sets under verifiable and unverifiable communication. To see this, consider first any equilibrium of the game with verifiable communication. The same outcome can be replicated in the hybrid game by having the sender use the verifiable message exactly as in the original equilibrium and append a fixed unverifiable message. The receiver conditions only on the verifiable message, and off-path beliefs can be chosen so that the sender has no incentive to deviate in the unverifiable message. Similarly, any equilibrium of the game with unverifiable communication can be replicated in the hybrid game by having the sender remain silent in the verifiable channel and use the unverifiable message exactly as in the original equilibrium. The receiver conditions only on the unverifiable message, and beliefs following an off-path disclosure can be chosen so that the receiver takes the same action as she would have taken in response to the accompanying unverifiable message. Thus, introducing the verifiable channel again creates no profitable deviations.

To understand the second part of the remark, suppose $M^U = [0, 1]$. When $b = 0$, the sender's and receiver's preferences are perfectly aligned, and a fully revealing equilibrium can already be sustained with unverifiable communication alone by having the sender report $m^U = \theta$. When $b \geq \bar{B}$, the sender's payoff is weakly increasing in the receiver's action over $[0, 1]$. Consequently, conditional on any realized signal, all sender types rank the available message pairs in the same way, so unverifiable communication cannot reveal any additional information about the state beyond the sender's evidence. Full evidence disclosure is therefore efficient in the hybrid game. Hence, when $M^U = [0, 1]$, the hybrid game can be strictly more informative than both communication modes in isolation only when $b \in (0, \bar{B})$.

6 Concluding Remarks

This paper studies communication when the sender knows more than her evidence can prove. Verifiability affects communication in two opposing ways. By restricting which messages each sender type can send, it limits imitation and lends credibility to the sender's claims. Yet the same restrictions constrain her ability to communicate information that her evidence does not capture. This credibility–flexibility tradeoff governs our results. When preferences are sufficiently aligned, receiver-optimal equilibria under unverifiable evidence are more informative

than those under verifiable evidence, which are in turn more informative than the full-evidence-disclosure outcome. When preferences are sufficiently misaligned, the comparison between the two regimes reverses. At intermediate biases, the ranking is not uniform and can depend on the evidence structure in unexpected ways. Finally, our model opens the door to studying the largely unexplored interactions between verifiable and unverifiable communication and the many ways in which the two may complement one another, of which disclose-and-refine equilibria provide one illustration.

We conclude by discussing three extensions and what our analysis suggests about each.

First, our analysis compares two polar communication regimes: verifiable communication, in which fabricating evidence inconsistent with the realized signal is prohibitively costly, and unverifiable communication, in which it is costless. In many applications, however, fabrication may be costly but not impossible. Introducing a finite fabrication cost, in the spirit of [Kartik \(2009\)](#), raises a natural design question: how much flexibility should the sender be granted to maximize the receiver’s welfare? Put differently, is strict verifiability *ever* optimal? We conjecture that it is not: the receiver always benefits from preserving some scope for the sender to fabricate evidence. Section [B.4](#) of the Online Appendix formalizes this problem and presents a result in support of this conjecture.

Second, our analysis assumed that type (θ, s) can either disclose s or remain silent, so that $M^V(\theta, s) = \{s, \circ\}$. Although this assumption is common in the literature (see, e.g., [Dye, 1985](#); [Jung and Kwon, 1988](#); [Shin, 2003](#); [Dziuda, 2011](#)), it is admittedly stylized. A natural extension, which nests our model, is to let $M^V(\theta, s) = \{m \subseteq S : s \in m\}$, so that the sender may disclose any subset of S containing the realized signal, as in much of the disclosure literature. When S is finite, this richer language gives the sender *additional* flexibility, thus attenuating the credibility–flexibility tradeoff, but does not overturn our main qualitative conclusions. Our baseline binary-disclosure model has the virtue of isolating this tradeoff in its simplest and starkest form.

Finally, our analysis assumes that the sender always has evidence to report. In practice, evidence may sometimes be unavailable, as in [Dye \(1985\)](#). Silence would then pool senders who withhold evidence with those who have none, combining the classic force that limits unraveling with the mechanism identified in this paper. The possibility of having no evidence would also further randomize which verifiable messages are available. By the logic of Section

4.2, this additional randomness can discipline the sender by limiting imitation, but it also dilutes the information conveyed by silence. Which effect dominates at intermediate biases remains an open question.

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Appendix

A Proofs

A.1 Equilibrium Characterization under Verifiable Evidence

Definition 3. A sender strategy under verifiable evidence is a generalized cutoff rule if there exist a partition (L, M, H) of S and cutoffs $(t_s)_{s \in L \cup H} \subseteq [0, 1]$ such that, for every type (θ, s) , the sender discloses s if and only if either $s \in L$ and $\theta \leq t_s$, or $s \in H$ and $\theta \geq t_s$. Signals in M are never disclosed.

Lemma A.1. Every equilibrium under verifiable evidence is outcome-equivalent to one in which the sender uses a generalized cutoff rule.

Proof. Fix an equilibrium (σ, α, μ) under verifiable evidence. Write $a_\circ = \alpha(\circ)$ and $a_s = \alpha(s)$ for each $s \in S$. If $\Pr_\sigma(m = \circ) = 0$, then the sender always discloses her realized signal. Thus the sender's strategy is already a generalized cutoff rule: take $H = S$, $L = M = \emptyset$, and $t_s = 0$ for every $s \in S$. Hence suppose $\Pr_\sigma(m = \circ) > 0$.

First consider any signal s such that $\Pr_\sigma(m = s) > 0$ and $a_s = a_\circ$. Replacing disclosure of s by silence for all types with signal s does not change the induced action, and hence does not change the outcome. Receiver optimality after silence is preserved because both messages induced the same action. Thus, without loss of outcome, we may assume that no disclosed signal induces the same action as silence.

Now fix a signal s such that $\Pr_\sigma(m = s) > 0$. By the previous paragraph, $a_s \neq a_\circ$. Conditional on signal s , the sender compares only the two actions a_s and $a_\circ$. If $a_s > a_\circ$, the payoff difference $u(a_s, \theta, b) - u(a_\circ, \theta, b)$ is strictly increasing in θ by $u_{12} > 0$. Hence the set of states for which disclosure is optimal is an upper interval, up to a cutoff at which the sender may be indifferent. Since $m = s$ is sent with positive probability, this cutoff can be chosen so that $t_s < 1$. If $a_s < a_\circ$, the same single-crossing argument implies that the disclosure set is a lower interval, again up to a cutoff; since $m = s$ is sent with positive probability, this cutoff can be chosen so that $t_s > 0$. Since F is atomless, behavior at cutoff states does not affect the outcome.

Define

$$L \triangleq \{s \in S : \Pr_{\sigma}(m = s) > 0 \text{ and } a_s < a_{\circ}\}, \quad H \triangleq \{s \in S : \Pr_{\sigma}(m = s) > 0 \text{ and } a_s > a_{\circ}\},$$

and let $M \triangleq S \setminus (L \cup H)$. For $s \in M$, set the off-path receiver action after s equal to $a_{\circ}$ and choose beliefs that rationalize this action. This is feasible by full support. Applying the cutoff construction signal by signal yields an outcome-equivalent equilibrium in generalized cutoff form. $\square$

The proof above motivates the following notion of a canonical representation. Whenever $\circ$ is on path, we will say that a generalized cutoff representation is *canonical* if every signal in M is never disclosed and induces the same action as silence, every $s \in L$ satisfies $\alpha(s) < \alpha(\circ)$ and has cutoff $t_s \in (0, 1]$, and every $s \in H$ satisfies $\alpha(s) > \alpha(\circ)$ and has cutoff $t_s \in [0, 1)$. Thus, in a canonical representation, signals in $L \cup H$ are exactly the disclosed signals whose induced actions differ from the silence action. The signals with interior cutoffs are those for which the relevant cutoff indifference equation binds; boundary cases correspond to disclosure with probability one conditional on the realized signal.

A.2 Proofs for Section 3

Throughout this subsection, denote $\bar{a}_S = \max_{s \in S} a^{\mathcal{R}}(F_s) = a^{\mathcal{R}}(F_{s_N})$, and $\underline{a}_S = \min_{s \in S} a^{\mathcal{R}}(F_s) = a^{\mathcal{R}}(F_{s_1})$.

Lemma A.2. *For every posterior belief $\mu \in \Delta(\Theta)$, $a^{\mathcal{R}}(\mu) \in [0, 1]$. Moreover, if μ' strictly first-order stochastically dominates μ , then $a^{\mathcal{R}}(\mu') > a^{\mathcal{R}}(\mu)$.*

Proof. For any posterior belief μ , the first-order condition for $a^{\mathcal{R}}(\mu)$ is $\int_{\Theta} u_1(a, \theta, 0) d\mu(\theta) = 0$. Since $u_{12} > 0$, we have $u_1(0, \theta, 0) \geq u_1(0, 0, 0) = 0$ for every θ , and the inequality is strict for $\theta > 0$. Similarly, $u_1(1, \theta, 0) \leq u_1(1, 1, 0) = 0$ for every θ , and the inequality is strict for $\theta < 1$. Since the receiver's expected payoff is strictly concave in a , its maximizer lies in $[0, 1]$.

Now suppose μ' strictly first-order stochastically dominates μ . Fix any a . Since $u_{12} > 0$, the function $\theta \mapsto u_1(a, \theta, 0)$ is strictly increasing. Therefore $\int u_1(a, \theta, 0) d\mu'(\theta) > \int u_1(a, \theta, 0) d\mu(\theta)$ whenever the dominance is strict. Evaluating at $a = a^{\mathcal{R}}(\mu)$ gives $\int u_1(a^{\mathcal{R}}(\mu), \theta, 0) d\mu'(\theta) > 0$. By strict concavity in a , the maximizer under μ' must be strictly larger than $a^{\mathcal{R}}(\mu)$. $\square$

Lemma A.3. *Both thresholds $\bar{b}$ and $\bar{B}$ are well defined and unique. Moreover, $\bar{b} < \bar{B}$.*

Proof. For every $s \in S$, the posterior F_s has full support on $[0, 1]$. Hence $\int_{\Theta} u_1(0, \theta, 0) dF_s(\theta) > 0$, so $a^{\mathcal{R}}(F_s) > 0$. In particular, $\bar{a}_S > 0$. Consider $g(b) \triangleq u(\bar{a}_S, 0, b) - u(0, 0, b)$. Since $\bar{a}_S > 0$, strict concavity gives $g(0) < 0$. By $u_{13} > 0$, g is strictly increasing in b , because $g'(b) = \int_0^{\bar{a}_S} u_{13}(x, 0, b) dx > 0$. Finally, by the large-bias assumption, $u_1(\bar{a}_S, 0, b) > 0$ for all sufficiently large b , and thus $u_1(a, 0, b) > 0$ for all $a \leq \bar{a}_S$ by $u_{11} < 0$. Hence $g(b) > 0$ for large b . Therefore there exists a unique $\bar{b}$ such that $g(\bar{b}) = 0$.

For $\bar{B}$, note that $a^{\mathcal{S}}(0, 0) = 0 < 1$. The sender's ideal action $a^{\mathcal{S}}(0, b)$ is strictly increasing in b by $u_{13} > 0$ and strict concavity in a . The large-bias assumption implies that $a^{\mathcal{S}}(0, b) > 1$ for all sufficiently large b . Hence there exists a unique $\bar{B}$ such that $a^{\mathcal{S}}(0, \bar{B}) = 1$.

At $b = \bar{B}$, the sender's ideal action at $\theta = 0$ is 1. Since $\bar{a}_S \in (0, 1]$, strict concavity implies $u(\bar{a}_S, 0, \bar{B}) > u(0, 0, \bar{B})$. Since $u(\bar{a}_S, 0, \bar{b}) = u(0, 0, \bar{b})$ and g is strictly increasing, we have $\bar{b} < \bar{B}$. $\square$

Proof of Proposition 1. We first prove sufficiency. Suppose $b \geq \bar{b}$. Consider the assessment in which the sender always discloses the realized signal, the receiver chooses $\alpha(s) = a^{\mathcal{R}}(F_s)$ after every $s \in S$, and the receiver chooses $\alpha(\circ) = 0$ after silence. Beliefs after disclosed signals are Bayesian, and the off-path belief after silence assigns probability one to $\theta = 0$.

Receiver optimality is immediate. We verify sender optimality. By Lemma A.2, $a^{\mathcal{R}}(F_s) \in [0, \bar{a}_S]$ for every s . Since $b \geq \bar{b}$, we have $u(\bar{a}_S, 0, b) \geq u(0, 0, b)$. By strict concavity of $u(\cdot, 0, b)$, this implies $u(a^{\mathcal{R}}(F_s), 0, b) \geq u(0, 0, b)$ for every s . Since $a^{\mathcal{R}}(F_s) \geq 0$ and $u_{12} > 0$, the payoff difference $u(a^{\mathcal{R}}(F_s), \theta, b) - u(0, \theta, b)$ is increasing in θ . Therefore every type weakly prefers disclosing s to remaining silent. Hence the assessment is a FED equilibrium.

We now prove necessity. Suppose an equilibrium (σ, α, μ) induces the FED outcome, which we denote by λ^{FED} , and let $a_{\circ} \triangleq \alpha(\circ)$. By Lemma A.2, $a_{\circ} \in [0, 1]$. Define

$$S_- \triangleq \{s \in S : a^{\mathcal{R}}(F_s) = \underline{a}_S\} \quad \text{and} \quad S_+ \triangleq \{s \in S : a^{\mathcal{R}}(F_s) = \bar{a}_S\}.$$

We first claim that $a_{\circ} \leq \underline{a}_S$. Suppose instead that $a_{\circ} > \underline{a}_S$. Since $a_{\circ} \leq 1 \leq a^{\mathcal{S}}(1, b)$, strict concavity implies $u(a_{\circ}, 1, b) > u(\underline{a}_S, 1, b)$. By continuity, there exists $\delta > 0$ such that $u(a_{\circ}, \theta, b) > u(\underline{a}_S, \theta, b)$ for every $\theta \in [1 - \delta, 1]$. Under the FED outcome,

$$\lambda^{\text{FED}}([1 - \delta, 1] \times \{\underline{a}_S\}) = \int_{1-\delta}^1 \sum_{s \in S_-} \pi(s \mid \theta) dF(\theta) > 0,$$

where the strict inequality follows from full support of F and π . Since the equilibrium induces the FED outcome, there is therefore a positive-probability set of types with states in $[1 - \delta, 1]$ who induce action $\underline{a}_S$. Each such type can deviate to silence and induce a_o , contradicting sender optimality. Hence $a_o \leq \underline{a}_S$.

Now suppose $b < \bar{b}$. Then $u(0, 0, b) > u(\bar{a}_S, 0, b)$. Since $a_o \in [0, \underline{a}_S]$ and $\underline{a}_S < \bar{a}_S$, strict concavity of $u(\cdot, 0, b)$ implies $u(a_o, 0, b) > u(\bar{a}_S, 0, b)$. By continuity, there exists $\delta > 0$ such that $u(a_o, \theta, b) > u(\bar{a}_S, \theta, b)$ for every $\theta \in [0, \delta]$. Under the FED outcome,

$$\lambda^{FED}([0, \delta] \times \{\bar{a}_S\}) = \int_0^\delta \sum_{s \in S_+} \pi(s | \theta) dF(\theta) > 0.$$

Since the equilibrium induces the FED outcome, there is therefore a positive-probability set of types with states in $[0, \delta]$ who induce action $\bar{a}_S$. Each such type can deviate to silence and induce a_o , again contradicting sender optimality. Hence $b \geq \bar{b}$. $\square$

Lemma A.4. *For every environment satisfying the maintained assumptions, when $b = 0$, every receiver-optimal equilibrium under verifiable evidence is strictly more informative than the FED outcome.*

Proof. At $b = 0$, sender and receiver preferences coincide. Let ρ denote the joint distribution of (θ, s) induced by (F, π) . Consider the auxiliary problem in which one chooses a verifiable communication rule and a receiver action rule to maximize the common expected payoff. Formally, let $\mathcal{P}$ be the set of probability measures φ on $\Theta \times S \times M$ such that the (θ, s) -marginal of φ is ρ and $\varphi(\{(\theta, s, m) : m \notin \{s, o\}\}) = 0$. Let $\mathcal{A} = [0, 1]^M$ be the set of receiver action rules with values in $[0, 1]$. For $(\varphi, \alpha) \in \mathcal{P} \times \mathcal{A}$, define $W(\varphi, \alpha) \triangleq \int u^R(\alpha(m), \theta) d\varphi(\theta, s, m)$.

The set $\mathcal{P} \times \mathcal{A}$ is compact, and W is continuous, so a maximizer (φ^*, α^*) exists. For every on-path message, optimality of α^* implies receiver optimality. For every off-path message, choose a belief over feasible types that rationalizes $\alpha^*(m)$; this is possible because every action in $[0, 1]$ is receiver-optimal for some posterior belief over Θ , and full support makes every state compatible with every evidence message.

We claim that the maximizer induces an equilibrium. Let σ^* be a regular conditional distribution induced by φ^* . If σ^* assigned positive probability to a feasible message that gave a type a strictly lower payoff than another feasible message on a set of positive ρ -measure, shifting that probability to a best feasible message would strictly increase W , contradicting opti-

mality. Hence, after modifying it on a ρ -null set if necessary, σ^* is sender-optimal.²¹ Receiver optimality and belief consistency hold by construction. Thus the auxiliary optimum is implementable as an equilibrium under verifiable evidence.

It remains to show that the auxiliary optimum strictly improves on the FED outcome. Fix any signal $\hat{s} \in S$ and any cutoff $r \in (0, 1)$. Consider the following feasible verifiable rule: every type with signal $s \neq \hat{s}$ discloses s , while a type with signal $\hat{s}$ sends $\circ$ if $\theta < r$ and discloses $\hat{s}$ if $\theta \geq r$. Since F has full support and $\pi(\hat{s} \mid \theta) > 0$ for all θ , both events $\{\theta < r, \hat{s}\}$ and $\{\theta \geq r, \hat{s}\}$ have positive probability. The posterior after $\{\theta \geq r, \hat{s}\}$ strictly first-order stochastically dominates the posterior after $\{\theta < r, \hat{s}\}$, so Lemma A.2 implies that their receiver-optimal actions are distinct. Therefore splitting the posterior induced by $\hat{s}$ strictly increases the receiver's expected payoff: the receiver could always ignore the split and choose $a^R(F_{\hat{s}})$ after both messages, thereby reproducing the contribution of signal $\hat{s}$ to the FED payoff, while choosing the distinct posterior-optimal actions after the two subevents yields a strict gain. Thus the auxiliary value, and hence the receiver-optimal equilibrium payoff at $b = 0$, is strictly above the FED payoff. $\square$

Before proving the continuation result, we record a simple implication of canonical generalized cutoff form at $b = 0$. Consider a canonical generalized cutoff representation of a $b = 0$ PBE in which $\circ$ is on path. Then every disclosed message whose action differs from silence has an interior cutoff. Indeed, suppose $s \in H$ and $t_s = 0$. Then s is disclosed with probability one conditional on signal s , and it induces an action $a_s > a_\circ$. Since both actions lie in $[0, 1]$ and sender and receiver preferences coincide at $b = 0$, types with θ close to 0 strictly prefer the lower action $a_\circ$ to a_s , contradicting sender optimality. Hence $t_s > 0$ for every $s \in H$. The argument for $s \in L$ is symmetric: if $t_s = 1$, then all types with signal s disclose and induce $a_s < a_\circ$, but types with θ close to 1 strictly prefer $a_\circ$ to a_s , again contradicting sender optimality. Hence $t_s < 1$ for every $s \in L$.

Lemma A.5. *Let $U^V(b)$ denote the receiver payoff in a receiver-optimal equilibrium under verifiable evidence at bias b . Then*

$$\lim_{b \downarrow 0} U^V(b) = U^V(0).$$

²¹On this null set, assign each type a feasible best reply; this leaves the induced distribution and expected payoff unchanged.

Proof. The auxiliary-problem argument in the proof of Lemma A.4 implies that $U^V(0)$ is the maximal receiver payoff over all verifiable communication rules, even without imposing sender incentive compatibility. Every equilibrium at bias $b \geq 0$ induces a feasible rule for this auxiliary problem. Hence $U^V(b) \leq U^V(0)$ for every $b \geq 0$. It remains to prove the reverse limiting inequality.

For each $s \in S$, define $R_s(t) \triangleq \int_0^t \pi(s \mid \theta) dF(\theta)$ and let $T_s : [0, R_s(1)] \rightarrow [0, 1]$ be its inverse. Since F is atomless with full support and $\pi(s \mid \theta) > 0$, each R_s is continuous and strictly increasing, and hence T_s is continuous. Using the compactness argument in the proof of Lemma B.5, choose a canonical generalized cutoff representation (L^0, M^0, H^0, t^0) of a $b = 0$ receiver-optimal equilibrium that first minimizes the number of active signals $|L^0 \cup H^0|$ and, subject to this, minimizes $\sum_{s \in L^0 \cup H^0} R_s(t_s^0)$. Such a representation exists: if a minimizing sequence lost an active message in the limit, its canonical limit would be a $b = 0$ receiver-optimal equilibrium with fewer active messages. Let $J^0 \triangleq L^0 \cup H^0$ and $x_s^0 \triangleq R_s(t_s^0)$ for $s \in J^0$. Write $x^0 \triangleq (x_s^0)_{s \in J^0}$. Lemma A.4 implies that silence is on path and $J^0 \neq \emptyset$, while the observation above implies that x^0 is interior to $\prod_{s \in J^0} [0, R_s(1)]$.

Fix the pattern (L^0, M^0, H^0) . For x near x^0 , use cutoff $T_s(x_s)$ for each $s \in J^0$, and let $a_s(x_s)$ and $a_o(x)$ denote the corresponding posterior-optimal actions after disclosure and silence. In these mass coordinates, the receiver's first-order conditions are integrals of $u_1(a, T_s(z), 0)$ over intervals whose endpoints are the coordinates of x . Since T_s is continuous and $u_{11} < 0$, the action maps are continuously differentiable. Let $W(x)$ denote the receiver payoff induced by this cutoff profile and these posterior-optimal actions. For each $s \in J^0$, define

$$G_s^V(x, b) \triangleq \begin{cases} u(a_s(x_s), T_s(x_s), b) - u(a_o(x), T_s(x_s), b), & s \in L^0, \\ u(a_o(x), T_s(x_s), b) - u(a_s(x_s), T_s(x_s), b), & s \in H^0. \end{cases}$$

Thus G_s^V is the cutoff type's payoff from the lower action minus that from the higher action. The envelope theorem gives

$$\frac{\partial W(x)}{\partial x_s} = G_s^V(x, 0) \quad \text{for every } s \in J^0. \quad (1)$$

We next establish local complementarity. At x^0 , cutoff indifference gives $G_s^V(x^0, 0) = 0$. Moreover, cutoff indifference and strict concavity imply

$$u_1(a_o(x^0), t_s^0, 0) < 0 \quad (s \in L^0), \quad u_1(a_o(x^0), t_s^0, 0) > 0 \quad (s \in H^0).$$

Let $\mathcal{S}(a, x) = 0$ denote the receiver's unnormalized first-order condition after silence, with $\mathcal{S}_a^0 \triangleq \mathcal{S}_a(a_\circ(x^0), x^0) < 0$. Implicit differentiation of the silence first-order condition yields

$$\frac{\partial a_\circ}{\partial x_r}(x^0) = \begin{cases} \frac{u_1(a_\circ(x^0), t_r^0, 0)}{\mathcal{S}_a^0} > 0, & r \in L^0, \\ -\frac{u_1(a_\circ(x^0), t_r^0, 0)}{\mathcal{S}_a^0} > 0, & r \in H^0. \end{cases}$$

Consequently, for $r \neq s$,

$$\frac{\partial G_s^V}{\partial x_r}(x^0, 0) = \begin{cases} -u_1(a_\circ(x^0), t_s^0, 0) \frac{\partial a_\circ}{\partial x_r}(x^0) > 0, & s \in L^0, \\ u_1(a_\circ(x^0), t_s^0, 0) \frac{\partial a_\circ}{\partial x_r}(x^0) > 0, & s \in H^0. \end{cases}$$

By continuity, there exists a closed rectangle $\mathcal{X}^0$ containing x^0 in its interior and $\varepsilon^0 > 0$ such that the cutoff pattern and action ordering remain canonical and every G_s^V is nondecreasing in every x_r , $r \neq s$, on $\mathcal{X}^0 \times [0, \varepsilon^0]$. By (1), W has increasing differences and hence is supermodular on $\mathcal{X}^0$.

The selected x^0 is the *least* maximizer of W on $\mathcal{X}^0$. Indeed, every $x \in \mathcal{X}^0$ defines a feasible rule for the $b = 0$ auxiliary problem, so $W(x) \leq U^V(0) = W(x^0)$. If $\hat{x}^0$ is another maximizer, supermodularity gives

$$W(x^0 \wedge \hat{x}^0) + W(x^0 \vee \hat{x}^0) \geq W(x^0) + W(\hat{x}^0) = 2U^V(0),$$

where $\wedge$ and $\vee$ are coordinatewise. Both terms on the left are at most $U^V(0)$, so both equal $U^V(0)$. By the auxiliary implementation argument in Lemma A.4, $x^0 \wedge \hat{x}^0$ is another $b = 0$ receiver-optimal equilibrium with the same active pattern. The minimizing choice of x^0 therefore implies $x^0 \wedge \hat{x}^0 = x^0$, and hence $x^0 \leq \hat{x}^0$.

For $\eta > 0$, let x^η maximize

$$W(x) - \eta \sum_{s \in J^0} x_s$$

over $\mathcal{X}^0$. Supermodularity and the optimality of x^0 imply $W(x^\eta \wedge x^0) \geq W(x^\eta)$, so the linear penalty gives $x^\eta \leq x^0$. Every limit point of x^η as $\eta \downarrow 0$ is a maximizer of W weakly below the least maximizer x^0 , and therefore equals x^0 . Thus $x^\eta \rightarrow x^0$. For sufficiently small η , x^η is interior to $\mathcal{X}^0$, so (1) and the first-order conditions give

$$G_s^V(x^\eta, 0) = \eta > 0, \quad s \in J^0. \quad (2)$$

Moreover, $x_s^\eta < x_s^0$ for every $s \in J^0$: otherwise, local complementarity and $x^\eta \leq x^0$ would imply $G_s^V(x^\eta, 0) \leq G_s^V(x^0, 0) = 0$.

Fix some $x \in \mathcal{X}^0$. For $s \in L^0$, we have

$$\frac{\partial G_s^V(x, b)}{\partial b} = u_3(a_s(x_s), T_s(x_s), b) - u_3(a_o(x), T_s(x_s), b) = - \int_{a_s(x_s)}^{a_o(x)} u_{13}(a, T_s(x_s), b) da < 0.$$

For $s \in H^0$, we have

$$\frac{\partial G_s^V(x, b)}{\partial b} = u_3(a_o(x), T_s(x_s), b) - u_3(a_s(x_s), T_s(x_s), b) = - \int_{a_o(x)}^{a_s(x_s)} u_{13}(a, T_s(x_s), b) da < 0.$$

Consequently, $G_s^V(x^0, b) < 0$ for every $b \in (0, \varepsilon^0]$, whereas (2) and continuity imply $G_s^V(x^\eta, b) > 0$ for all sufficiently small $b > 0$.

Fix such η and b , and consider the rectangle $[x^\eta, x^0]$. On its lower s -face, local complementarity gives $G_s^V(x, b) \geq G_s^V(x^\eta, b) > 0$, while on its upper s -face, $G_s^V(x, b) \leq G_s^V(x^0, b) < 0$. The Poincaré–Miranda theorem therefore yields $x(b) \in [x^\eta, x^0]$ such that $G_s^V(x(b), b) = 0$ for every $s \in J^0$.²²

Use the pattern (L^0, M^0, H^0) and cutoffs $t_s(b) = T_s(x_s(b))$, and let the receiver choose the corresponding posterior-optimal actions. For each $s \in M^0$, assign the off-path message s the silence action and choose beliefs accordingly; this is feasible by full support. The preserved action ordering, cutoff indifference, and $u_{12} > 0$ imply sender optimality by single crossing; receiver optimality and belief consistency hold by construction. Hence this is an equilibrium under verifiable evidence, with receiver payoff $W(x(b))$.

Finally, fix $\delta > 0$. Since $x^\eta \rightarrow x^0$ and W is continuous, η can be chosen so that $W(x) > U^V(0) - \delta$ throughout $[x^\eta, x^0]$. The equilibrium above then implies $U^V(b) > U^V(0) - \delta$ for every sufficiently small $b > 0$. Thus $\liminf_{b \downarrow 0} U^V(b) \geq U^V(0)$. Together with the initial upper bound, this proves the result. $\square$

Proof of Proposition 2. We first prove the large-bias clause. Suppose $b \geq \bar{B}$. Then $a^S(0, b) \geq a^S(0, \bar{B}) = 1$. Since sender ideal actions are increasing in θ , every type's ideal action is at least 1. Therefore every sender type weakly prefers higher receiver actions throughout $[0, 1]$.

Fix any equilibrium under verifiable evidence. Conditional on a signal s , the sender compares only $\alpha(s)$ and $\alpha(o)$. Both actions lie in $[0, 1]$ by Lemma A.2. Since all types with signal s rank

²²The Poincaré–Miranda theorem is the multidimensional analogue of the intermediate value theorem and is equivalent to Brouwer's fixed point theorem.

these two actions in the same direction, the induced action is a function only of the realized signal s . Thus every equilibrium outcome is a deterministic garbling of the realized signal. The receiver can weakly improve on any garbling by observing the full signal and choosing $a^{\mathcal{R}}(F_s)$ after each s . Hence no equilibrium can give the receiver a payoff above the FED payoff. Since $b \geq \bar{B} > \bar{b}$ whenever evidence is informative, Proposition 1 implies that a FED equilibrium exists in that case. If evidence is uninformative, the FED outcome is uninformative and is trivially induced by an uninformative equilibrium. Therefore the FED outcome is induced by a receiver-optimal equilibrium under verifiable evidence.

The second clause follows from Lemmas A.4 and A.5. The former implies that $U^V(0)$ is strictly above the FED payoff, while the latter gives $U^V(b) \rightarrow U^V(0)$ as $b \downarrow 0$. Since the FED payoff is independent of b , $U^V(b)$ remains strictly above it for all sufficiently small $b \geq 0$.

We now prove the third clause. Fix $b < \bar{B}$. Since $a^{\mathcal{S}}(0, b) < 1$, choose an atomless full-support prior F sufficiently concentrated near 1 that $a^{\mathcal{R}}(F) > a^{\mathcal{S}}(0, b)$. Let the evidence structure be statistically uninformative: choose some signal $\hat{s} \in S$ with probability $p \in (0, 1)$, independent of θ . In the FED outcome, the receiver learns no payoff-relevant information, so the FED payoff is the no-information payoff.

For each $t \in (0, 1)$, consider the following strategy: a type discloses $\hat{s}$ iff her signal is $\hat{s}$ and $\theta \geq t$; all other types send $\circ$. Let $A_1(t) \triangleq a^{\mathcal{R}}(F(\cdot \mid \theta \geq t))$ be the receiver's action after $\hat{s}$, and let $A_0(t)$ be the receiver's posterior-optimal action after silence. The posterior after $\hat{s}$ strictly first-order stochastically dominates the posterior after silence, so $A_1(t) > A_0(t)$ by Lemma A.2. Define $D(t) \triangleq u(A_1(t), t, b) - u(A_0(t), t, b)$. As $t \downarrow 0$, both $A_1(t)$ and $A_0(t)$ converge to $a^{\mathcal{R}}(F)$, which is strictly above $a^{\mathcal{S}}(0, b)$; hence for small t , both actions are above the cutoff type's ideal action, and since $A_1(t) > A_0(t)$, we have $D(t) < 0$. As $t \uparrow 1$, $A_1(t) \rightarrow 1$ while $A_0(t) \rightarrow a^{\mathcal{R}}(F) < 1$; since $a^{\mathcal{S}}(1, b) \geq 1$, we have $D(t) > 0$ for t close to 1. By continuity, there exists $t^* \in (0, 1)$ such that $D(t^*) = 0$.

At t^* , the cutoff type with signal $\hat{s}$ is indifferent between disclosure and silence. Since $A_1(t^*) > A_0(t^*)$ and $u_{12} > 0$, types with signal $\hat{s}$ above t^* prefer disclosure and those below t^* prefer silence. For every other signal, set the off-path disclosed-signal action equal to $A_0(t^*)$ and choose beliefs accordingly; full support makes such beliefs feasible. Thus the constructed assessment is an informative equilibrium under verifiable evidence. Because the two on-path receiver actions are distinct, the receiver obtains a strict value of information relative to the

no-information FED outcome. Therefore every receiver-optimal equilibrium under verifiable evidence is strictly more informative than the FED outcome in this environment. $\square$

A.3 Proofs for Section 4.1

For any Borel set $B \subseteq \Theta$ with $F(B) > 0$, write $a^{\mathcal{R}}(B) \triangleq a^{\mathcal{R}}(F(\cdot | B))$. Let

$$\mathcal{B}^{CT} \triangleq \left\{ b \geq 0 : u(a^{\mathcal{R}}([t, 1]), t, b) > u(a^{\mathcal{R}}([0, t]), t, b) \text{ for every } t \in (0, 1) \right\},$$

and define

$$b^{CT} \triangleq \inf \mathcal{B}^{CT}. \quad (3)$$

Since $u_{13} > 0$, the set $\mathcal{B}^{CT}$ is an upper set; hence $b > b^{CT}$ implies $b \in \mathcal{B}^{CT}$. The boundary case $b = b^{CT}$ may or may not belong to $\mathcal{B}^{CT}$.

Let $K \triangleq N + 1$.

Lemma A.6. *Every equilibrium under unverifiable evidence is outcome-equivalent to one in which the sender ignores the realized signal and uses an ordered interval partition of the state space.*

Proof. By the cheap-talk equivalence established in Appendix B.2, we may view any equilibrium under unverifiable evidence as an equilibrium of the finite-message cheap-talk problem with state θ and message space M .

Let X be the finite set of on-path receiver actions. Suppose $x < x'$ are two actions in X and there exist $\theta < \theta'$ such that type θ sends a message inducing x' with positive probability and type θ' sends a message inducing x with positive probability. Sender optimality implies $u(x', \theta, b) \geq u(x, \theta, b)$ and $u(x, \theta', b) \geq u(x', \theta', b)$. But the difference $u(x', \cdot, b) - u(x, \cdot, b)$ is strictly increasing because $u_{12} > 0$, a contradiction. Hence higher actions are chosen only by weakly higher states. Since F is atomless, indifference points can be assigned arbitrarily. Therefore the outcome is equivalent to one induced by ordered actions $x_1 < \dots < x_J$ and cutoffs $0 = t_0 < t_1 < \dots < t_J = 1$, where action x_j is induced on $[t_{j-1}, t_j]$. Receiver optimality gives $x_j = a^{\mathcal{R}}([t_{j-1}, t_j])$ for each on-path cell. $\square$

Lemma A.7. *If $b \in \mathcal{B}^{CT}$, every equilibrium under unverifiable evidence is uninformative.*

Proof. Suppose, toward a contradiction, that there is an informative equilibrium under unverifiable evidence. By Lemma A.6, it is outcome-equivalent to an ordered interval equilibrium with at least two on-path actions. Let $t \in (0, 1)$ be the first nontrivial cutoff. The lowest action is $x_1 = a^{\mathcal{R}}([0, t])$. Let x_2 be the next higher on-path action. Then $x_2 \leq a^{\mathcal{R}}([t, 1])$, because the posterior on $[t, 1]$ first-order stochastically dominates the posterior on the second cell. Moreover, $x_2 > x_1$.

Since $b \in \mathcal{B}^{CT}$, we have $u(a^{\mathcal{R}}([t, 1]), t, b) > u(a^{\mathcal{R}}([0, t]), t, b) = u(x_1, t, b)$. Since $x_1 < x_2 \leq a^{\mathcal{R}}([t, 1])$ and $u(\cdot, t, b)$ is strictly concave, it follows that $u(x_2, t, b) > u(x_1, t, b)$. Thus the cutoff type strictly prefers the higher action. By continuity, types just below t also prefer the higher action, contradicting sender optimality. Hence no informative equilibrium exists. $\square$

For $L \geq 1$, let Q_L denote the maximal receiver payoff attainable by any measurable action rule that takes at most L values: $Q_L \triangleq \sup_h \int_{\Theta} u^{\mathcal{R}}(h(\theta), \theta) dF(\theta)$, where the supremum is over all measurable $h : \Theta \rightarrow A$ whose image has cardinality at most L .

Lemma A.8. *For every $L \geq 1$, $Q_{L+1} > Q_L$. Moreover, when $b = 0$, the receiver payoff of a receiver-optimal equilibrium under unverifiable evidence is Q_K .*

Proof. By strict single crossing, any finite-valued action rule can be weakly improved by assigning each state to the action that gives the receiver the highest payoff among the finitely many available actions. The resulting cells are ordered intervals. Hence the maximization defining Q_L can be restricted to ordered interval partitions with at most L cells and actions in $[0, 1]$. Compactness of the cutoff simplex and continuity imply that Q_L is attained.

Let h attain Q_L . Since F is atomless and h takes finitely many values, at least one cell $C = \{\theta : h(\theta) = x\}$ has positive probability and a nondegenerate conditional distribution. Choose $r \in (0, 1)$ such that both $C_\ell = C \cap [0, r]$ and $C_h = C \cap (r, 1]$ have positive probability. Let $x_\ell = a^{\mathcal{R}}(F(\cdot \mid C_\ell))$ and $x_h = a^{\mathcal{R}}(F(\cdot \mid C_h))$. The posterior on C_h strictly first-order stochastically dominates the posterior on C_ℓ , so $x_h > x_\ell$. Replacing the single action x on C by x_ℓ on C_ℓ and x_h on C_h strictly increases the receiver's payoff: the receiver can always choose x after both subevents, and because the two posterior-optimal actions are distinct, choosing the posterior-optimal action after each subevent is strictly better. Therefore $Q_{L+1} > Q_L$.

Now set $b = 0$. Sender and receiver preferences coincide. Any equilibrium under unverifiable evidence induces at most $K = N + 1$ on-path actions, so its receiver payoff is at most

Q_K . Conversely, let h attain Q_K . Without loss, for each θ , $h(\theta)$ maximizes $u^{\mathcal{R}}(x, \theta)$ among the finite set of actions in the image of h ; otherwise one could reassign θ to a better action and weakly increase the objective. Also, each action in the image of h is receiver-optimal conditional on the cell on which it is assigned; otherwise one could change that action and strictly improve. Thus the sender can send one message for each cell of h , and the receiver can choose the corresponding action. Sender optimality follows from the pointwise optimality of the assigned action and common interests; receiver optimality follows from the definition of the actions. Hence this is an equilibrium under unverifiable evidence with payoff Q_K . Therefore the receiver-optimal equilibrium payoff under unverifiable evidence is Q_K . $\square$

Lemma A.9. *When $b = 0$, every receiver-optimal equilibrium under unverifiable evidence is strictly more informative than every receiver-optimal equilibrium under verifiable evidence.*

Proof. By Lemma A.8, the receiver-optimal equilibrium payoff under unverifiable evidence is Q_K . We show that every equilibrium under verifiable evidence gives the receiver strictly less than Q_K .

Fix an equilibrium under verifiable evidence at $b = 0$. Let A denote the induced receiver action, and let $X \triangleq \{\alpha(\circ)\} \cup \{\alpha(s) : s \in S\}$ be the finite set of receiver actions that can be induced. Note that $|X| \leq K$.

First suppose the induced action is not a function of θ . For each θ , let $\hat{h}(\theta) \in \arg \max_{x \in X} u^{\mathcal{R}}(x, \theta)$. Since $u^{\mathcal{R}}(x', \theta) - u^{\mathcal{R}}(x, \theta)$ is strictly increasing in θ whenever $x' > x$, ties between two distinct actions occur at at most one state. Thus ties occur only on an F -null set. It follows that $\mathbb{E}[u^{\mathcal{R}}(\hat{h}(\theta), \theta)] > \mathbb{E}[u^{\mathcal{R}}(A, \theta)]$ whenever the conditional distribution of A given θ is nondegenerate on a positive-measure set of states. Since $\hat{h}$ takes at most K values, $\mathbb{E}[u^{\mathcal{R}}(\hat{h}(\theta), \theta)] \leq Q_K$. Hence the original equilibrium payoff is strictly below Q_K .

Now suppose instead that the induced action is a function of θ ; write it as $h(\theta)$. Let $a_\circ \triangleq \alpha(\circ)$ and $a_s \triangleq \alpha(s)$ for each $s \in S$. By full support, every signal s occurs with positive probability conditional on every θ . A type (θ, s) can induce only $a_\circ$ or a_s . Since the realized action is $h(\theta)$ almost surely conditional on θ , we must have $h(\theta) \in \{a_\circ, a_s\}$ for every $s \in S$ and for F -almost every θ . Hence $h(\theta) \in \bigcap_{s \in S} \{a_\circ, a_s\}$ for F -almost every θ . This intersection contains at most two distinct actions. Therefore the equilibrium payoff is at most Q_2 . Since $K = N + 1 \geq 3$, Lemma A.8 implies $Q_2 < Q_K$. Thus the equilibrium payoff is strictly below Q_K .

Because a receiver-optimal equilibrium exists under verifiable evidence, its payoff is also strictly below Q_K . The result follows. $\square$

Lemma A.10. *Let $U^{CT}(b)$ denote the receiver payoff in a receiver-optimal equilibrium under unverifiable evidence at bias b . Then*

$$\lim_{b \downarrow 0} U^{CT}(b) = U^{CT}(0) = Q_K.$$

Proof. By Lemma A.6, every equilibrium under unverifiable evidence is outcome-equivalent to an ordered partition with at most K induced actions. Hence $U^{CT}(b) \leq Q_K$ for every $b \geq 0$. Moreover, Lemma A.8 gives $U^{CT}(0) = Q_K$. It remains to prove the reverse limiting inequality.

Let $T : [0, 1] \rightarrow [0, 1]$ be the inverse of the CDF F . Since F is atomless with full support, T is continuous and strictly increasing. For $q = (q_1, \dots, q_{K-1})$ with $0 = q_0 < q_1 < \dots < q_{K-1} < q_K = 1$, define

$$x_j(q) \triangleq a^{\mathcal{R}}([T(q_{j-1}), T(q_j)]), \quad j = 1, \dots, K,$$

and let

$$W(q) \triangleq \sum_{j=1}^K \int_{q_{j-1}}^{q_j} u(x_j(q), T(z), 0) dz$$

denote the receiver payoff from the corresponding K -cell quantization. Using the compactness argument in the proof of Lemma A.8, choose an optimal K -cell quantization q^0 that minimizes $\sum_{j=1}^{K-1} q_j$ among all optimal K -cell quantizations. Since $Q_K > Q_{K-1}$, every cell of q^0 is non-degenerate.

For q near q^0 , the receiver's first-order condition in cell j is $\int_{q_{j-1}}^{q_j} u_1(x_j(q), T(z), 0) dz = 0$. Since T is continuous and the derivative of this expression with respect to x_j is strictly negative, the implicit function theorem implies that the action maps $x_j(\cdot)$ are continuously differentiable. For $j = 1, \dots, K-1$, define

$$G_j^{CT}(q, b) \triangleq u(x_j(q), T(q_j), b) - u(x_{j+1}(q), T(q_j), b).$$

Thus G_j^{CT} is the cutoff type's payoff from the lower action minus that from the higher action. The envelope theorem gives

$$\frac{\partial W(q)}{\partial q_j} = G_j^{CT}(q, 0), \quad j = 1, \dots, K-1. \quad (4)$$

Since q^0 is an interior maximizer, $G_j^{CT}(q^0, 0) = 0$ for every j .

We next establish local complementarity. For each cell, let

$$D_j(q) \triangleq \int_{q_{j-1}}^{q_j} u_{11}(x_j(q), T(z), 0) dz < 0.$$

Lemma A.2 and implicit differentiation of the receiver's first-order condition give

$$\frac{\partial x_j}{\partial q_{j-1}}(q) = \frac{u_1(x_j(q), T(q_{j-1}), 0)}{D_j(q)} > 0, \quad \frac{\partial x_j}{\partial q_j}(q) = -\frac{u_1(x_j(q), T(q_j), 0)}{D_j(q)} > 0.$$

Moreover, $x_j(q^0) < a^{\mathcal{R}}(T(q_j^0)) < x_{j+1}(q^0)$ so $u_1(x_j(q^0), T(q_j^0), 0) > 0 > u_1(x_{j+1}(q^0), T(q_j^0), 0)$. Consequently, at $(q^0, 0)$, the only nonzero cross-partial of G_j^{CT} with respect to q_r , $r \neq j$, are

$$\frac{\partial G_j^{CT}}{\partial q_{j-1}}(q^0, 0) = u_1(x_j(q^0), T(q_j^0), 0) \frac{\partial x_j}{\partial q_{j-1}}(q^0) > 0$$

whenever $j \geq 2$, and

$$\frac{\partial G_j^{CT}}{\partial q_{j+1}}(q^0, 0) = -u_1(x_{j+1}(q^0), T(q_j^0), 0) \frac{\partial x_{j+1}}{\partial q_{j+1}}(q^0) > 0$$

whenever $j \leq K-2$. By continuity, there exist a closed rectangle $\mathcal{Q}^0$ containing q^0 in its interior and contained in the strictly ordered region, and $\varepsilon^0 > 0$, such that every G_j^{CT} is nondecreasing in every q_r , $r \neq j$, on $\mathcal{Q}^0 \times [0, \varepsilon^0]$. By (4), W is supermodular on $\mathcal{Q}^0$.

The selected q^0 is the *least* maximizer of W on $\mathcal{Q}^0$. Indeed, if $\hat{q}^0$ is another maximizer, supermodularity gives

$$W(q^0 \wedge \hat{q}^0) + W(q^0 \vee \hat{q}^0) \geq W(q^0) + W(\hat{q}^0) = 2Q_K.$$

Both terms on the left are at most Q_K , so both equal Q_K . Thus $q^0 \wedge \hat{q}^0$ is another optimal K -cell quantization. The minimizing choice of q^0 therefore implies $q^0 \wedge \hat{q}^0 = q^0$, and hence $q^0 \leq \hat{q}^0$.

For $\eta > 0$, let q^η maximize

$$W(q) - \eta \sum_{j=1}^{K-1} q_j$$

over $\mathcal{Q}^0$. Supermodularity and the optimality of q^0 imply $W(q^\eta \wedge q^0) \geq W(q^\eta)$, so the linear penalty gives $q^\eta \leq q^0$. Optimality relative to q^0 implies $W(q^\eta) \rightarrow W(q^0)$ as $\eta \downarrow 0$. Hence every limit point of q^η is a maximizer of W weakly below the least maximizer q^0 , and therefore

equals q^0 . Thus $q^\eta \rightarrow q^0$. For sufficiently small η , q^η is interior to $\mathcal{Q}^0$, so (4) and the first-order conditions give

$$G_j^{CT}(q^\eta, 0) = \eta > 0, \quad j = 1, \dots, K-1. \quad (5)$$

Moreover, $q_j^\eta < q_j^0$ for every j : otherwise, local complementarity and $q^\eta \leq q^0$ would imply $G_j^{CT}(q^\eta, 0) \leq G_j^{CT}(q^0, 0) = 0$.

Since $x_j(q) < x_{j+1}(q)$ throughout $\mathcal{Q}^0$ and $u_{13} > 0$,

$$\frac{\partial G_j^{CT}(q, b)}{\partial b} = u_3(x_j(q), T(q_j), b) - u_3(x_{j+1}(q), T(q_j), b) = - \int_{x_j(q)}^{x_{j+1}(q)} u_{13}(a, T(q_j), b) da < 0.$$

Consequently, $G_j^{CT}(q^0, b) < 0$ for every $b \in (0, \varepsilon^0]$, whereas (5) and continuity imply $G_j^{CT}(q^\eta, b) > 0$ for all sufficiently small $b > 0$.

Fix such η and b , and consider the rectangle $[q^\eta, q^0]$. On its lower j -face, local complementarity gives $G_j^{CT}(q, b) \geq G_j^{CT}(q^\eta, b) > 0$, while on its upper j -face, $G_j^{CT}(q, b) \leq G_j^{CT}(q^0, b) < 0$. The Poincaré–Miranda theorem therefore yields $q(b) \in [q^\eta, q^0]$ such that $G_j^{CT}(q(b), b) = 0$ for every $j = 1, \dots, K-1$.

Set $t_j(b) = T(q_j(b))$, with $t_0(b) = 0$ and $t_K(b) = 1$. Assign a distinct message to each cell $[t_{j-1}(b), t_j(b)]$, and let the receiver choose $x_j(q(b))$ after the corresponding message. Receiver optimality holds by construction. At each cutoff, $G_j^{CT}(q(b), b) = 0$ gives adjacent indifference. Since the actions are strictly ordered and $u_{12} > 0$, single crossing implies that every type weakly prefers the action assigned to her cell to every other induced action. Hence the resulting assessment is an equilibrium under unverifiable evidence, with receiver payoff $W(q(b))$.

Finally, fix $\delta > 0$. Since $q^\eta \rightarrow q^0$ and W is continuous, choose $\eta > 0$ sufficiently small that all the preceding conclusions hold and $W(q) > Q_K - \delta$ throughout $[q^\eta, q^0]$. For this fixed η , the preceding continuity and Poincaré–Miranda arguments give, for every sufficiently small $b > 0$, an equilibrium under unverifiable evidence with cutoff vector $q(b) \in [q^\eta, q^0]$. Hence $U^{CT}(b) \geq W(q(b)) > Q_K - \delta$ for every sufficiently small $b > 0$. Thus $\liminf_{b \downarrow 0} U^{CT}(b) \geq Q_K$. Together with the initial upper bound, this proves the result. $\square$

Lemma A.11. *Suppose evidence is uninformative. There exist $\hat{b} \in (b^{CT}, \bar{B})$ such that:*

1. *For every $b > \hat{b}$, every equilibrium is uninformative regardless of whether evidence is verifiable or unverifiable.*

2. For every $b \in (b^{CT}, \hat{b})$, the game with verifiable evidence admits a strictly informative equilibrium.

At $b = \hat{b}$, every equilibrium under unverifiable evidence is uninformative, while verifiable evidence may or may not admit an informative equilibrium.

Proof. We construct the threshold $\hat{b}$. Write $\pi(s \mid \theta) = p_s$ for every (θ, s) , where $p_s \in (0, 1)$. For each $s \in S$ and $t \in (0, 1)$, consider the following sender strategy: a type with signal s discloses s if and only if $\theta \geq t$; all other types send $\circ$. Let $A_1(t) \triangleq a^{\mathcal{R}}(F(\cdot \mid \theta \geq t))$ be the receiver's posterior-optimal action after s , and let $A_{0,s}(t)$ be the receiver's posterior-optimal action after silence. The silence posterior is a strict mixture of $F(\cdot \mid \theta \leq t)$ and $F(\cdot \mid \theta \geq t)$. Hence, by Lemma A.2,

$$a^{\mathcal{R}}([0, t]) < A_{0,s}(t) < A_1(t) = a^{\mathcal{R}}([t, 1]).$$

By atomlessness of F , continuity of u , and uniqueness of posterior-optimal actions, $A_{0,s}$ and A_1 are continuous, and admit continuous extensions $A_{0,s}(0) = A_1(0) = a^{\mathcal{R}}(F)$, $A_{0,s}(1) = a^{\mathcal{R}}(F)$, and $A_1(1) = 1$.

For every $s \in S$, $t \in [0, 1]$, and $b \geq 0$, define

$$\Gamma_s(t, b) \triangleq \int_0^1 u_1((1-z)A_{0,s}(t) + zA_1(t), t, b) dz.$$

For every $t \in (0, 1)$, $u(A_1(t), t, b) - u(A_{0,s}(t), t, b) = (A_1(t) - A_{0,s}(t))\Gamma_s(t, b)$. Thus an interior cutoff t implements the proposed single-signal upper-tail disclosure strategy if and only if $\Gamma_s(t, b) = 0$. Moreover, $\Gamma_s(1, b) > 0$ for every s and $b \geq 0$, because $a^{\mathcal{R}}(F) < 1 \leq a^{\mathcal{S}}(1, b)$.

Define

$$\Psi(b) \triangleq \min_{s \in S} \min_{t \in [0, 1]} \Gamma_s(t, b).$$

The function Ψ is continuous and strictly increasing. Continuity follows from the joint continuity of $\Gamma_s(t, b)$ and minimization over the compact set $S \times [0, 1]$. On every compact bias interval, continuity and $u_{13} > 0$ imply that u_{13} has a strictly positive minimum on the relevant compact set, so every $\Gamma_s(t, \cdot)$ increases at a uniformly positive rate. Hence Ψ is strictly increasing. Moreover, $\Psi(0) \leq \Gamma_s(0, 0) = u_1(a^{\mathcal{R}}(F), 0, 0) < 0$. At $b = \bar{B}$, we have $a^{\mathcal{S}}(0, \bar{B}) = 1$ and $a^{\mathcal{S}}(t, \bar{B}) > 1$ for every $t > 0$. Since $A_{0,s}(t) \leq A_1(t) \leq 1$ and $A_{0,s}(0) = A_1(0) =$

$a^{\mathcal{R}}(F) < 1$, it follows that $\Gamma_s(t, \bar{B}) > 0$ for every (s, t) . Hence $\Psi(\bar{B}) > 0$. There is therefore a unique $\hat{b} \in (0, \bar{B})$ satisfying

$$\Psi(\hat{b}) = 0.$$

We show that $\hat{b} > b^{CT}$. If $\Gamma_s(0, b^{CT}) < 0$, then $\Psi(b^{CT}) < 0$, and the conclusion follows. Suppose instead that $\Gamma_s(0, b^{CT}) \geq 0$. Then $a^{\mathcal{S}}(0, b^{CT}) \geq a^{\mathcal{R}}(F)$; moreover, since $\Gamma_s(0, 0) < 0$, we have $b^{CT} > 0$. For $t \in (0, 1)$, define the cheap-talk cutoff difference

$$\Delta(t, b) \triangleq u(a^{\mathcal{R}}([t, 1]), t, b) - u(a^{\mathcal{R}}([0, t]), t, b),$$

and extend Δ continuously to $t = 0$ and $t = 1$. Because $b > b^{CT}$ implies $b \in \mathcal{B}^{CT}$, continuity in b gives $\Delta(t, b^{CT}) \geq 0$ for every $t \in (0, 1)$. Moreover, under the present case, $\Delta(0, b^{CT}) > 0$ and $\Delta(1, b^{CT}) > 0$. There must nevertheless be some $t^0 \in (0, 1)$ such that $\Delta(t^0, b^{CT}) = 0$. Otherwise, compactness and joint continuity would imply that $\Delta(t, b^{CT} - \eta) > 0$ for every $t \in [0, 1]$ and some $\eta > 0$, so $b^{CT} - \eta \in \mathcal{B}^{CT}$, contradicting the definition of b^{CT} as the infimum of $\mathcal{B}^{CT}$. At this critical cutoff t^0 , $u(a^{\mathcal{R}}([t^0, 1]), t^0, b^{CT}) = u(a^{\mathcal{R}}([0, t^0]), t^0, b^{CT})$. Since $a^{\mathcal{R}}([0, t^0]) < A_{0,s}(t^0) < A_1(t^0) = a^{\mathcal{R}}([t^0, 1])$, strict concavity implies $u(A_{0,s}(t^0), t^0, b^{CT}) > u(A_1(t^0), t^0, b^{CT})$, and hence $\Gamma_s(t^0, b^{CT}) < 0$. Thus $\Psi(b^{CT}) < 0$ in either case, which proves that $\hat{b} > b^{CT}$.

We next show that, for every $b > b^{CT}$, an informative equilibrium under verifiable evidence exists if and only if $\Gamma_s(t, b) = 0$ for some $s \in S$ and $t \in (0, 1)$. The sufficiency direction follows from the single-signal strategy above. If $\Gamma_s(t, b) = 0$, the cutoff type is indifferent between disclosure and silence; since $A_1(t) > A_{0,s}(t)$ and $u_{12} > 0$, types with signal s above t prefer disclosure and those below t prefer silence. Assign every other signal message the silence action and choose beliefs accordingly. This yields a strictly informative equilibrium.

For necessity, suppose an informative equilibrium exists under verifiable evidence at some $b > b^{CT}$. Silence must be on path; otherwise every type discloses her realized signal, and statistical uninformative implies that every on-path message induces $a^{\mathcal{R}}(F)$. Take a canonical generalized cutoff representation $(L, M, H, (t_s)_{s \in L \cup H})$, and write $a_\circ$ for the silence action.

We first show that $L = \emptyset$. No signal in L can have cutoff $t_s = 1$, because then it is always disclosed and induces an action below $a_\circ$, while types sufficiently close to state 1 strictly prefer the higher action $a_\circ$. Choose $s^* \in L$ with $t^* \triangleq t_{s^*} = \max_{s \in L} t_s \in (0, 1)$. The disclosed action after s^* is $a^{\mathcal{R}}([0, t^*])$. Every component of the silence posterior is either an upper-tail

posterior $F(\cdot \mid [t_s, 1])$ with $s \in L$, a lower-tail posterior $F(\cdot \mid [0, t_s])$ with $s \in H$, or the prior F for $s \in M$. By the choice of t^* , $F(\cdot \mid [t^*, 1])$ first-order stochastically dominates every such component. Hence $a_o \leq a^{\mathcal{R}}([t^*, 1])$. Since $s^* \in L$, $a^{\mathcal{R}}([0, t^*]) < a_o \leq a^{\mathcal{R}}([t^*, 1])$. Because $b > b^{CT}$ implies $b \in \mathcal{B}^{CT}$, $u(a^{\mathcal{R}}([t^*, 1]), t^*, b) > u(a^{\mathcal{R}}([0, t^*]), t^*, b)$. Strict concavity then implies $u(a_o, t^*, b) > u(a^{\mathcal{R}}([0, t^*]), t^*, b)$, contradicting cutoff indifference. Therefore $L = \emptyset$.

Since the equilibrium is informative, $H \neq \emptyset$, and some signal $s \in H$ must have an interior cutoff $t_s \in (0, 1)$. Otherwise every signal in H would always be disclosed, and statistical uninformative would make every disclosed-signal action and the silence action equal to $a^{\mathcal{R}}(F)$. For the selected signal s , $u(A_1(t_s), t_s, b) = u(a_o, t_s, b)$. Now consider the single-signal strategy in which only signal s is disclosed above t_s . Relative to the original silence pool, the new silence pool additionally includes the upper-tail observations previously disclosed by signals in $H \setminus \{s\}$. Each associated upper-tail posterior first-order stochastically dominates every component of the original silence posterior. Hence $a_o \leq A_{0,s}(t_s) < A_1(t_s)$. If $A_{0,s}(t_s) = a_o$, then $\Gamma_s(t_s, b) = 0$. If $A_{0,s}(t_s) > a_o$, strict concavity and cutoff indifference imply $u(A_{0,s}(t_s), t_s, b) > u(A_1(t_s), t_s, b)$, so $\Gamma_s(t_s, b) < 0$. Since $\Gamma_s(1, b) > 0$, continuity then gives some $t \in (t_s, 1)$ such that $\Gamma_s(t, b) = 0$. This proves the characterization.

We can now establish the two clauses. Suppose first that $b > \hat{b}$. Then $b > b^{CT}$ and $\Psi(b) > 0$, so $\Gamma_s(t, b) > 0$ for every $s \in S$ and $t \in [0, 1]$. The preceding characterization implies that every equilibrium under verifiable evidence is uninformative. Since $b > b^{CT}$ implies $b \in \mathcal{B}^{CT}$, Lemma A.7 implies that every equilibrium under unverifiable evidence is also uninformative. This proves the first clause.

Now suppose $b \in (b^{CT}, \hat{b})$. Then $\Psi(b) < 0$, so $\Gamma_s(t^0, b) < 0$ for some s and some $t^0 \in [0, 1]$. If $t^0 = 0$, the same strict inequality holds at some sufficiently small positive cutoff by continuity. Since $\Gamma_s(1, b) > 0$, there exists $t^* \in (0, 1)$ such that $\Gamma_s(t^*, b) = 0$. The resulting single-signal equilibrium is strictly informative. Its two on-path messages induce distinct posterior-optimal actions, so strict concavity gives the receiver a strict value of information relative to the no-information outcome. By contrast, Lemma A.7 implies that every equilibrium under unverifiable evidence is uninformative. Hence every receiver-optimal equilibrium under verifiable evidence is strictly more informative than every receiver-optimal equilibrium under unverifiable evidence. This proves the second clause.

At $b = \widehat{b}$, every equilibrium under unverifiable evidence is uninformative. The characterization above implies that verifiable evidence admits an informative equilibrium if and only if the minimum defining $\Psi(\widehat{b})$ is attained at some interior cutoff $t \in (0, 1)$. Thus verifiable evidence may or may not admit an informative equilibrium at the boundary. $\square$

Proof of Proposition 3. Let $U^V(b)$ and $U^{CT}(b)$ denote the receiver-optimal equilibrium payoffs under verifiable and unverifiable evidence, respectively, at bias b .

We first prove the low-bias comparison. By Lemmas A.8 and A.9,

$$U^{CT}(0) = Q_K > U^V(0).$$

Lemmas A.5 and A.10 imply that $U^V(b) \rightarrow U^V(0)$ and $U^{CT}(b) \rightarrow Q_K$ as $b \downarrow 0$. Hence there exists $\varepsilon > 0$ such that $U^{CT}(b) > U^V(b)$ for every $b \in [0, \varepsilon)$.

Moreover, $b^{CT} > 0$. Indeed, Lemma A.10 and $Q_K > Q_1$ imply that an informative equilibrium under unverifiable evidence exists for every sufficiently small positive b . By Lemma A.7, such biases cannot belong to $\mathcal{B}^{CT}$. Hence $b^{CT} > 0$.

Set $\beta_1 \triangleq b^{CT}$ and choose $\beta_0 \in (0, \min\{\varepsilon, \beta_1\})$. Then, for every $b \in [0, \beta_0]$, every receiver-optimal equilibrium under verifiable evidence is strictly less informative than every receiver-optimal equilibrium under unverifiable evidence. This proves the first clause.

We now prove the second clause. Suppose first that evidence is informative. Set $\beta_2 \triangleq \infty$. Fix any $b > \beta_1 = b^{CT}$. Since $\mathcal{B}^{CT}$ is an upper set, $b \in \mathcal{B}^{CT}$. By Lemma A.7, every equilibrium under unverifiable evidence is uninformative. Thus every receiver-optimal equilibrium under unverifiable evidence gives the receiver the no-information payoff. It remains to construct an informative equilibrium under verifiable evidence.

Choose $\bar{s} \in \arg \max_{s \in S} a^{\mathcal{R}}(F_s)$ so that $a^{\mathcal{R}}(F_{\bar{s}}) = \bar{a}_S$. We first note that $\bar{a}_S > a^{\mathcal{R}}(F)$. If $a^{\mathcal{R}}(F_s) \leq a^{\mathcal{R}}(F)$ for every $s \in S$, then strict concavity implies $\int_{\Theta} u_1(a^{\mathcal{R}}(F), \theta, 0) dF_s(\theta) \leq 0$ for every s , with strict inequality for some s because evidence is informative. Averaging over s would contradict $\int_{\Theta} u_1(a^{\mathcal{R}}(F), \theta, 0) dF(\theta) = 0$.

For each $t \in [0, 1)$, consider the following sender strategy: a type with signal $\bar{s}$ discloses $\bar{s}$ if and only if $\theta \geq t$; all other types send $\circ$. Let $A_1(t) \triangleq a^{\mathcal{R}}(F_{\bar{s}}(\cdot \mid \theta \geq t))$ be the receiver's posterior-optimal action after $\bar{s}$, and let $A_0(t)$ be the receiver's posterior-optimal action after silence. We have

$$A_0(t) < a^{\mathcal{R}}(F) < A_1(t)$$

for every $t \in [0, 1]$. To see this, note that $A_1(0) = \bar{a}_S > a^{\mathcal{R}}(F)$, while for $t > 0$, $F_{\bar{s}}(\cdot \mid \theta \geq t)$ strictly first-order stochastically dominates $F_{\bar{s}}$, so Lemma A.2 gives $A_1(t) > \bar{a}_S$. Moreover, the prior is a positive mixture of the posteriors after $\bar{s}$ and silence. Since the expected marginal receiver payoff at $a^{\mathcal{R}}(F)$ is strictly positive under the former posterior and equals zero under the prior, it must be strictly negative under the silence posterior. Strict concavity therefore implies $A_0(t) < a^{\mathcal{R}}(F)$.

By dominated convergence and uniqueness of posterior-optimal actions, A_0 and A_1 admit continuous extensions to $t = 1$, with $A_0(1) = a^{\mathcal{R}}(F)$ and $A_1(1) = 1$.

Define

$$D(t) \triangleq u(A_1(t), t, b) - u(A_0(t), t, b), \quad t \in [0, 1].$$

At the upper endpoint, $D(1) = u(1, 1, b) - u(a^{\mathcal{R}}(F), 1, b) > 0$, because $a^{\mathcal{R}}(F) < 1 \leq a^{\mathcal{S}}(1, b)$.

If $D(0) \geq 0$, set $t^* = 0$. Since $A_1(0) > A_0(0)$ and $u_{12} > 0$, the difference $u(A_1(0), \theta, b) - u(A_0(0), \theta, b)$ is strictly increasing in θ . Hence every type with signal $\bar{s}$ weakly prefers disclosure, and disclosure is optimal for all such types.

If instead $D(0) < 0$, continuity and $D(1) > 0$ imply that there exists $t^* \in (0, 1)$ such that $D(t^*) = 0$. Since $A_1(t^*) > A_0(t^*)$ and $u_{12} > 0$, types with signal $\bar{s}$ above t^* prefer disclosure and those below t^* prefer silence.

In either case, assign every other off-path signal message the action $A_0(t^*)$ and choose beliefs accordingly; full support makes such beliefs feasible. The resulting assessment is an equilibrium under verifiable evidence. Both on-path messages occur with positive probability and induce distinct posterior-optimal actions, so strict concavity gives the receiver a strict value of information relative to the no-information outcome. Hence every receiver-optimal equilibrium under verifiable evidence is strictly more informative than every receiver-optimal equilibrium under unverifiable evidence. Thus, when evidence is informative, the second clause holds for every $b > \beta_1$, so $\beta_2 = \infty$.

Suppose instead that evidence is uninformative. Let $\beta_2 \triangleq \hat{b}$, where $\hat{b}$ is the threshold constructed in Lemma A.11. That lemma gives $b^{CT} < \hat{b} < \bar{B}$, and hence $\beta_1 < \beta_2 < \infty$. Fix $b \in (\beta_1, \beta_2) = (b^{CT}, \hat{b})$. By Lemma A.11, the game with verifiable evidence admits a strictly informative equilibrium. By Lemma A.7, every equilibrium under unverifiable evidence is uninformative. Therefore every receiver-optimal equilibrium under verifiable evidence is strictly

more informative than every receiver-optimal equilibrium under unverifiable evidence. This proves the second clause. $\square$

A.4 Proofs for Section 4.3

We say that (S, π) is Blackwell more informative than (S', π') if there exists a garbling $\kappa : S \rightarrow \Delta(S')$ such that for every θ and every $s' \in S'$, $\pi'(s' | \theta) = \sum_{s \in S} \kappa(s' | s) \pi(s | \theta)$.

Proof of Remark 1. We first prove the first clause. Suppose $b \geq \bar{B}$. By Proposition 2, the FED outcome is induced by a receiver-optimal equilibrium under verifiable evidence for every evidence structure. If π is Blackwell more informative than π' , the receiver can reproduce the information generated by π' after observing the signal generated by π by applying the garbling κ and then choosing the posterior-optimal action under the garbled signal. Since the receiver can always ignore information, observing the signal generated by π gives weakly higher receiver payoff than observing the signal generated by π' . Hence the receiver-optimal equilibrium under (S, π) is more informative than the receiver-optimal equilibrium under (S', π') .

Now suppose $b = 0$. The sender and the receiver have common interests. For an evidence structure (S, π) , let $U_0^*(S, \pi)$ denote the receiver payoff of a receiver-optimal equilibrium under verifiable evidence at $b = 0$. By the common-interest argument used in the proof of Lemma A.4, $U_0^*(S, \pi)$ is also the value of the auxiliary problem in which the sender chooses a verifiable communication rule and the receiver chooses an action rule to maximize receiver payoff:

$$U_0^*(S, \pi) = \sup_{\sigma, \alpha} \int_{\Theta} \sum_{s \in S} \pi(s | \theta) \sum_{m \in \{s, \circ\}} \sigma(m | \theta, s) u^R(\alpha(m), \theta) dF(\theta),$$

where $\sigma(\cdot | \theta, s) \in \Delta(\{s, \circ\})$ and $\alpha : S \cup \{\circ\} \rightarrow [0, 1]$. It therefore suffices to show that U_0^* is monotone in the Blackwell order.

First suppose the garbling is deterministic, given by a map $g : S \rightarrow S'$. Let π^g denote the induced coarse experiment: $\pi^g(s' | \theta) = \sum_{s: g(s)=s'} \pi(s | \theta)$. We claim that $U_0^*(S, \pi) \geq U_0^*(S', \pi^g)$. Take any feasible disclosure policy σ' and action rule α' under the coarse experiment (S', π^g) . Construct a feasible pair (σ, α) under the fine experiment (S, π) by setting $\sigma(s | \theta, s) = \sigma'(g(s) | \theta, g(s))$, $\sigma(\circ | \theta, s) = \sigma'(\circ | \theta, g(s))$, $\alpha(s) = \alpha'(g(s))$, and $\alpha(\circ) = \alpha'(\circ)$.

Thus, after observing the fine signal s , the sender behaves exactly as the coarse policy would behave after observing the coarse signal $g(s)$, and the receiver assigns to the disclosed fine signal s the same action that the coarse receiver would assign to $g(s)$. A direct substitution shows that the receiver payoff is the same. Since (σ', α') was arbitrary, $U_0^*(S, \pi) \geq U_0^*(S', \pi^g)$.

Next, fix a finite alphabet S' . For any fixed pair (σ, α) , the payoff induced by (σ, α) is linear in the evidence structure. Hence $\pi' \mapsto U_0^*(S', \pi')$, being the pointwise supremum of linear functionals, is convex.

Finally, let $\kappa : S \rightarrow \Delta(S')$ be an arbitrary stochastic garbling. Let $\mathcal{G} = \{g : S \rightarrow S'\}$ be the finite set of deterministic maps. For each $g \in \mathcal{G}$, define $\omega_g \triangleq \prod_{s \in S} \kappa(g(s) \mid s)$. Then $\omega_g \geq 0$, $\sum_{g \in \mathcal{G}} \omega_g = 1$, and $\pi' = \sum_{g \in \mathcal{G}} \omega_g \pi^g$. Therefore,

$$U_0^*(S', \pi') = U_0^* \left(S', \sum_{g \in \mathcal{G}} \omega_g \pi^g \right) \leq \sum_{g \in \mathcal{G}} \omega_g U_0^*(S', \pi^g) \leq \sum_{g \in \mathcal{G}} \omega_g U_0^*(S, \pi) = U_0^*(S, \pi).$$

Thus the receiver-optimal equilibrium payoff at $b = 0$ is weakly higher under the Blackwell-more-informative evidence structure.

We now prove the second clause. Consider the pair of evidence structures $(S, \pi_{\delta, \varepsilon})$ and (S, π^0) constructed below, and set $b = \frac{1}{4}$. By construction, $\pi_{\delta, \varepsilon}$ is Blackwell more informative than π^0 . Let U^0 denote the receiver payoff of a receiver-optimal equilibrium under (S, π^0) at $b = \frac{1}{4}$. By Remark B.1, $U^0 > U^{CT2}(\frac{1}{4}) = -\frac{1}{12}$. Lemma A.12 below implies that the limsup of the receiver-optimal equilibrium payoff under $(S, \pi_{\delta, \varepsilon})$ as $(\delta, \varepsilon) \rightarrow (0, 0)$ is at most $-\frac{1}{12}$. Therefore, for sufficiently small $\delta, \varepsilon > 0$, the receiver-optimal equilibrium under $(S, \pi_{\delta, \varepsilon})$ gives the receiver strictly lower payoff than the receiver-optimal equilibrium under (S, π^0) . Thus the Blackwell-more-informative evidence structure induces a strictly less informative receiver-optimal equilibrium than the Blackwell-less-informative evidence structure. This proves the second clause. $\square$

Counterexample for the second clause of Remark 1. Consider the uniform-quadratic environment. Let the prior F be uniform, let $b = \frac{1}{4}$, and let $S = \{s_1, s_2\}$. We compare an informative evidence structure $(S, \pi_{\delta, \varepsilon})$ with a statistically uninformative structure (S, π^0) . The former is defined by

$$\pi_{\delta, \varepsilon}(s_2 \mid \theta) = \begin{cases} \varepsilon, & \text{if } \theta \in [0, 1 - \delta), \\ 1 - \varepsilon, & \text{if } \theta \in [1 - \delta, 1]. \end{cases} \quad (6)$$

where $\delta, \varepsilon \in (0, \frac{1}{2})$. The latter is given by $\pi^0(s_2 \mid \theta) = 1 - p$ for some $p \in (0, 1)$. So $\pi_{\delta, \varepsilon}$ is Blackwell more informative than π^0 .²³

Lemma A.12. *Fix the uniform-quadratic environment and $b \geq 0$. Let $\pi_{\delta, \varepsilon}$ be as defined in (6), and let $U^*(\pi; b)$ be the receiver payoff of a receiver-optimal equilibrium under verifiable evidence and evidence structure π . Then $\limsup_{(\delta, \varepsilon) \rightarrow (0, 0)} U^*(\pi_{\delta, \varepsilon}; b) \leq U^{CT2}(b)$.*

Proof. Take any sequence $(\delta_n, \varepsilon_n) \rightarrow (0, 0)$. For each n , choose an equilibrium under $\pi_{\delta_n, \varepsilon_n}$ whose receiver payoff U_n satisfies $U_n \geq U^*(\pi_{\delta_n, \varepsilon_n}; b) - 1/n$. Since $0 \leq U^*(\pi_{\delta_n, \varepsilon_n}; b) - U_n \leq 1/n$, it suffices to show $\limsup_n U_n \leq U^{CT2}(b)$.

Let $I_n \triangleq \text{Var}(\mathbb{E}[\theta \mid m])$ in the chosen equilibrium. Because the prior is uniform and receiver actions are posterior means, $U_n = -1/12 + I_n$. Hence it suffices to show $\limsup_n I_n \leq U^{CT2}(b) + 1/12$.

By Lemma A.1, whenever silence is on path, we represent the equilibrium in canonical generalized cutoff form. Thus, conditional on s_1 , either s_1 is never disclosed or it is disclosed on a lower or upper tail with cutoff $t_{1,n} \in [0, 1]$.

Passing to a subsequence if necessary and relabeling, assume that I_n converges to the limsup of the original sequence and that the form of disclosure of s_1 is the same for every n : either s_1 is never disclosed, it is disclosed on a lower tail, or it is disclosed on an upper tail. If s_1 is never disclosed, then $\Pr(m = s_1) = 0$, and $\Pr(m = s_2) \leq \Pr_{\pi_{\delta_n, \varepsilon_n}}(s_2) \rightarrow 0$. Hence $I_n \rightarrow 0$, and the desired bound follows. Thus it remains to consider the case in which s_1 is disclosed on the same tail for every n . Since $t_{1,n} \in [0, 1]$, after passing to a further subsequence if needed, assume $t_{1,n} \rightarrow t_1 \in [0, 1]$.

If $t_1 \in \{0, 1\}$, then one of the messages $m = s_1$ or $m = \circ$ has probability tending to zero. Indeed, if disclosure of s_1 is on the lower tail, then $\Pr(m = s_1) \leq t_{1,n} \rightarrow 0$ when $t_1 = 0$, while $\Pr(m = \circ) \leq 1 - t_{1,n} + \Pr_{\pi_{\delta_n, \varepsilon_n}}(s_2) \rightarrow 0$ when $t_1 = 1$. If disclosure of s_1 is on the upper tail, the analogous bounds are $\Pr(m = \circ) \leq t_{1,n} + \Pr_{\pi_{\delta_n, \varepsilon_n}}(s_2) \rightarrow 0$ when $t_1 = 0$ and $\Pr(m = s_1) \leq 1 - t_{1,n} \rightarrow 0$ when $t_1 = 1$. Since also $\Pr(m = s_2) \rightarrow 0$, it follows that $I_n \rightarrow 0$ along this subsequence.

Now suppose $t_1 \in (0, 1)$. Then for all large n , both $m = s_1$ and $m = \circ$ are on path. Let

²³Every statistically uninformative experiment is a garbling of any experiment: set $\kappa(s_2 \mid s) = 1 - p$ for every $s \in S$.

$a_{1,n} \triangleq \mathbb{E}[\theta \mid m = s_1]$ and $a_{\circ,n} \triangleq \mathbb{E}[\theta \mid m = \circ]$. We claim that along this subsequence, $\{a_{1,n}, a_{\circ,n}\} \rightarrow \{t_1/2, (1+t_1)/2\}$. To see this, note first that $\Pr_{\pi_{\delta_n, \varepsilon_n}}(s_2) \rightarrow 0$. Hence $\Pr(m = s_2) \leq \Pr_{\pi_{\delta_n, \varepsilon_n}}(s_2) \rightarrow 0$. Next, disclosure of s_1 can only occur when s_1 is observed. Since $\pi_{\delta_n, \varepsilon_n}(s_1 \mid \theta) \rightarrow 1$ for every fixed $\theta < 1$ and the region $\{\theta \geq 1 - \delta_n\}$ has vanishing mass, Bayes' rule implies that conditioning on $m = s_1$ becomes asymptotically equivalent to conditioning on θ lying in the disclosure tail for s_1 , and similarly conditioning on $m = \circ$ becomes asymptotically equivalent to conditioning on θ lying in the complementary tail. Concretely, if disclosure of s_1 is on the lower tail, then $a_{1,n} \rightarrow \mathbb{E}[\theta \mid \theta \leq t_1] = t_1/2$ and $a_{\circ,n} \rightarrow \mathbb{E}[\theta \mid \theta > t_1] = (1+t_1)/2$; if disclosure is on the upper tail, the two limits are swapped.

Sender optimality at the cutoff implies

$$t_{1,n} = \Pi_{[0,1]} \left(\frac{a_{1,n} + a_{\circ,n}}{2} - b \right) \rightarrow \Pi_{[0,1]} \left(\frac{1+2t_1}{4} - b \right) = t_1,$$

where $\Pi_{[0,1]}(x) = \max\{0, \min\{1, x\}\}$. This fixed-point equation has the unique solution $t_1 = \max\{1/2 - 2b, 0\}$. Since we are in the case $t_1 \in (0, 1)$, it follows that $t_1 = 1/2 - 2b$. The contribution of $m = s_2$ to I_n is asymptotically negligible. Moreover, $\{a_{1,n}, a_{\circ,n}\} \rightarrow \{t_1/2, (1+t_1)/2\}$, and the probability that $m = s_1$ converges to the mass of the disclosure tail, which is either t_1 or $1 - t_1$ depending on the tail choice. Since I_n is invariant to swapping the labels of s_1 and $\circ$, it follows that

$$\limsup_{n \rightarrow \infty} I_n \leq \frac{t_1(1-t_1)}{4} = \max \left\{ \frac{1}{16} - b^2, 0 \right\} = U^{CT2}(b) + \frac{1}{12}.$$

Therefore $\limsup_n U_n \leq U^{CT2}(b)$, as desired. $\square$

A.5 Proofs for Section 5

Proof of Proposition 4. Fix $b \geq 0$. We distinguish two cases.

First suppose $b > 0$. For each signal realization s , consider the standard cheap-talk problem with prior F_s , receiver payoff $u^R = u(\cdot, \cdot, 0)$, and sender payoff $u^S = u(\cdot, \cdot, b)$. Since F_s is atomless and has full support on $[0, 1]$, the argument of Theorem 1 of Crawford and Sobel (1982) yields a finite maximal number of induced actions and an equilibrium attaining it. Moreover, Proposition 1 of Chen et al. (2008) implies that every such maximal-step equilib-

rium satisfies NITS.²⁴

For each $s \in S$, select such a maximal-step equilibrium. Let its induced actions be $x_1^s < \dots < x_{j_s}^s$. By NITS,

$$u(x_1^s, 0, b) \geq u(0, 0, b)$$

for every $s \in S$. Identify the finitely many messages used in each selected equilibrium with distinct elements of M^U . Extend the assessment to every remaining $m^U \in M^U$ by assigning it the belief and action associated with the lowest-action message of the selected equilibrium.

Construct an assessment in the hybrid game as follows. Every type (θ, s) sends $m^V = s$ in the verifiable component and, conditional on s , follows the selected maximal-step cheap-talk equilibrium under posterior F_s in the unverifiable component. After every message (s, m^U) , the receiver follows the corresponding extended conditional cheap-talk assessment under F_s . After every message $(\circ, m^U)$, assign beliefs concentrated on $\theta = 0$, so that the receiver chooses 0.

Receiver optimality and belief consistency hold by construction. Sender optimality against deviations that continue to disclose s follows from the extended conditional cheap-talk assessment under F_s . It remains to rule out deviations to $m^V = \circ$. Fix s . By NITS, type $\theta = 0$ weakly prefers x_1^s to 0. Since $x_1^s \geq 0$ and $u_{12} > 0$, the payoff difference $u(x_1^s, \theta, b) - u(0, \theta, b)$ is increasing in θ , so every type weakly prefers x_1^s to 0. In addition, incentive compatibility in the selected conditional cheap-talk equilibrium implies that each type weakly prefers her own equilibrium action to x_1^s . Hence no type benefits from deviating to any message with $m^V = \circ$. Therefore the constructed assessment is a disclose-and-refine equilibrium.

Now suppose $b = 0$. Since $M^U = [0, 1]$, construct an assessment in which every type (θ, s) sends $m^V = s$ and $m^U = \theta$. After observing (s, m^U) , the receiver assigns probability one to $(\theta, s) = (m^U, s)$ and chooses $a^R(m^U)$. After every message $(\circ, m^U)$, assign beliefs concentrated on $\theta = 0$, so that the receiver chooses 0. Receiver optimality and belief consistency hold by construction. Since sender and receiver preferences coincide at $b = 0$, each type obtains her first-best action on path and therefore has no profitable deviation. Hence the constructed assessment is a disclose-and-refine equilibrium. $\square$

²⁴Crawford and Sobel (1982) and Chen et al. (2008) formulate their baseline model with a differentiable CDF admitting a density. As far as the prior is concerned, the proofs of Theorem 1 in the former and Proposition 1 in the latter require only atomlessness and full support.

Lemma A.13. *The set of equilibrium outcomes of the hybrid game contains the union of the equilibrium outcome sets under verifiable evidence and under unverifiable evidence.*

Proof. First take any equilibrium outcome under verifiable evidence, induced by an assessment (σ, α, μ) . Fix any $\bar{m}^U \in M^U$. In the hybrid game, let each type attach $\bar{m}^U$ to the verifiable message prescribed by σ . Let the receiver's action after every hybrid message (m^V, m^U) be $\alpha(m^V)$, independent of m^U . After $(m^V, \bar{m}^U)$, use the original belief following m^V . After every (m^V, m^U) with $m^U \neq \bar{m}^U$, choose a feasible belief with the same Θ -marginal as the original belief following m^V . These beliefs are feasible because the feasibility restrictions on the verifiable component are the same as in the original verifiable-evidence game. Deviations in the unverifiable component do not change the induced action, and deviations in the verifiable component are already unprofitable in the original equilibrium. Hence the same outcome is induced by a hybrid equilibrium.

Next take any equilibrium outcome under unverifiable evidence, induced by an assessment (σ, α, μ) with message space M^U . In the hybrid game, whenever the original strategy sends m^U , let the sender send $(\circ, m^U)$. Let the receiver choose $\alpha^H(m^V, m^U) \triangleq \alpha(m^U)$ after every hybrid message (m^V, m^U) . Beliefs after on-path messages $(\circ, m^U)$ are given by Bayes' rule and reproduce the original posterior after m^U . For every off-path message (m^V, m^U) , choose a feasible belief with the same Θ -marginal as the original belief after m^U ; full support makes this possible after any disclosed signal and after silence. Under this construction, every hybrid deviation induces the same action as some message available in the original unverifiable-evidence equilibrium. Hence no sender type obtains a profitable new deviation. Therefore the original unverifiable-evidence outcome is replicated by a hybrid equilibrium. $\square$

Proof of Remark 3. Let Λ^H denote the set of hybrid equilibrium outcomes, and define $\bar{U}^H \triangleq \sup_{\lambda \in \Lambda^H} U(\lambda)$. By Lemma A.13, Λ^H contains the equilibrium outcome sets under verifiable evidence and under unverifiable evidence. Under the maintained assumption that M^U is finite or $M^U = [0, 1]$, a receiver-optimal equilibrium exists in the hybrid game (Online Appendix B.3), so $\bar{U}^H$ is attained. Receiver-optimal equilibria also exist under the two communication technologies in isolation (Online Appendix B.2).²⁵ Hence every receiver-optimal equilibrium

²⁵When $M^U = [0, 1]$, existence for the game with unverifiable evidence follows from the argument of Remark B.3, with the verifiable component suppressed.

of the hybrid game is weakly more informative than every receiver-optimal equilibrium under either mode of communication alone.

It remains to prove the second statement. Suppose $M^U = [0, 1]$. When $b = 0$, the sender's and receiver's preferences coincide, and full revelation can be sustained under unverifiable communication alone by having the sender report $m^U = \theta$. This attains the receiver's first-best payoff, so the hybrid game cannot be strictly more informative.

Now suppose $b \geq \bar{B}$. Every sender type strictly prefers a higher receiver action throughout $[0, 1]$. Fix any equilibrium of the hybrid game. If two types with the same signal s induced distinct actions, the type inducing the lower action could imitate the message pair of the type inducing the higher action, since both types have the same feasible set $\{\circ, s\} \times M^U$. This would be a profitable deviation. Hence the induced action is a function only of the realized signal s . Every hybrid equilibrium outcome is therefore a garbling of the evidence, and cannot give the receiver a higher payoff than the full-evidence-disclosure outcome. By Proposition 2, receiver-optimal equilibria under verifiable communication alone are as informative as the full-evidence-disclosure outcome when $b \geq \bar{B}$. Thus the hybrid game cannot be strictly more informative than both communication modes in isolation. Strictness can therefore occur only if $b \in (0, \bar{B})$. $\square$

B Online Appendix for The Limits of Verifiability: Credibility and Flexibility in Communication

B.1 Additional Proofs for Section 4.2

Throughout this subsection, we work in the uniform-quadratic environment $u^R(a, \theta) = -(a - \theta)^2$, and $u^S(a, \theta, b) = -(a - \theta - b)^2$. Given an outcome λ , recall that $U(\lambda) = \mathbb{E}_\lambda[u^R(a, \theta)]$. Thus all comparisons below are receiver-payoff comparisons.

Lemma B.1. *Suppose F is uniform. Under unverifiable evidence, an equilibrium with $2 \leq n \leq N + 1$ on-path actions exists if and only if $b < 1/(2n(n - 1))$. Moreover, for each such n , this equilibrium outcome is unique.*

Proof. This follows directly from the explicit equilibrium characterization in Section 4 of Crawford and Sobel (1982) for the uniform prior. Since under unverifiable evidence the realized signal is strategically irrelevant, the environment is outcome-equivalent to a standard cheap-talk problem with message alphabet of cardinality $N + 1$. $\square$

Lemma B.2. *Suppose F is uniform and $b < 1/(2n(n - 1))$ with $2 \leq n \leq N + 1$. Under unverifiable evidence, the equilibrium with n on-path actions gives the receiver strictly higher payoff than the equilibrium with $n - 1$ on-path actions.*

Proof. This follows directly from Theorem 3 of Crawford and Sobel (1982). $\square$

Lemma B.3. *Suppose F is uniform and (S, π) is statistically uninformative. If $b \geq 1/2$, then every equilibrium under verifiable evidence is uninformative.*

Proof. Let $b \geq 1/2$ and suppose, toward a contradiction, that there exists an informative equilibrium under verifiable evidence. If $\Pr(m = \circ) = 0$, then the receiver's action depends only on the realized signal. Since the evidence structure is statistically uninformative, the outcome is uninformative, a contradiction. Hence $\Pr(m = \circ) > 0$.

By Lemma A.1, take a canonical outcome-equivalent generalized cutoff representation. Write $a_\circ = \alpha(\circ)$ and $a_s = \alpha(s)$ for each $s \in S$.

We first claim that some signal must be partially disclosed. If not, every signal is either always disclosed or never disclosed. Since evidence is statistically uninformative, any always-

disclosed signal induces posterior mean $1/2$. Silence pools only whole signal realizations, each of which is statistically independent of θ , and therefore also induces posterior mean $1/2$. Thus every on-path message induces the same receiver action, contradicting informativeness.

Hence there exists a partially disclosed signal s . If $a_s < a_o$, then disclosure of s occurs on a lower interval $[0, t]$ for some $t \in (0, 1)$. Since evidence is statistically uninformative and F is uniform, $a_s = t/2$. The cutoff condition is $t + b = (a_s + a_o)/2$, so $a_o = 3t/2 + 2b > 1$, contradicting $a_o \in [0, 1]$.

If $a_s > a_o$, then disclosure of s occurs on an upper interval $[t, 1]$ for some $t \in (0, 1)$. Since evidence is statistically uninformative and F is uniform, $a_s = (1 + t)/2$. The cutoff condition gives $a_o = 3t/2 + 2b - 1/2$. Therefore $a_o - a_s = t + 2b - 1 \geq 0$, contradicting $a_s > a_o$. Hence every equilibrium under verifiable evidence is uninformative. $\square$

For later reference, let $U^{CT2}(b)$ denote the receiver payoff from the most informative equilibrium under unverifiable evidence with at most two on-path messages in the uniform-quadratic environment. Its value is computed in the proof below.

Proposition B.1. *Suppose (S, π) is statistically uninformative, that is, $\pi(s \mid \theta) = p_s$ for all $(\theta, s) \in [0, 1] \times S$, with $p_s > 0$ and $\sum_{s \in S} p_s = 1$, and let F be uniform on $[0, 1]$. There exists $\beta^\circ \in (1/8, 1/4)$ such that:*

- *for all $b \leq \beta^\circ$, the receiver-optimal equilibrium payoff under verifiable evidence is $U^{CT2}(b)$;*
- *for $b \in (\beta^\circ, 1/2)$, every receiver-optimal equilibrium under verifiable evidence gives the receiver strictly higher payoff than every receiver-optimal equilibrium under unverifiable evidence.*

Proof. For any equilibrium outcome in the uniform-quadratic environment, the receiver payoff can be written as $U = -1/12 + \text{Var}(\mathbb{E}[\theta \mid m]) = -1/3 + \mathbb{E}[(\mathbb{E}[\theta \mid m])^2]$. Thus, within this proof, comparing receiver payoffs is equivalent to comparing $\text{Var}(\mathbb{E}[\theta \mid m])$ or $\mathbb{E}[(\mathbb{E}[\theta \mid m])^2]$.

Two-message benchmark under unverifiable evidence. For $b < 1/4$, the unique nonbabbling two-message equilibrium under unverifiable evidence is characterized by a cutoff $r \in (0, 1)$, actions $x_\ell = r/2$ and $x_h = (1 + r)/2$, and cutoff indifference $r + b = (x_\ell + x_h)/2 =$

$(2r + 1)/4$. Hence $r = 1/2 - 2b$ and $U^{CT2}(b) = -1/48 - b^2$. For $b \geq 1/4$, every two-message equilibrium under unverifiable evidence is babbling, so $U^{CT2}(b) = -1/12$.

A verifiable equilibrium achieving $U^{CT2}(b)$. For $b < 1/4$, fix the same cutoff $r = 1/2 - 2b$. Consider the following strategy under verifiable evidence: for every (θ, s) , disclose s if and only if $\theta \geq r$, and send $\circ$ otherwise. Since evidence is statistically uninformative, after any disclosed signal the receiver's posterior is the uniform distribution conditional on $\theta \geq r$, so the induced action is $(1 + r)/2$; after silence the receiver's posterior is the uniform distribution conditional on $\theta < r$, so the induced action is $r/2$. The cutoff type is indifferent by the same equation as in the two-message equilibrium under unverifiable evidence, and single crossing gives sender optimality. Thus the receiver-optimal equilibrium under verifiable evidence achieves at least $U^{CT2}(b)$.

Canonical structure of equilibria under verifiable evidence. Fix any equilibrium under verifiable evidence. If $\Pr(m = \circ) = 0$, then only evidence messages are observed and, because evidence is statistically uninformative, $\mathbb{E}[\theta \mid m] = 1/2$ almost surely, so $U = -1/12$. Hence assume $\Pr(m = \circ) > 0$.

By Lemma A.1, take a canonical outcome-equivalent generalized cutoff representation. Write $a_\circ = \alpha(\circ)$ and $a_s = \alpha(s)$. Let $L \triangleq \{s \in S : s \text{ is disclosed with positive probability and } a_s < a_\circ\}$, $H \triangleq \{s \in S : s \text{ is disclosed with positive probability and } a_s > a_\circ\}$, and M be the remaining signals, which are never disclosed and induce the same action as silence.

Genuine three-action verifiable equilibria never beat $U^{CT2}(b)$. Suppose both L and H are nonempty. We first show that every signal in $L \cup H$ has an interior cutoff. If $s \in L$ were always disclosed conditional on s , then statistical uninformativeness would imply $a_s = 1/2 < a_\circ$. The indifference cutoff at which the sender is indifferent between a_s and $a_\circ$ is $(a_s + a_\circ)/2 - b \leq 3/4 - b < 1$. Hence disclosure of s cannot be optimal for all θ , implying $t_s < 1$. By canonicity, we also have $t_s > 0$. Therefore, $t_s \in (0, 1)$.

Now suppose some $s \in H$ were always disclosed conditional on s . Then $a_s = 1/2 > a_\circ$. Since L is nonempty and every lower-tail cutoff is interior, for any $s' \in L$ the cutoff indifference condition gives $a_\circ = 3t_{s'}/2 + 2b$. Hence $a_\circ < 1/2$ implies $b < 1/4$. But disclosure of $s \in H$ for all θ requires $(1/2 + a_\circ)/2 - b \leq 0$, which implies $b \geq (1/2 + a_\circ)/2 > 1/4$, a contradiction. Hence every $s \in H$ also has an interior cutoff.

Therefore, for every $s \in L \cup H$, the cutoff is interior and the cutoff indifference condition binds. If $s \in L$, disclosure occurs iff $\theta \leq t_s$, so $a_s = t_s/2$ and $t_s = (a_s + a_o)/2 - b$; hence $t_s = (2a_o - 4b)/3 \triangleq t_\ell$. If $s \in H$, disclosure occurs iff $\theta \geq t_s$, so $a_s = (1 + t_s)/2$ and the indifference condition gives $t_s = (1 + 2a_o - 4b)/3 = t_\ell + 1/3 \triangleq t_h$. Thus the equilibrium induces exactly three on-path actions

$$x_\ell = \frac{t_\ell}{2}, \quad a_o = \frac{3t_\ell + 4b}{2}, \quad x_h = \frac{t_\ell}{2} + \frac{2}{3},$$

where $t_\ell \in (0, 2/3)$. This conclusion does not require M to be nonempty: because the cutoffs in both L and H are interior, silence is on path even when $M = \emptyset$.

Let $\lambda \triangleq \sum_{s \in L} p_s$, $\rho \triangleq \sum_{s \in H} p_s$, and $\tau \triangleq \sum_{s \in M} p_s = 1 - \lambda - \rho$. For fixed $t \triangleq t_\ell$, the on-path probabilities are $\Pr(x_\ell) = \lambda t$, $\Pr(x_h) = \rho(2/3 - t)$, and $\Pr(a_o) = \lambda(1 - t) + \rho(t + 1/3) + \tau$. Bayes' rule after silence gives

$$a_o = \frac{\lambda(1 - t)^{\frac{1+t}{2}} + \rho(t + \frac{1}{3})^{\frac{t+\frac{1}{3}}{2}} + \tau \frac{1}{2}}{\lambda(1 - t) + \rho(t + \frac{1}{3}) + \tau}.$$

Since the silence posterior is a mixture of components with means $(1 + t)/2$, $(t + 1/3)/2$, and $1/2$, we have $a_o \leq (1 + t)/2$. Together with $a_o = (3t + 4b)/2$, this implies $b < 1/4$.

For fixed t , relax the subset-sum restriction on (λ, ρ, τ) and allow these variables to range continuously over the simplex subject to the Bayes equation. This relaxed set contains every triple generated by a partition (L, M, H) of S . Since $a_o = (3t + 4b)/2$ is fixed, the Bayes equation together with $\lambda + \rho + \tau = 1$ uniquely determines $\lambda = \lambda(\tau)$ and $\rho = \rho(\tau)$ as affine functions of τ . A direct calculation shows that both $\lambda(\tau)$ and $\rho(\tau)$ are strictly decreasing in τ in the genuine three-action region. Hence, if $\lambda, \rho > 0$ at some $\tau > 0$, reducing τ to zero preserves $\lambda, \rho > 0$ and remains feasible in the relaxed problem.

Let

$$\begin{aligned} W(\tau) &\triangleq \mathbb{E} \left[(\mathbb{E}[\theta \mid m])^2 \right] \\ &= \lambda(\tau) t x_\ell^2 + \rho(\tau) \left(\frac{2}{3} - t \right) x_h^2 + \left(\lambda(\tau)(1 - t) + \rho(\tau) \left(t + \frac{1}{3} \right) + \tau \right) a_o^2. \end{aligned}$$

Substituting for $\lambda(\tau)$ and $\rho(\tau)$ and differentiating gives

$$W'(\tau) = -\frac{t(2b + t)(3t - 2)(6b + 3t - 2)}{3(18bt - 6b + 9t^2 - 6t + 2)} < 0.$$

Indeed, the denominator is positive in the relevant region, as verified below; moreover, $t > 0$, $2b + t > 0$, and $3t - 2 < 0$, while $x_h > a_o$ is equivalent to $6b + 3t - 2 < 0$. Thus receiver payoff is strictly decreasing in τ . Consequently, every genuine three-action configuration with $\tau > 0$ gives strictly lower receiver payoff than its relaxed counterpart with $\tau = 0$, and it suffices to consider $\tau = 0$ with $\lambda, \rho \in (0, 1)$.

Suppose $\tau = 0$. Then $\rho = 1 - \lambda$, and Bayes' rule after silence gives

$$\lambda = \frac{(3t + 1)(12b + 6t - 1)}{4(18bt - 6b + 9t^2 - 6t + 2)}.$$

For the two-message unverifiable benchmark, $U^{CT2}(b) = -1/48 - b^2$, so the corresponding value of $\mathbb{E}[(\mathbb{E}[\theta | m])^2]$ is $U^{CT2}(b) + 1/3 = 5/16 - b^2$. A direct simplification yields

$$\left(\frac{5}{16} - b^2\right) - \mathbb{E}[(\mathbb{E}[\theta | m])^2] = \frac{(4b + 2t - 1)(12b + 6t - 1)(-18bt + 6b + 15t^2 - 10t - 2)}{48(18bt - 6b + 9t^2 - 6t + 2)}.$$

For $t \in (0, 2/3)$ and $b < 1/4$, the denominator is positive. Feasibility with $\lambda \in (0, 1)$ implies $12b + 6t - 1 > 0$ and $4b + 2t - 1 < 0$. Moreover, $-18bt + 6b + 15t^2 - 10t - 2 \leq -2 + 6b < 0$. Hence the right-hand side is strictly positive, so every genuine three-action verifiable equilibrium gives strictly lower receiver payoff than $U^{CT2}(b)$.

Two-action verifiable equilibria with lower-tail disclosure. Suppose $L \neq \emptyset$ and $H = \emptyset$. Let $\lambda \triangleq \sum_{s \in L} p_s$ and let $t \in (0, 1)$ be the common cutoff for all $s \in L$.² Thus the equilibrium has exactly two on-path actions. The disclosed action is $x_\ell = t/2$. Bayes' rule after silence gives $a_o = (1 - \lambda t^2)/(2(1 - \lambda t))$. The cutoff indifference gives $a_o = 3t/2 + 2b$. Combining these equations yields $\lambda = (3t + 4b - 1)/(2t^2 + 4bt)$. Feasibility requires $\lambda \leq 1$, equivalently $(t - 1)(4b + 2t - 1) \geq 0$. Since $t < 1$, this implies $4b + 2t - 1 \leq 0$, so in particular $b < 1/4$. The receiver payoff is

$$U^{L2}(b; \lambda) = -\frac{1}{12} + \lambda t(1 - \lambda t)(t + 2b)^2.$$

Using $U^{CT2}(b) = -1/48 - b^2$, a direct simplification gives

$$U^{CT2}(b) - U^{L2}(b; \lambda) = \frac{(-4b + 6t - 5)(4b + 2t - 1)}{16}.$$

²If instead some $s \in L$ were always disclosed, then $a_s = 1/2 < a_o$ and the indifference cutoff $(a_s + a_o)/2 - b$ would be at most $3/4 - b < 1$, a contradiction. Hence every signal in L is partially disclosed, and the cutoff indifference condition implies that all signals in L have the same cutoff.

Since $4b + 2t - 1 \leq 0$, we have $t \leq (1 - 4b)/2$, and hence $-4b + 6t - 5 \leq -4b + 3(1 - 4b) - 5 = -2 - 16b < 0$. Thus $U^{L2}(b; \lambda) \leq U^{CT2}(b)$. Equality can occur only when $4b + 2t - 1 = 0$, which, upon substitution into the expression for λ , gives $\lambda = 1$. In that case the outcome coincides with the two-message equilibrium under unverifiable evidence.

Two-action verifiable equilibria with upper-tail disclosure. Now suppose $L = \emptyset$ and $H \neq \emptyset$. Let $\rho \triangleq \sum_{s \in H} p_s \in (0, 1]$ and let $t \in (0, 1)$ be the common cutoff for all $s \in H$.³ Thus the equilibrium has exactly two on-path actions. The disclosed action is $x_h = (1 + t)/2$. Bayes' rule after silence gives $a_o = (1 - \rho + \rho t^2)/(2(1 - \rho + \rho t))$. Indifference at $\theta = t$ gives $t = (x_h + a_o)/2 - b$, equivalently,

$$g(t) \triangleq 2\rho t^2 + (4\rho(b - 1) + 3)t + 2(1 - \rho)(2b - 1) = 0.$$

If $\rho < 1$ and $b < 1/2$, then $g(0) < 0 < g(1)$, so $g(t) = 0$ has a unique positive root $t = t(b, \rho) \in (0, 1)$. Moreover, $g(1 - 2b) = 1 - 2b > 0$, so uniqueness of the positive root implies $t < 1 - 2b$. Thus $x_h - a_o = 1 - t - 2b > 0$, and single crossing implies that this root indeed implements an upper-tail equilibrium. If $\rho = 1$ and $b < 1/4$, the nonzero root is $1/2 - 2b$ and gives exactly the two-message benchmark under unverifiable evidence.

For such an equilibrium, the receiver payoff is

$$U^{H2}(b; \rho) = -\frac{1}{12} + \rho(1 - t)(1 - \rho(1 - t))(1 - t - 2b)^2,$$

where we use $x_h + a_o = 2(t + b)$ and therefore $x_h - a_o = 1 - t - 2b$. For $\rho \in (0, 1)$ and $b \leq 1/4$, a direct simplification shows that $U^{H2}(b; \rho) = U^{CT2}(b)$ if and only if

$$Q(b, \rho) \triangleq 64\rho b^2 + (108 - 100\rho)b + (25\rho - 27) = 0,$$

and that $U^{H2}(b; \rho) > U^{CT2}(b)$ if and only if $Q(b, \rho) > 0$. Moreover, $Q(1/8, \rho) = 27(\rho - 1)/2 < 0$ and $Q(1/4, \rho) = 4\rho > 0$. Since $\partial Q(b, \rho)/\partial b = 128\rho b + 108 - 100\rho > 0$ for every $b \geq 0$, for each $\rho \in (0, 1)$ there is a unique $b(\rho) \in (1/8, 1/4)$ such that $U^{H2}(b(\rho); \rho) =$

³Suppose first that some signal in H has an interior cutoff $t > 0$. Its cutoff indifference condition gives $a_o = (3t + 4b - 1)/2$. If another signal in H were always disclosed, its disclosed action would be $1/2$, and its indifference cutoff would be $(1/2 + a_o)/2 - b = 3t/4 > 0$, a contradiction. It remains to rule out the case in which every signal in H is always disclosed. Because silence is on path, M is then nonempty. Statistical uninformaticity implies that every always-disclosed signal induces action $1/2$, while silence, which pools whole signal realizations in M , also induces action $1/2$, contradicting the definition of H . Hence every signal in H has an interior cutoff, and cutoff indifference implies that all signals in H have a common cutoff.

$U^{CT2}(b(\rho))$. For $b \leq 1/4$, $b < b(\rho)$ implies $U^{H2}(b; \rho) < U^{CT2}(b)$, while $b > b(\rho)$ implies $U^{H2}(b; \rho) > U^{CT2}(b)$.

Now let $R(\pi) \triangleq \{\sum_{s \in H} p_s : \emptyset \neq H \subseteq S\} \subseteq (0, 1]$, and define

$$\beta^\circ(\pi) \triangleq \min\{b(\rho) : \rho \in R(\pi) \cap (0, 1)\}.$$

Because $N \geq 2$ and $p_s > 0$ for all s , the set $R(\pi) \cap (0, 1)$ is nonempty and finite, so $\beta^\circ(\pi) \in (1/8, 1/4)$.

Comparison with unverifiable evidence. If $b \leq \beta^\circ(\pi)$, then $b < 1/4$. Consider any equilibrium under verifiable evidence. If $\Pr(m = \circ) = 0$, its payoff is $-1/12 \leq U^{CT2}(b)$. Otherwise, take the canonical representation above. If $L = H = \emptyset$, the equilibrium is babbling and gives payoff $-1/12 < U^{CT2}(b)$. If both L and H are nonempty, the equilibrium has exactly three on-path actions and the preceding analysis gives payoff strictly below $U^{CT2}(b)$. If $L \neq \emptyset$ and $H = \emptyset$, the equilibrium has exactly two on-path actions with lower-tail disclosure and gives payoff at most $U^{CT2}(b)$. If $L = \emptyset$ and $H \neq \emptyset$, the equilibrium has exactly two on-path actions with upper-tail disclosure and is characterized by some $\rho \in R(\pi)$. If $\rho = 1$, its payoff is $U^{CT2}(b)$. If $\rho \in (0, 1)$, then $b \leq \beta^\circ(\pi) \leq b(\rho)$, so $U^{H2}(b; \rho) \leq U^{CT2}(b)$. Since $\rho = 1$ is feasible, the construction above attains $U^{CT2}(b)$, and the receiver-optimal equilibrium payoff under verifiable evidence is exactly $U^{CT2}(b)$.

If $b \in (\beta^\circ(\pi), 1/4]$, choose $\rho^* \in R(\pi) \cap (0, 1)$ such that $\beta^\circ(\pi) = b(\rho^*)$. Then $U^{H2}(b; \rho^*) > U^{CT2}(b)$. Since $\beta^\circ(\pi) > 1/8 > 1/12$, the receiver-optimal equilibrium under unverifiable evidence on $(\beta^\circ(\pi), 1/4)$ is exactly the two-message equilibrium, while at $b = 1/4$ every equilibrium under unverifiable evidence is babbling. Therefore, for every $b \in (\beta^\circ(\pi), 1/4]$, every receiver-optimal equilibrium under verifiable evidence is strictly more informative than every receiver-optimal equilibrium under unverifiable evidence.

Finally, if $b \in (1/4, 1/2)$, then every equilibrium under unverifiable evidence is babbling. Choose any $\rho \in R(\pi) \cap (0, 1)$. Since $g(0) < 0 < g(1)$, the upper-tail analysis above gives a unique cutoff $t = t(b, \rho) \in (0, 1)$. Moreover, $g(1 - 2b) = 1 - 2b > 0$, so $t < 1 - 2b$ and hence $x_h - a_\circ = 1 - t - 2b > 0$. Thus this is a strictly informative two-action equilibrium under verifiable evidence. Therefore every receiver-optimal equilibrium under verifiable evidence is strictly more informative than every receiver-optimal equilibrium under unverifiable evidence for every $b \in (1/4, 1/2)$. Combined with Lemma B.3, the threshold $b = 1/2$ is sharp. $\square$

Remark B.1. *Under the assumptions of Example 3, there exists $\beta^\circ \in (\frac{1}{8}, \frac{1}{4})$ such that:*

1. *If $b \geq \frac{1}{2}$, receiver-optimal equilibria are uninformative regardless of whether evidence is verifiable.*
2. *If $b \in (\beta^\circ, \frac{1}{2})$, receiver-optimal equilibria under verifiable evidence are strictly more informative than those under unverifiable evidence.*
3. *If $b \in [\frac{1}{12}, \beta^\circ]$, receiver-optimal equilibria are equally informative regardless of whether evidence is verifiable.*
4. *If $b < \frac{1}{12}$, receiver-optimal equilibria under verifiable evidence are strictly less informative than those under unverifiable evidence.*

Proof. The result follows from Lemmas B.1, B.2, and B.3, together with Proposition B.1. In particular, for $b < 1/12$, the receiver-optimal equilibrium under unverifiable evidence uses at least three on-path messages and gives the receiver strictly higher payoff than $U^{CT2}(b)$; by Proposition B.1, $U^{CT2}(b)$ is the receiver-optimal equilibrium payoff under verifiable evidence when $b \leq \beta^\circ$. For $b \in [1/12, \beta^\circ]$, the receiver-optimal equilibrium under unverifiable evidence is the two-message equilibrium and hence gives the same receiver payoff as the receiver-optimal equilibrium under verifiable evidence. For $b \in (\beta^\circ, 1/2)$, Proposition B.1 gives the strict ranking in favor of verifiable evidence. For $b \geq 1/2$, Lemma B.3 implies that every equilibrium under verifiable evidence is uninformative, while Lemmas B.1 and B.2 imply that every equilibrium under unverifiable evidence is also uninformative. $\square$

B.2 Equilibrium and Existence under Verifiable and Unverifiable Communication

B.2.1 Equilibrium Definition

First, we define a perfect Bayesian equilibrium in our setting. Recall that a belief system is a map $\mu : M \rightarrow \Delta(\Theta \times S)$. For each message m , let $\mu_\Theta(\cdot \mid m)$ denote the marginal of $\mu(\cdot \mid m)$ on Θ .

Definition 4 (PBE). *We define the following properties of an assessment (σ, α, μ) .*

1. (Receiver optimality). For every message $m \in M$, $\alpha(m) = a^{\mathcal{R}}(\mu_{\Theta}(\cdot | m))$.

2a. (Sender optimality under verifiable evidence). For every sender type (θ, s) and every message $m \in M$,

$$\sigma(m | \theta, s) > 0 \implies m \in \arg \max_{m' \in \{s, \circ\}} u(\alpha(m'), \theta, b).$$

2b. (Sender optimality under unverifiable evidence). For every sender type (θ, s) and every message $m \in M$,

$$\sigma(m | \theta, s) > 0 \implies m \in \arg \max_{m' \in M} u(\alpha(m'), \theta, b).$$

3. (Belief consistency). Beliefs are pinned down by Bayes' rule whenever possible. Off path, beliefs are arbitrary subject to feasibility: under verifiable evidence, after message $s \in S$, beliefs must be supported on $\Theta \times \{s\}$; after message $\circ$, every type is feasible. Under unverifiable evidence, every message is feasible for every type.

An assessment is a *Perfect Bayesian Equilibrium under verifiable evidence* if it satisfies properties 1, 2a, and 3. It is a *Perfect Bayesian Equilibrium under unverifiable evidence* if it satisfies properties 1, 2b, and 3.

We will use the following observation repeatedly. By the same argument as in Lemma A.2, for every posterior belief $\mu \in \Delta(\Theta)$, $a^{\mathcal{R}}(\mu) \in [0, 1]$. Conversely, every action $a \in [0, 1]$ is rationalizable by some posterior belief over Θ : since $\theta \mapsto a^{\mathcal{R}}(\theta)$ is continuous and strictly increasing, there exists $\theta_a \in [0, 1]$ such that $a^{\mathcal{R}}(\theta_a) = a$. Thus a is optimal under the degenerate belief at θ_a . Full support of the signal distribution implies that, under verifiable evidence, this posterior can be paired with any signal realization s after an off-path evidence message s .

B.2.2 Equilibrium Existence

We first show that equilibria exist under both communication technologies.

Lemma B.4. *For every $b \geq 0$, an uninformative PBE exists under both verifiable and unverifiable evidence.*

Proof. Consider the following assessment. The sender sends $m = \circ$ for every type (θ, s) . The receiver chooses $\alpha(m) = a^{\mathcal{R}}(F)$ after every message $m \in M$. After $\circ$, beliefs are given by Bayes' rule. After any off-path message $s \in S$, choose a belief supported on $\Theta \times \{s\}$ whose marginal on Θ is F .

Receiver optimality holds by construction. Belief feasibility after off-path evidence messages follows from full support. Sender optimality is immediate because every message induces the same receiver action. Thus the assessment is a PBE under verifiable evidence. The same construction is also a PBE under unverifiable evidence. $\square$

B.2.3 Existence of Receiver-Optimal Equilibria under Verifiable Evidence

We next prove that receiver-optimal equilibria exist under verifiable evidence. The argument uses the generalized-cutoff characterization in Lemma A.1.

For each signal $s \in S$, define the finite measure ρ_s on Θ by

$$\rho_s(B) \triangleq \int_B \pi(s \mid \theta) dF(\theta)$$

for every Borel set $B \subseteq \Theta$. Since F is atomless and ρ_s is absolutely continuous with respect to F , each ρ_s is atomless.

Fix a partition $P = (L, M, H)$ of S . For each cutoff vector $t = (t_s)_{s \in S} \in [0, 1]^S$, define

$$D_s^P(t) \triangleq \begin{cases} [0, t_s], & \text{if } s \in L, \\ \emptyset, & \text{if } s \in M, \\ [t_s, 1], & \text{if } s \in H. \end{cases}$$

The set $D_s^P(t)$ is the set of states in which signal s is disclosed. Let $C_s^P(t) \triangleq \Theta \setminus D_s^P(t)$ be the set of states in which a type with signal s sends silence.

Let $a = (a_\circ, (a_s)_{s \in S}) \in [0, 1]^{N+1}$. Define the receiver payoff induced by (P, t, a) as

$$U^V(P, t, a) \triangleq \sum_{s \in S} \left[\int_{D_s^P(t)} u(a_s, \theta, 0) d\rho_s(\theta) + \int_{C_s^P(t)} u(a_\circ, \theta, 0) d\rho_s(\theta) \right].$$

For each partition P , let $\mathcal{K}^V(P)$ be the set of all $(t, a) \in [0, 1]^S \times [0, 1]^{N+1}$ satisfying the following conditions:

$$\int_{D_s^P(t)} u_1(a_s, \theta, 0) d\rho_s(\theta) = 0 \quad \text{for all } s \in S, \quad (7)$$

$$\sum_{s \in S} \int_{C_s^P(t)} u_1(a_\circ, \theta, 0) d\rho_s(\theta) = 0, \quad (8)$$

$$a_s = a_\circ \quad \text{for all } s \in M, \quad (9)$$

$$\int_{D_s^P(t)} [u(a_\circ, \theta, b) - u(a_s, \theta, b)]_+ d\rho_s(\theta) = 0 \quad \text{for all } s \in S, \quad (10)$$

$$\int_{C_s^P(t)} [u(a_s, \theta, b) - u(a_\circ, \theta, b)]_+ d\rho_s(\theta) = 0 \quad \text{for all } s \in S. \quad (11)$$

Here $[x]_+ \triangleq \max\{x, 0\}$.

Lemma B.5. *For every $b \geq 0$, a receiver-optimal equilibrium exists under verifiable evidence.*

Proof. By Lemma A.1, every equilibrium outcome under verifiable evidence is outcome-equivalent to one induced by a generalized cutoff rule. We show that the set of such equilibrium outcomes is compact, and that the receiver's expected payoff attains a maximum on it.

Fix a partition $P = (L, M, H)$. The set $\mathcal{K}^V(P)$ is closed in the compact set $[0, 1]^S \times [0, 1]^{N+1}$. Indeed, the constraint maps in (7)–(11) are continuous in (t, a) . This follows from dominated convergence, atomlessness of the measures ρ_s , and continuity of u and u_1 on the compact set $[0, 1] \times \Theta \times \{0, b\}$. Hence $\mathcal{K}^V(P)$ is compact. Since there are only finitely many partitions P , the union $\mathcal{K}^V \triangleq \bigcup_P (\{P\} \times \mathcal{K}^V(P))$ is compact.

The set $\mathcal{K}^V$ is nonempty by Lemma B.4. The map U^V is continuous on $\mathcal{K}^V$, again by dominated convergence. Hence U^V attains a maximum on $\mathcal{K}^V$.

It remains to verify that $\mathcal{K}^V$ parameterizes exactly the relevant equilibrium outcomes. First, any generalized-cutoff PBE yields an element of $\mathcal{K}^V$. Conditions (7) and (8) are the receiver's first-order conditions after disclosed messages and silence; condition (9) is the canonical normalization for never-disclosed signals; and conditions (10) and (11) are precisely sender optimality on the disclosure and silence regions.

Conversely, take any $(P, t, a) \in \mathcal{K}^V$. Let the sender disclose s exactly on $D_s^P(t)$ and send $\circ$ on $C_s^P(t)$, assigning cutoff types to any best reply. Let the receiver choose a_s after message s and $a_\circ$ after silence. Beliefs are Bayesian after on-path messages. After off-path messages, choose beliefs that rationalize the prescribed actions; this is possible because all prescribed actions lie in $[0, 1]$ and full support makes every state compatible with every signal realization. Conditions (7) and (8) imply receiver optimality on path. Conditions (10) and (11), together with continuity and atomlessness, imply sender optimality at all interior types; cutoff types can

be assigned best replies without changing the outcome. Therefore the constructed assessment is a PBE under verifiable evidence.

Thus the maximizer of U^V over $\mathcal{K}^V$ induces a PBE whose receiver payoff is at least as high as that of any other PBE under verifiable evidence. This equilibrium is receiver-optimal. $\square$

B.2.4 Unverifiable Evidence and Cheap-Talk Equivalence

As discussed in the main text, the environment with unverifiable evidence is a standard cheap-talk environment with a restricted message alphabet: the sender can use at most $N + 1$ messages. The evidence structure (S, π) plays no role beyond determining the size of this alphabet.

To see this, note that when evidence is unverifiable, the sender can choose any message $m \in M$ regardless of the realized signal s , and payoffs depend on (a, θ) but not directly on s . Thus the realized signal is strategically inessential. Formally, for any sender strategy $\sigma : \Theta \times S \rightarrow \Delta(M)$, define the reduced strategy $\hat{\sigma} : \Theta \rightarrow \Delta(M)$ by

$$\hat{\sigma}(m \mid \theta) \triangleq \sum_{s \in S} \pi(s \mid \theta) \sigma(m \mid \theta, s).$$

For any receiver strategy $\alpha : M \rightarrow A$, the induced distribution of (θ, m) and the receiver's posterior after each message depend on σ only through $\hat{\sigma}$. Therefore, every PBE outcome under unverifiable evidence can be replicated by a PBE in which the sender ignores the realized signal and uses the reduced strategy $\hat{\sigma}$. The converse is immediate. Consequently, the game under unverifiable evidence is outcome-equivalent to a finite-message cheap-talk game with message space M .

B.2.5 Existence of Receiver-Optimal Equilibria under Unverifiable Evidence

We now prove existence of receiver-optimal equilibria under unverifiable evidence.

Lemma B.6. *For every $b \geq 0$, a receiver-optimal equilibrium exists under unverifiable evidence.*

Proof. By the cheap-talk equivalence above, it suffices to prove the result for the finite-message cheap-talk problem with message space M . Let $K \triangleq N + 1$.

First suppose $b > 0$. By Lemma A.6, every equilibrium under unverifiable evidence is outcome-equivalent to an ordered interval partition with at most K cells.

It remains to verify that the receiver's payoff attains a maximum over the feasible partition equilibria. For each $J \in \{1, \dots, K\}$, let

$$T_J \triangleq \{(t_0, \dots, t_J) \in [0, 1]^{J+1} : 0 = t_0 \leq t_1 \leq \dots \leq t_{J-1} \leq t_J = 1\}.$$

For $t \in T_J$, write $I_j(t) \triangleq [t_{j-1}, t_j]$ for $j = 1, \dots, J$, and let $x = (x_1, \dots, x_J) \in [0, 1]^J$. Define

$$U_J^{CT}(t, x) \triangleq \sum_{j=1}^J \int_{I_j(t)} u(x_j, \theta, 0) dF(\theta).$$

Let $\mathcal{K}_J^{CT}$ be the set of all $(t, x) \in T_J \times [0, 1]^J$ satisfying

$$\int_{I_j(t)} u_1(x_j, \theta, 0) dF(\theta) = 0 \quad \text{for all } j = 1, \dots, J, \quad (12)$$

$$\int_{I_j(t)} [u(x_k, \theta, b) - u(x_j, \theta, b)]_+ dF(\theta) = 0 \quad \text{for all } j, k \in \{1, \dots, J\}. \quad (13)$$

Here $[z]_+ \triangleq \max\{z, 0\}$. Conditions (12) and (13) are receiver optimality and sender optimality for an ordered partition equilibrium. They also allow for degenerate cells, which simply represent limits of equilibria with fewer effective cells.

For each J , the set $\mathcal{K}_J^{CT}$ is closed in the compact set $T_J \times [0, 1]^J$. The constraint maps are continuous by dominated convergence and atomlessness of F . Hence $\mathcal{K}_J^{CT}$ is compact. Since there are finitely many $J \leq K$, the union $\mathcal{K}^{CT} \triangleq \bigcup_{J=1}^K (\{J\} \times \mathcal{K}_J^{CT})$ is compact. It is nonempty because the babbling equilibrium belongs to $\mathcal{K}_1^{CT}$. The receiver payoff U_J^{CT} is continuous on each $\mathcal{K}_J^{CT}$, so it attains a maximum over $\mathcal{K}^{CT}$. By Lemma A.6, every equilibrium outcome under unverifiable evidence is represented in $\mathcal{K}^{CT}$, and every point in $\mathcal{K}^{CT}$ induces a PBE outcome after omitting degenerate cells. Therefore a maximizer is a receiver-optimal equilibrium.

Now suppose $b = 0$. Sender and receiver preferences coincide. Any equilibrium under unverifiable evidence induces at most K receiver actions. Conversely, any finite-valued action rule that is receiver-optimal conditional on each induced cell and pointwise optimal among its induced actions can be supported as an equilibrium, because sender and receiver have common interests. By strict single crossing, the receiver-payoff-maximizing rule among all rules with at most K induced actions can be chosen to be an ordered interval partition. The set of ordered interval partitions with at most K cells is compact, and the receiver payoff induced by posterior-optimal actions on the cells is continuous in the cutoffs. Therefore a maximizer exists. Since

preferences coincide at $b = 0$, the sender is willing to follow the induced partition rule, and the maximizer is a receiver-optimal equilibrium. $\square$

Combining Lemmas B.5 and B.6 yields the existence claim used in the main text.

Proposition B.2. *For every $b \geq 0$, receiver-optimal equilibria exist under both verifiable and unverifiable evidence.*

Proof. This follows immediately from Lemmas B.5 and B.6. $\square$

B.2.6 What if the Sender Does Not Observe θ ?

Finally, we record a benchmark result used in the main text. Suppose the sender observes only the realized signal s , not the payoff-relevant state θ . Then disclosure exhausts the sender's private information.

Proposition B.3. *Suppose the sender observes only s , not θ . Then for every $b \geq 0$, a FED equilibrium exists and is receiver-optimal.*

Proof. Consider the assessment in which the sender discloses s , the receiver chooses $a^{\mathcal{R}}(F_s)$ after every signal message s , and the receiver chooses 0 after silence. Beliefs after disclosed signals are given by Bayes' rule, and the off-path belief after silence assigns probability one to $\theta = 0$.

Receiver optimality is immediate. We verify sender optimality. Conditional on signal s , the sender's expected payoff as a function of the receiver's action is strictly concave and is maximized at $a^{\mathcal{S}}(F_s, b)$. Since $a^{\mathcal{S}}(F_s, b) \geq a^{\mathcal{R}}(F_s)$ and $a^{\mathcal{R}}(F_s) \geq 0$, the sender weakly prefers $a^{\mathcal{R}}(F_s)$ to 0. Hence disclosure is optimal for every signal type, and the constructed assessment is a FED equilibrium.

To see receiver optimality, observe that if the sender observes only s , any verifiable-communication outcome is a garbling of the signal s : after each realized signal, the sender can only disclose s or remain silent, and there is no additional payoff-relevant private information to encode. The receiver can weakly improve on any garbling by observing the full signal and choosing the posterior-optimal action $a^{\mathcal{R}}(F_s)$ after each s . Therefore the FED outcome is receiver-optimal. $\square$

B.3 Equilibrium Structure and Existence in the Hybrid Game

Definition 5. A sender strategy is a **hybrid generalized cutoff rule** if, for every $s \in S$, there exist an integer $J_s \geq 1$, cutoffs $0 = t_{s,0} < t_{s,1} < \dots < t_{s,J_s} = 1$, and message pairs $z_{s,j} = (m_{s,j}^V, m_{s,j}^U) \in \{\circ, s\} \times M^U$ for $j = 1, \dots, J_s$, such that a type (θ, s) sends $z_{s,j}$ whenever $\theta \in (t_{s,j-1}, t_{s,j})$. Behavior at the cutoff states may be assigned arbitrarily among best replies. The same concealed pair $(\circ, m^U)$ may be used on cells associated with different signals.

Lemma B.7. Suppose M^U is finite. Every equilibrium of the hybrid game is outcome-equivalent to one in which the sender uses a hybrid generalized cutoff rule. Moreover, the representation can be chosen so that, for every $s \in S$,

$$\alpha(z_{s,1}) < \dots < \alpha(z_{s,J_s}) \quad \text{and} \quad J_s \leq 2|M^U|.$$

Proof. Fix an equilibrium (σ, α, μ) of the hybrid game. By strict concavity of the receiver's payoff, we may take α to be pure. For each $s \in S$, let

$$X_s \triangleq \{\alpha(\circ, m^U) : m^U \in M^U\} \cup \{\alpha(s, m^U) : m^U \in M^U\}$$

be the finite set of actions available to a type with signal s . Clearly, $|X_s| \leq 2|M^U|$.

For any distinct $x, y \in X_s$, the equation $u(x, \theta, b) = u(y, \theta, b)$ has at most one solution by strict single crossing. Thus, except at finitely many states, the sender has a unique optimal action in X_s . Since F is atomless, we may assign each exceptional type an arbitrary best reply and thereby obtain a pure induced-action rule without changing the outcome. Strict single crossing also implies that this rule is nondecreasing in θ .

After omitting actions used only on a null set, write the actions induced conditional on s as $a_{s,1} < \dots < a_{s,J_s}$. Their inverse images are ordered intervals separated by cutoffs $0 = t_{s,0} < \dots < t_{s,J_s} = 1$. For each $a_{s,j}$, choose one feasible message pair $z_{s,j} \in \{\circ, s\} \times M^U$ that induces it, and let every type in the corresponding interval send this pair. This reassignment preserves sender optimality, receiver optimality, and the induced outcome. Vacated pairs may retain their original actions and beliefs.

At every interior cutoff, the adjacent actions are both best replies by continuity, so the cutoff type may be assigned arbitrarily between them. The resulting strategy is therefore a hybrid generalized cutoff rule satisfying $\alpha(z_{s,j}) = a_{s,j}$ and

$$J_s \leq |X_s| \leq 2|M^U|.$$

□

Remark B.2. Lemma B.7 extends to $M^U = [0, 1]$ whenever $b > 0$. Define

$$\delta_b \triangleq \min_{\theta \in \Theta} \left\{ a^S(\theta, b) - a^R(\theta) \right\} > 0 \quad \text{and} \quad K_b \triangleq \left\lceil \frac{1}{\delta_b} \right\rceil.$$

Following any fixed verifiable message, any two distinct on-path actions $x < y$ must satisfy $y - x > \delta_b$. Hence at most K_b distinct actions can be induced following each fixed verifiable message, and every equilibrium is outcome-equivalent to one in hybrid generalized cutoff form with $J_s \leq 2K_b$ for every $s \in S$.

We leave the straightforward verification of the remark to the reader.

B.3.1 Existence of Receiver-Optimal Equilibria in the Hybrid Game

Suppose M^U is finite, and let $K \triangleq |M^U|$. Write

$$Z \triangleq (S \cup \{\circ\}) \times M^U \quad \text{and} \quad Z_s \triangleq \{\circ, s\} \times M^U$$

for the hybrid message space and the set of messages available to a type with signal s , respectively.

A hybrid cutoff pattern P specifies, for every $s \in S$, an integer $J_s \in \{1, \dots, 2K\}$ and message pairs $z_{s,j} \in Z_s$ for $j = 1, \dots, J_s$. For each pattern P , let $T(P)$ be the set of cutoff vectors satisfying

$$0 = t_{s,0} \leq t_{s,1} \leq \dots \leq t_{s,J_s} = 1 \quad \text{for every } s \in S.$$

For $t \in T(P)$, write $I_{s,j}(t) \triangleq [t_{s,j-1}, t_{s,j}]$. Let $a = (a_z)_{z \in Z} \in [0, 1]^Z$, and define

$$U^H(P, t, a) \triangleq \sum_{s \in S} \sum_{j=1}^{J_s} \int_{I_{s,j}(t)} u(a_{z_{s,j}}, \theta, 0) d\rho_s(\theta).$$

For each pattern P , let $\mathcal{K}^H(P)$ be the set of all $(t, a) \in T(P) \times [0, 1]^Z$ satisfying

$$\sum_{s \in S} \sum_{\substack{j=1, \dots, J_s \\ z_{s,j}=z}} \int_{I_{s,j}(t)} u_1(a_z, \theta, 0) d\rho_s(\theta) = 0 \quad \text{for every } z \in Z, \tag{14}$$

$$\int_{I_{s,j}(t)} \left[u(a_{z'}, \theta, b) - u(a_{z_{s,j}}, \theta, b) \right]_+ d\rho_s(\theta) = 0 \quad \text{for every } s \in S, j = 1, \dots, J_s, z' \in Z_s. \tag{15}$$

Lemma B.8. *Suppose M^U is finite. For every $b \geq 0$, a receiver-optimal equilibrium exists in the hybrid game.*

Proof. By Lemma B.7, every hybrid equilibrium outcome is outcome-equivalent to one represented by some hybrid cutoff pattern P . We show that the set of such equilibrium outcomes is compact.

Fix a pattern P . The set $\mathcal{K}^H(P)$ is closed in the compact set $T(P) \times [0, 1]^Z$. Indeed, all the constraint maps in (14)–(15) are continuous in (t, a) by dominated convergence, atomlessness of the measures ρ_s , and continuity of u and u_1 . Hence $\mathcal{K}^H(P)$ is compact. Since Z is finite, there are only finitely many hybrid cutoff patterns, so

$$\mathcal{K}^H \triangleq \bigcup_P (\{P\} \times \mathcal{K}^H(P))$$

is compact. It is nonempty because the hybrid game admits a babbling equilibrium in which every type sends a fixed concealed pair and every message induces $a^R(F)$. The receiver payoff U^H is continuous on $\mathcal{K}^H$ and therefore attains a maximum.

It remains to verify that $\mathcal{K}^H$ parameterizes the relevant equilibrium outcomes. Any equilibrium in hybrid generalized cutoff form satisfies (14) by receiver optimality and (15) by sender optimality. Conversely, take any $(P, t, a) \in \mathcal{K}^H(P)$. After omitting degenerate cells, let each type (θ, s) send $z_{s,j}$ on the interior of $I_{s,j}(t)$, and let the receiver choose a_z after every $z \in Z$. Beliefs are Bayesian after on-path messages, while off-path beliefs are chosen to rationalize the prescribed actions. Condition (14) implies receiver optimality. Condition (15), together with continuity and full support, implies sender optimality at every interior type; cutoff types may be assigned arbitrary best replies without changing the outcome. Thus every point in $\mathcal{K}^H$ induces a hybrid equilibrium outcome.

A maximizer of U^H over $\mathcal{K}^H$ therefore induces a receiver-optimal equilibrium. $\square$

Remark B.3. *The conclusion of Lemma B.8 also holds when $M^U = [0, 1]$. If $b > 0$, Remark B.2 gives a uniform bound of K_b on the number of distinct actions induced following each fixed verifiable message. Since unverifiable messages are payoff-irrelevant labels, the unverifiable messages used to induce these actions may be assigned to a fixed finite subset of $[0, 1]$, after which the preceding compactness argument applies. If $b = 0$, the sender can fully reveal θ through the unverifiable message, and the receiver chooses $a^R(\theta)$. This equilibrium attains the first-best receiver payoff and is therefore receiver-optimal.*

We leave the straightforward verification of the remark to the reader.

B.4 Optimal Flexibility

We conclude the paper by briefly considering a more general version of our model that nests the two extremes studied so far—verifiable and unverifiable evidence—as special cases. In this more general formulation, the sender can send any message regardless of her type, but incurs a cost whenever she sends a message that is neither the realized signal nor silence. To keep the analysis simple, we specialize to the quadratic payoff environment of Example 1. Thus the only departure is that the sender’s payoff becomes

$$u^S(a, \theta, s, m) = -(a - \theta - b)^2 - c \cdot \mathbb{1}\{m \notin \{s, \circ\}\},$$

where $c \in [0, \infty]$ is a *fabrication cost*, which is common knowledge. Thus, when the sender is either silent or discloses the realized signal, she bears no cost. If instead she announces any other signal realization, she incurs the cost c . This formulation nests the two benchmark communication models studied in the paper. When $c = 0$, this model reduces to the game with unverifiable evidence. When $c \geq 1 + 2b$, it reduces to the game with verifiable evidence.

This way of modeling the fabrication cost is rather simple and stark. In a more general model, the cost could depend more finely on s and m .⁴ Yet, this formulation will deliver a rather robust point: strictly enforcing verifiability, i.e., setting a large penalty for fabrication, is not optimal when the sender’s bias is large.

Before stating this result formally, note first that the notion of an outcome λ must now be understood as the joint distribution of (θ, s, m, a) . For each cost c , let $\Lambda^{\text{PBE}}(c)$ denote the set of equilibrium outcomes in the game with cost c , and define $\mathcal{U}(c) \triangleq \sup_{\lambda \in \Lambda^{\text{PBE}}(c)} \mathbb{E}_\lambda[u^R(a, \theta)]$. We say that an equilibrium is *receiver-optimal* if it attains $\mathcal{U}(c)$. We say that a fabrication cost c^* is *optimal* if $\mathcal{U}(c^*) \geq \mathcal{U}(c)$ for all $c \in [0, \infty]$.

Lemma B.9. *Suppose payoffs are quadratic with an additive sender bias b . For every $c \geq 0$, a receiver-optimal equilibrium exists. Moreover, an optimal fabrication cost exists.*

⁴Kartik et al. (2007) and Kartik (2009) study models of lying costs in cheap talk models. Perez-Richet and Skreta (2022) study falsification costs in a setting with designed information.

Proof. It is without loss to restrict c to $[0, 1 + 2b]$. The argument is then the same finite-pattern compactness argument as in Lemma B.8. Conditional on each s , the sender chooses from the finite message space $M = S \cup \{\circ\}$. Since message costs are independent of θ , single crossing implies that every equilibrium outcome is outcome-equivalent to one generated by an ordered interval rule with at most $|M|$ cells conditional on each signal. There are finitely many such patterns, and the corresponding cutoff, action, receiver-optimality, and sender-optimality constraints are continuous in c . Hence the joint feasible set over $c \in [0, 1 + 2b]$ is compact and nonempty, and the receiver's payoff is continuous on it. Maximizing at a fixed c , and then jointly over c , proves the two claims. $\square$

Proposition B.4. *Assume payoffs are quadratic with an additive sender bias b , and (S, π) is informative.⁵ Suppose $b \geq 1$. Then every optimal cost satisfies $c^* < 1 + 2b$, so verifiable communication is suboptimal. Moreover, in a receiver-optimal equilibrium at c^* , the sender fabricates evidence with strictly positive probability.*

Despite the sender's bias being large, it is optimal to grant the sender some flexibility. The receiver is strictly better off in a regime that allows some amount of fabrication. Full verifiability is not optimal.⁶

To understand why allowing some fabrication can improve the receiver's payoff, consider the senders with the strongest incentive to lie under full evidence disclosure: those who observe the lowest signal realization s_1 but know that the state θ is above some threshold t close to 1. If the fabrication cost c is sufficiently small, these senders would prefer to report $m = s_N$ rather than s_1 . When c is chosen appropriately, one can make these senders the only ones who have an incentive to fabricate the signal. This selective lying benefits the receiver. By reassigning these high- θ senders away from message s_1 , it makes the receiver's action following s_1 more pessimistic. At the same time, if the threshold t is chosen appropriately, the action following s_N becomes more optimistic. The reason is that the additional senders who pool with s_N are precisely those with very high values of θ , so their presence worsens the interpretation of s_N

⁵For this result, MLRP is substantive rather than merely expositional: it orders the signal posteriors used in the construction.

⁶It is easy to show that, for every prior F and evidence structure (S, π) , the optimal fabrication cost at $b = 0$ is $c^* = 0$. In other words, unverifiable communication is optimal when preferences are aligned. To keep the paper concise, we omit this result.

only slightly and can in fact raise the posterior mean after s_N .

Proof of Proposition B.4. Consider any equilibrium outcome with no lying, that is, an outcome $\lambda \in \Lambda^{\text{PBE}}(c)$ such that $\Pr_\lambda(m \notin \{s, \circ\}) = 0$. Then λ is also an equilibrium outcome of the game with verifiable evidence. Indeed, on path the sender uses only messages in $\{s, \circ\}$, so the same on-path behavior is feasible in the game with verifiable evidence; sender optimality is preserved because the sender's feasible deviations in the game with verifiable evidence are a subset of those in the lying-cost game; and off-path beliefs in the game with verifiable evidence can be chosen so that the receiver takes the same off-path actions as in the lying-cost game. By Proposition 2, when $b \geq 1$, the FED outcome is efficient and maximizes the receiver's expected payoff. Hence, every zero-lying equilibrium outcome gives the receiver at most $U^{\text{FED}} \triangleq -\sum_{i=1}^N p_i \mathbb{E}[(\mathbb{E}(\theta | s_i) - \theta)^2 | s_i]$, where $p_i \triangleq \Pr(s_i)$.

We now show that there exists a finite cost and an equilibrium with positive lying that gives the receiver strictly more than U^{FED} . Write $\bar{a}_i \triangleq \mathbb{E}[\theta | s_i]$ for $i = 1, \dots, N$. By MLRP, $\bar{a}_1 \leq \dots \leq \bar{a}_N$, and because the evidence structure is informative, $\bar{a}_1 < \bar{a}_N$. Let $\mathcal{I} \triangleq \{i \in \{1, \dots, N\} : \bar{a}_i = \bar{a}_1\}$. For every $i \in \mathcal{I}$, by MLRP the posterior under s_i first-order stochastically dominates that under s_1 , and since $\bar{a}_i = \bar{a}_1$, the two posterior distributions coincide.

Fix $t \in (\bar{a}_N, 1)$ and consider the sender strategy

$$\sigma(m | \theta, s) = \begin{cases} s_N, & \text{if } s = s_i \text{ for some } i \in \mathcal{I} \text{ and } \theta > t, \\ s, & \text{otherwise.} \end{cases}$$

Denote $q(t) \triangleq \Pr(\theta > t | s_1)$, $\mu(t) \triangleq \mathbb{E}[\theta | s_1, \theta > t]$, and $p_{\mathcal{I}} \triangleq \sum_{i \in \mathcal{I}} p_i$. Under this strategy, the receiver's posterior means after on-path messages are $a_i(t) = a_1(t) = \mathbb{E}[\theta | s_1, \theta \leq t]$ for $i \in \mathcal{I}$, $a_i = \bar{a}_i$ for all $i \in \{1, \dots, N-1\} \setminus \mathcal{I}$, and $a_N(t) = \frac{p_N \bar{a}_N + p_{\mathcal{I}} q(t) \mu(t)}{p_N + p_{\mathcal{I}} q(t)}$. Because $t > \bar{a}_N$ and $\mu(t) > t$, we have $a_N(t) > \bar{a}_N \geq \bar{a}_i$ for every i . Finally, let the off-path action $a_\circ = 0$. Since $b \geq 1$, $a_\circ$ is the worst action for all senders, and thus no one has a strict incentive to deviate to it. Define the lying cost

$$c(t) \triangleq (a_N(t) - a_1(t)) (2(t + b) - a_N(t) - a_1(t)) < 1 + 2b.$$

For a type (θ, s_i) with $i \in \mathcal{I}$, the gain from sending $m = s_N$ instead of $m = s_i$ is

$$\begin{aligned} \Delta_1(\theta; t) &\triangleq (- (a_N(t) - \theta - b)^2 - c(t)) - (- (a_1(t) - \theta - b)^2) \\ &= 2(a_N(t) - a_1(t))(\theta - t), \end{aligned}$$

which is positive if and only if $\theta > t$. So exactly the intended types, namely those with $i \in \mathcal{I}$ and $\theta > t$, have an incentive to lie under cost $c(t)$.

Next, we show no other types have an incentive to lie. Since $b \geq 1$, every sender strictly prefers higher receiver actions. Because $a_N(t) > \bar{a}_i$ for every $i \neq N$, and every lie costs the same amount $c(t)$, any profitable lie would have to be a deviation to $m = s_N$, which induces the highest action. Fix $i \in \{1, \dots, N-1\} \setminus \mathcal{I}$. The gain from sending $m = s_N$ rather than $m = s_i$ is $\Delta_i(\theta; t) \triangleq (a_N(t) - \bar{a}_i)(2(\theta + b) - a_N(t) - \bar{a}_i) - c(t)$. Since $a_N(t) > \bar{a}_i$, $\Delta_i(\theta; t)$ is increasing in θ , so the most profitable such deviation occurs at $\theta = 1$. By continuity, each $\Delta_i(1; t)$ is continuous at $t = 1$, and $\Delta_i(1; 1) = g(\bar{a}_i) - g(\bar{a}_1)$ with $g(a) \triangleq (\bar{a}_N - a)(2(1 + b) - \bar{a}_N - a)$. Moreover, $g'(a) = -2(1 + b - a) < 0$, so g is strictly decreasing. Since $i \notin \mathcal{I}$, we have $\bar{a}_i > \bar{a}_1$, and thus $\Delta_i(1; 1) < 0$. Therefore for t sufficiently close to 1, no such type wants to deviate. Finally, a type with signal s_N already obtains the highest action without paying cost, so never deviates. Thus for t close to 1, this defines a PBE with outcome λ_t such that $\Pr_{\lambda_t}(m \notin \{s, \circ\}) > 0$.

It remains to compare the receiver's payoff under λ_t with U^{FED} . Note that only the messages associated with s_i for $i \in \mathcal{I}$ and s_N are altered. Under posterior-mean actions, we have $\mathbb{E}[-(a - \theta)^2] = -\mathbb{E}[\theta^2] + \mathbb{E}[a^2]$, so

$$U(\lambda_t) - U^{\text{FED}} = p_{\mathcal{I}}(1 - q(t))a_1(t)^2 + (p_N + p_{\mathcal{I}}q(t))a_N(t)^2 - p_{\mathcal{I}}\bar{a}_1^2 - p_N\bar{a}_N^2.$$

Add and subtract $p_{\mathcal{I}}q(t)\mu(t)^2$:

$$\begin{aligned} U(\lambda_t) - U^{\text{FED}} &= \left[p_{\mathcal{I}}(1 - q(t))a_1(t)^2 + p_{\mathcal{I}}q(t)\mu(t)^2 - p_{\mathcal{I}}\bar{a}_1^2 \right] \\ &\quad - \left[p_{\mathcal{I}}q(t)\mu(t)^2 + p_N\bar{a}_N^2 - (p_N + p_{\mathcal{I}}q(t))a_N(t)^2 \right]. \end{aligned}$$

The first bracket equals $p_{\mathcal{I}}q(t)(1 - q(t))(\mu(t) - a_1(t))^2$, and the second equals $\frac{p_N p_{\mathcal{I}}q(t)}{p_N + p_{\mathcal{I}}q(t)}(\mu(t) - \bar{a}_N)^2$. Hence

$$U(\lambda_t) - U^{\text{FED}} = p_{\mathcal{I}}q(t) \left[(1 - q(t))(\mu(t) - a_1(t))^2 - \frac{p_N}{p_N + p_{\mathcal{I}}q(t)}(\mu(t) - \bar{a}_N)^2 \right].$$

As $t \rightarrow 1$, we have $q(t) \rightarrow 0$, $a_1(t) \rightarrow \bar{a}_1$, and $\mu(t) \rightarrow 1$. Therefore,

$$\frac{U(\lambda_t) - U^{\text{FED}}}{p_{\mathcal{I}}q(t)} \rightarrow (1 - \bar{a}_1)^2 - (1 - \bar{a}_N)^2 = (\bar{a}_N - \bar{a}_1)(2 - \bar{a}_1 - \bar{a}_N) > 0.$$

Hence for t sufficiently close to 1, $U(\lambda_t) > U^{\text{FED}}$.

Finally, suppose λ^* is receiver-optimal at optimal c^* and has $\Pr_{\lambda^*}(m \notin \{s, \circ\}) = 0$. Then $\mathcal{U}(c^*) \leq U^{\text{FED}}$. But we constructed $c(t)$ with $\mathcal{U}(c(t)) \geq U(\lambda_t) > U^{\text{FED}}$, a contradiction. Therefore $\Pr_{\lambda^*}(m \notin \{s, \circ\}) > 0$. $\square$